



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/11

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Blank pages are indicated.



- 1 Nitrogen is heated in a balloon, which expands slightly.

Which statements about the molecules of nitrogen are correct?

- 1 They move further apart.
- 2 They move more quickly.
- 3 They remain the same distance apart.
- 4 Their speed remains unchanged.

**A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

- 2 Which piece of apparatus should be used to measure exactly  $21.4 \text{ cm}^3$  of water?

- A**  $25 \text{ cm}^3$  beaker  
**B**  $25 \text{ cm}^3$  pipette  
**C**  $50 \text{ cm}^3$  burette  
**D**  $50 \text{ cm}^3$  measuring cylinder

- 3 Which method of separation is used to separate a soluble solid from its solution?

- A** chromatography  
**B** condensation  
**C** crystallisation  
**D** filtration

- 4 The atomic number and nucleon number of a potassium atom are shown.

|                | potassium atom |
|----------------|----------------|
| atomic number  | 19             |
| nucleon number | 39             |

How many protons, neutrons and electrons are in a potassium ion,  $\text{K}^+$ ?

|          | protons | neutrons | electrons |
|----------|---------|----------|-----------|
| <b>A</b> | 19      | 20       | 18        |
| <b>B</b> | 19      | 20       | 20        |
| <b>C</b> | 20      | 19       | 18        |
| <b>D</b> | 20      | 19       | 19        |

- 5 Sodium is in Group I of the Periodic Table.

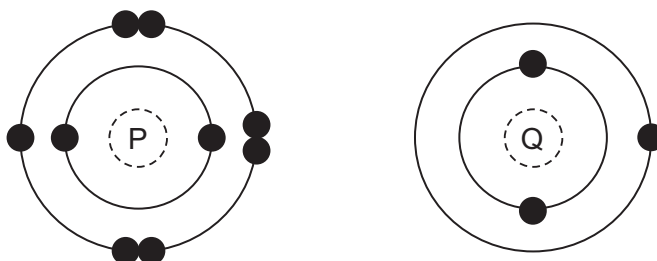
Chlorine is in Group VII of the Periodic Table.

Sodium and chlorine combine to form a compound.

Which statement about the combination of sodium and chlorine atoms is correct?

- A Both sodium and chlorine lose electrons.
- B Both sodium and chlorine gain electrons.
- C Sodium loses electrons and chlorine gains electrons.
- D Sodium gains electrons and chlorine loses electrons.

- 6 The electronic structures of two atoms, P and Q, are shown.

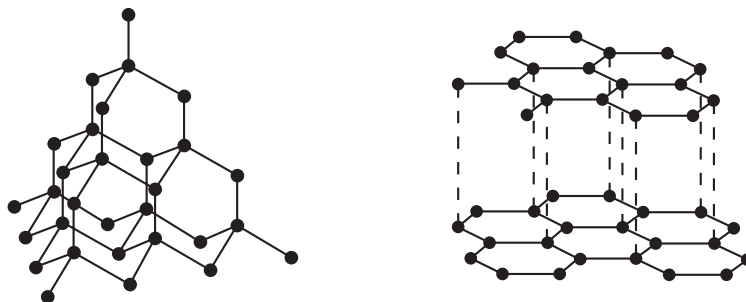


P and Q combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

|   | type of bonding | formula         |
|---|-----------------|-----------------|
| A | ionic           | PQ              |
| B | ionic           | PQ <sub>2</sub> |
| C | covalent        | PQ <sub>2</sub> |
| D | covalent        | PQ              |

- 7 The structures of diamond and graphite are shown.



Which statement about diamond and graphite is correct?

- A Diamond and graphite have low melting points.
  - B Diamond and graphite have mobile electrons.
  - C Diamond and graphite have layered structures.
  - D Diamond and graphite contain strong covalent bonds between carbon atoms.
- 8 Aluminium oxide has the formula  $Al_2O_3$ .

Which statement about aluminium oxide is correct?

- A 2 g of aluminium atoms are combined with 3 g of oxygen atoms.
  - B 2 g of aluminium atoms are combined with 3 g of oxygen molecules.
  - C Aluminium oxide has a relative formula mass of 102.
  - D Pure aluminium oxide contains a higher mass of oxygen than of aluminium.
- 9 Which products are formed when dilute sulfuric acid undergoes electrolysis?

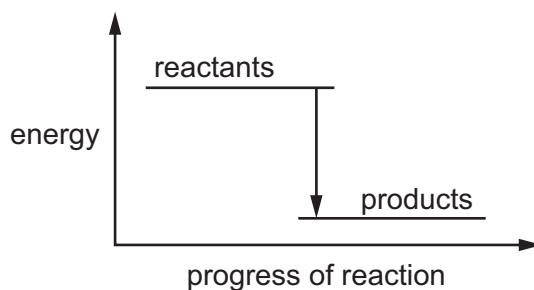
|   | at the anode   | at the cathode |
|---|----------------|----------------|
| A | oxygen         | hydrogen       |
| B | hydrogen       | oxygen         |
| C | sulfur dioxide | hydrogen       |
| D | oxygen         | sulfur dioxide |

- 10 Which element is **not** used as a fuel?

- A carbon
- B helium
- C hydrogen
- D uranium



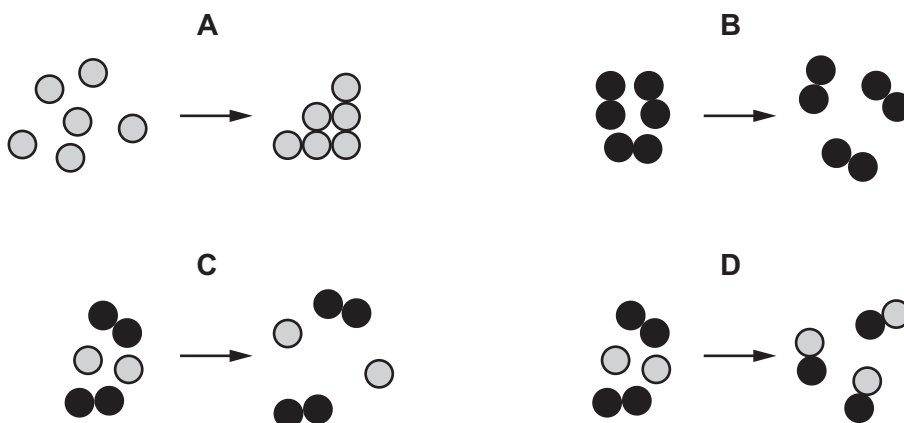
11 The energy level diagram shows the energy of the reactants and products in a chemical reaction.



Which row correctly describes the energy change and the type of reaction shown?

|          | description of energy change             | type of reaction |
|----------|------------------------------------------|------------------|
| <b>A</b> | energy is given out to the surroundings  | endothermic      |
| <b>B</b> | energy is given out to the surroundings  | exothermic       |
| <b>C</b> | energy is taken in from the surroundings | endothermic      |
| <b>D</b> | energy is taken in from the surroundings | exothermic       |

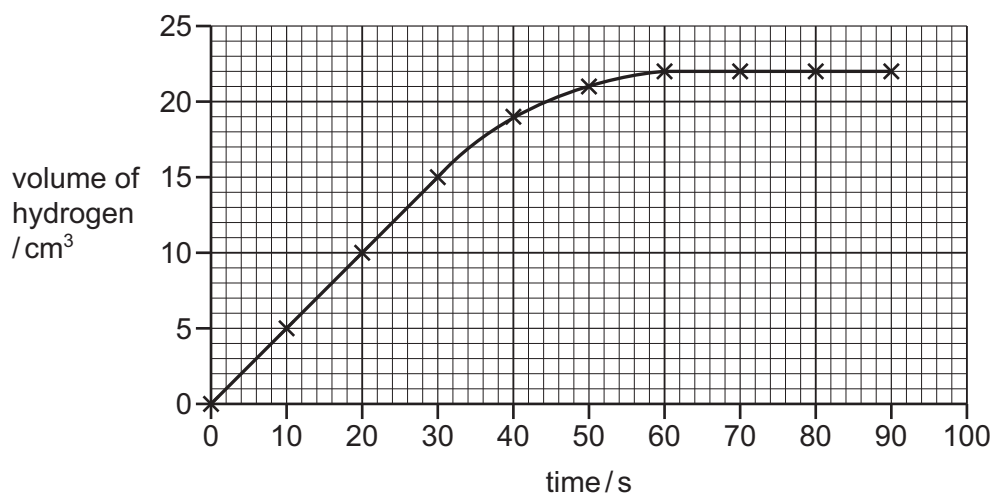
12 Which diagram represents a chemical change?



**13** The rate of reaction between magnesium and hydrochloric acid is investigated.

The volume of hydrogen given off at different times is measured.

The results are shown.



Which conclusions are correct?

- 1 The rate is fastest between 0 and 20 seconds.
- 2 The maximum volume of hydrogen given off is 22 cm<sup>3</sup>.
- 3 At 40 seconds, 20 cm<sup>3</sup> of hydrogen is given off.

**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

**14** Which reaction can be easily reversed?

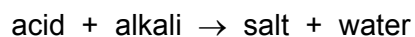
- A** dissolving zinc in hydrochloric acid
- B** fermenting glucose with yeast
- C** heating hydrated cobalt(II) chloride
- D** the rusting of an iron nail

**15** Carbon reacts with silver oxide to form carbon dioxide and silver.

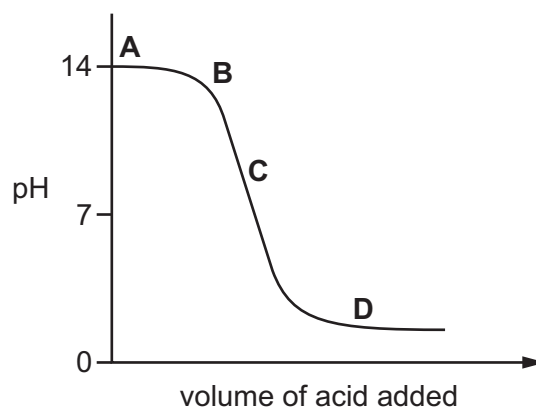
Which substance is reduced?

- A** carbon
- B** carbon dioxide
- C** silver
- D** silver oxide

- 16 The graph shows how the pH of a solution changes as an acid is added to an alkali.



Which letter represents the area of the graph where both acid and salt are present?



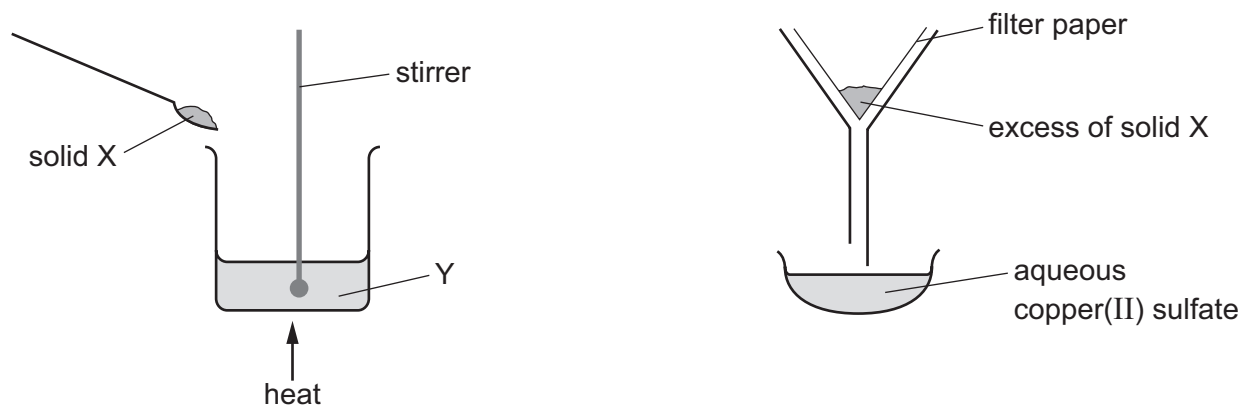
- 17 Phosphorus is an element in Group V of the Periodic Table.

It burns in air to form an oxide, which dissolves in water to form a solution with a pH of 1.

Which row describes this oxide of phosphorus?

|          | metal<br>oxide | non-metal<br>oxide | acidic<br>oxide | basic<br>oxide |
|----------|----------------|--------------------|-----------------|----------------|
| <b>A</b> | ✓              | x                  | ✓               | x              |
| <b>B</b> | ✓              | x                  | x               | ✓              |
| <b>C</b> | x              | ✓                  | ✓               | x              |
| <b>D</b> | x              | ✓                  | x               | ✓              |

18 The apparatus shown is used to prepare aqueous copper(II) sulfate.



What are X and Y?

|          | X                   | Y                           |
|----------|---------------------|-----------------------------|
| <b>A</b> | copper              | aqueous iron(II) sulfate    |
| <b>B</b> | copper(II) chloride | dilute sulfuric acid        |
| <b>C</b> | copper(II) oxide    | dilute sulfuric acid        |
| <b>D</b> | sulfur              | aqueous copper(II) chloride |

19 Two tests are carried out on substance Z.

test 1 A flame test produces a red flame.

test 2 Z is dissolved in water and dilute nitric acid is added, followed by aqueous silver nitrate. A yellow precipitate is produced.

What is substance Z?

- A** lithium bromide
- B** lithium iodide
- C** sodium bromide
- D** sodium iodide

20 The elements in Period 3 of the Periodic Table are shown.

|    |    |    |    |   |   |    |    |
|----|----|----|----|---|---|----|----|
| Na | Mg | Al | Si | P | S | Cl | Ar |
|----|----|----|----|---|---|----|----|

Which statements about the elements in Period 3 are correct?

- 1 Na, Mg and Al are metals.
- 2 S, Cl and Ar are non-metals.
- 3 Si, P and S are metals.

**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

21 A Group I metal (lithium, sodium or potassium) is reacted with a Group VII element (chlorine, bromine or iodine).

Which compound is formed when the Group I metal of highest density reacts with the Group VII element of lowest density?

- A** lithium chloride  
**B** potassium chloride  
**C** potassium iodide  
**D** lithium iodide

22 The properties of the element titanium, Ti, can be predicted from its position in the Periodic Table.

Which row identifies the properties of titanium?

|          | can be used<br>as a catalyst | conducts electricity<br>when solid | has low density | forms coloured<br>compounds |
|----------|------------------------------|------------------------------------|-----------------|-----------------------------|
| <b>A</b> | ✓                            | ✓                                  | ✓               | x                           |
| <b>B</b> | ✓                            | ✓                                  | x               | ✓                           |
| <b>C</b> | ✓                            | x                                  | ✓               | ✓                           |
| <b>D</b> | x                            | ✓                                  | ✓               | ✓                           |

- 23** A balloon is filled with helium. Helium is a noble gas and makes the balloon rise up in the air.

The density of air is  $1.23 \text{ g/dm}^3$ .

Which gas is helium?

|          | density in $\text{g/dm}^3$ | reaction with oxygen       |
|----------|----------------------------|----------------------------|
| <b>A</b> | 0.0899                     | burns rapidly              |
| <b>B</b> | 0.179                      | does not react with oxygen |
| <b>C</b> | 1.78                       | does not react with oxygen |
| <b>D</b> | 3.75                       | does not react with oxygen |

- 24** Which property is shown by **all** metals?

- A** They are extracted from their ores by heating with carbon.
- B** They conduct electricity.
- C** They form acidic oxides.
- D** They react with hydrochloric acid to form hydrogen.

- 25** The properties of four metals, W, X, Y and Z, are shown.

W It does not react with cold water but reacts with steam.

X It does not react with water or dilute acid but the oxide of X is reduced by carbon.

Y The oxide of Y is not reduced by carbon but Y reacts vigorously with cold water.

Z It does not react with water or steam but reacts with dilute acid.

What is the order of reactivity of the elements starting with the most reactive?

|          | most<br>reactive |  |   |  | least<br>reactive |
|----------|------------------|-------------------------------------------------------------------------------------|---|--|-------------------|
| <b>A</b> | X                | W                                                                                   | Z |  | Y                 |
| <b>B</b> | X                | Z                                                                                   | W |  | Y                 |
| <b>C</b> | Y                | W                                                                                   | Z |  | X                 |
| <b>D</b> | Y                | Z                                                                                   | W |  | X                 |

- 26 Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

What are the reasons for blowing in oxygen and adding calcium oxide?

|          | blowing in oxygen                         | adding calcium oxide                      |
|----------|-------------------------------------------|-------------------------------------------|
| <b>A</b> | carbon is removed by reacting with oxygen | reacts with acidic impurities making slag |
| <b>B</b> | carbon is removed by reacting with oxygen | reacts with slag and so removes it        |
| <b>C</b> | iron reacts with the oxygen               | reacts with acidic impurities making slag |
| <b>D</b> | iron reacts with the oxygen               | reacts with slag and so removes it        |

- 27 Which row describes two uses of the named steel?

|          | type of steel   | uses                          |
|----------|-----------------|-------------------------------|
| <b>A</b> | mild steel      | cutlery and car bodies        |
| <b>B</b> | mild steel      | car bodies and chemical plant |
| <b>C</b> | stainless steel | cutlery and chemical plant    |
| <b>D</b> | stainless steel | car bodies and cutlery        |

- 28 Which statement shows that a liquid is pure water?

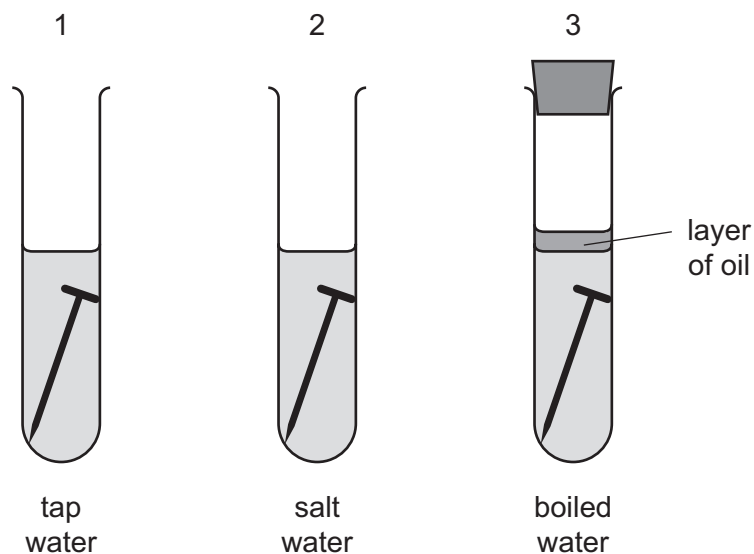
- A** It boils at 100 °C.  
**B** It has a pH value of 7.  
**C** It turns blue cobalt(II) chloride pink.  
**D** It turns white copper(II) sulfate blue.

- 29 Some gases are present in clean air while other gases are only present in polluted air.

Which row is correct?

|          | a gas present in clean air | a gas <b>only</b> present in polluted air |
|----------|----------------------------|-------------------------------------------|
| <b>A</b> | argon                      | carbon dioxide                            |
| <b>B</b> | argon                      | nitrogen dioxide                          |
| <b>C</b> | sulfur dioxide             | carbon dioxide                            |
| <b>D</b> | sulfur dioxide             | nitrogen dioxide                          |

30 The diagrams show experiments to investigate rusting of iron nails.



In which test-tubes do the nails rust?

- A** 1 only      **B** 1 and 2 only      **C** 1 and 3 only      **D** 1, 2 and 3

31 Which mixture contains all of the elements in a typical fertiliser?

- A** ammonium nitrate and calcium phosphate  
**B** ammonium phosphate and potassium chloride  
**C** potassium nitrate and ammonium chloride  
**D** potassium carbonate and ammonium nitrate

32 Which processes produce methane?

- 1 complete combustion of carbon-containing compounds  
2 decomposition of vegetation  
3 digestion in animals  
4 respiration in animals

- A** 1 and 4      **B** 1 and 3      **C** 2 and 3      **D** 2 and 4



**33** The list shows four methods that were suggested for the formation of carbon dioxide.

- 1 cracking methane using steam
- 2 action of heat on a carbonate
- 3 complete combustion of methane
- 4 reaction of a carbonate with oxygen

Which methods would result in the production of carbon dioxide?

- A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

**34** A student suggests three uses of calcium carbonate (limestone).

- 1 manufacture of cement
- 2 manufacture of iron
- 3 treating alkaline soils

Which suggestions are correct?

- A** 1 and 2 only      **B** 1 and 3 only      **C** 2 and 3 only      **D** 1, 2 and 3

**35** Which list shows the fractions obtained from distilling petroleum, in order of increasing boiling point?

- A** bitumen → diesel oil → fuel oil → lubricating oil
- B** diesel oil → gasoline → naphtha → kerosene
- C** gasoline → naphtha → kerosene → diesel oil
- D** kerosene → lubricating oil → naphtha → refinery gas

**36** Which statement about members of a homologous series is correct?

- A** They are elements with the same chemical properties.
- B** They are compounds with the same functional group.
- C** They are atoms with the same number of outer electrons.
- D** They are molecules with the same boiling point.

- 37 Increasing the number of atoms in one molecule of a hydrocarbon increases the amount of energy released when it burns.

What is the correct order?

|          | less energy released | →       | more energy released |
|----------|----------------------|---------|----------------------|
| <b>A</b> | ethene               | ethane  | methane              |
| <b>B</b> | ethene               | methane | ethane               |
| <b>C</b> | methane              | ethane  | ethene               |
| <b>D</b> | methane              | ethene  | ethane               |

- 38 Which statements about ethanol are correct?

- 1 Ethanol is made by reacting steam with ethene at 300 °C.
- 2 Ethanol is made by fermentation at 55 °C.
- 3 Ethanol burns to produce carbon dioxide and water.
- 4 Ethanol contains a carbon-carbon double bond.

- A** 1 and 2      **B** 1 and 3      **C** 2 and 3      **D** 3 and 4

- 39 Some properties of an organic compound J are listed.

- It is a liquid at room temperature.
- It is soluble in water.
- A solution of J reacts with calcium carbonate to form carbon dioxide.
- A solution of J has a pH of 3.

In which homologous series does J belong?

- A** alkane  
**B** alkene  
**C** alcohol  
**D** carboxylic acid

**40** Which polymers or types of polymer are synthetic?

- 1 carbohydrates
- 2 nylon
- 3 proteins
- 4 *Terylene*

**A** 1 and 3

**B** 1 and 4

**C** 2 and 3

**D** 2 and 4

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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                             |                           |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|------------------------------|-----------------------------|---------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             | III                          | IV                           | V                              | VI                            | VII                           | VIII                        |                               |                             |                              |                             |                           |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             | 5<br>B<br>boron<br>11        | 6<br>C<br>carbon<br>12       | 7<br>N<br>nitrogen<br>14       | 8<br>O<br>oxygen<br>16        | 9<br>F<br>fluorine<br>19      | 10<br>Ne<br>neon<br>20      |                               |                             |                              |                             |                           |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             | 12<br>Mg<br>magnesium<br>24   | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28    | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32   |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73   | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79   | 35<br>Br<br>bromine<br>80   | 36<br>Kr<br>krypton<br>84 |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119        | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127    | 54<br>Xe<br>xenon<br>131  |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207       | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—    | 85<br>At<br>astatine<br>—   | 86<br>Rn<br>radon<br>—    |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— | 116<br>Lv<br>livermorium<br>— |                             |                              |                             |                           |

lanthanoids

actinoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2020**

MARK SCHEME

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | A      | 1     |
| 2        | C      | 1     |
| 3        | C      | 1     |
| 4        | A      | 1     |
| 5        | C      | 1     |
| 6        | A      | 1     |
| 7        | D      | 1     |
| 8        | C      | 1     |
| 9        | A      | 1     |
| 10       | B      | 1     |
| 11       | B      | 1     |
| 12       | D      | 1     |
| 13       | A      | 1     |
| 14       | C      | 1     |
| 15       | D      | 1     |
| 16       | D      | 1     |
| 17       | C      | 1     |
| 18       | C      | 1     |
| 19       | B      | 1     |
| 20       | A      | 1     |
| 21       | B      | 1     |
| 22       | B      | 1     |
| 23       | B      | 1     |
| 24       | B      | 1     |
| 25       | C      | 1     |
| 26       | A      | 1     |
| 27       | C      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | B      | 1     |
| 30       | B      | 1     |
| 31       | B      | 1     |
| 32       | C      | 1     |
| 33       | C      | 1     |
| 34       | A      | 1     |
| 35       | C      | 1     |
| 36       | B      | 1     |
| 37       | D      | 1     |
| 38       | B      | 1     |
| 39       | D      | 1     |
| 40       | D      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/12

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

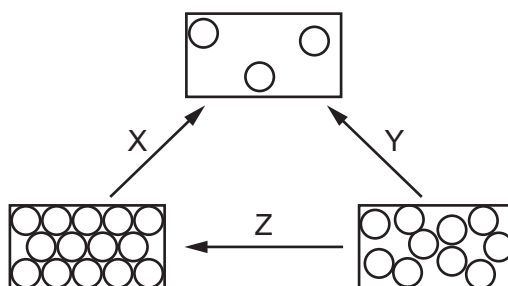
This document has **16** pages. Blank pages are indicated.





- 1 Each rectangle shows the arrangement of particles in each of the three states of matter.

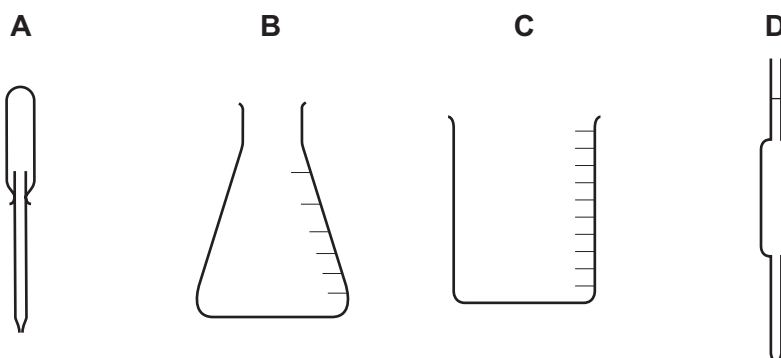
X, Y and Z represent the processes needed to change from one state to another.



What are the processes X, Y and Z?

|          | X           | Y           | Z          |
|----------|-------------|-------------|------------|
| <b>A</b> | evaporating | subliming   | condensing |
| <b>B</b> | evaporating | subliming   | freezing   |
| <b>C</b> | subliming   | evaporating | condensing |
| <b>D</b> | subliming   | evaporating | freezing   |

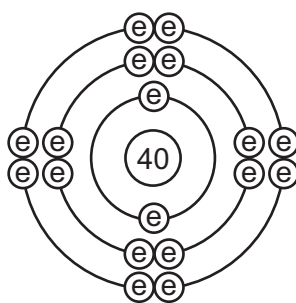
- 2 Which piece of apparatus is used to measure  $25.0\text{ cm}^3$  of aqueous sodium hydroxide?



- 3 Which process is used to produce drinking water from sea water?

- A** crystallisation
- B** distillation
- C** filtration
- D** chlorination

- 4 The diagram shows the electronic structure of a particle with a nucleon number (mass number) of 40.

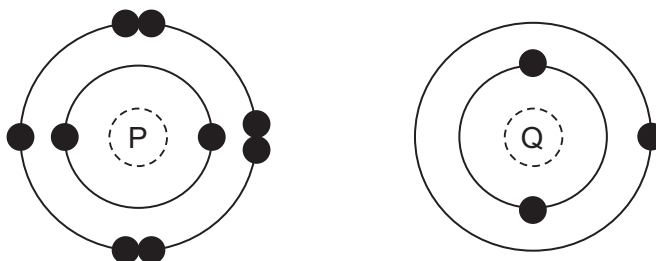


The table shows the suggestions that three students, 1, 2 and 3, made to identify the particle.

|          | student |    |                  |
|----------|---------|----|------------------|
|          | 1       | 2  | 3                |
| particle | Ar      | Cl | Ca <sup>2+</sup> |

Which students are correct?

- A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3
- 5 The Group I element sodium forms an ionic bond with the Group VII element fluorine.
- Which two ions are produced?
- A** Na<sup>+</sup> and F<sup>+</sup>    **B** Na<sup>+</sup> and F<sup>-</sup>    **C** Na<sup>-</sup> and F<sup>-</sup>    **D** Na<sup>-</sup> and F<sup>+</sup>
- 6 The electronic structures of two atoms, P and Q, are shown.



P and Q combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

|          | type of bonding | formula         |
|----------|-----------------|-----------------|
| <b>A</b> | ionic           | PQ              |
| <b>B</b> | ionic           | PQ <sub>2</sub> |
| <b>C</b> | covalent        | PQ <sub>2</sub> |
| <b>D</b> | covalent        | PQ              |

7 Graphite is a macromolecule.

Which statements about graphite are correct?

- 1 Carbon atoms form four covalent bonds with neighbouring atoms.
- 2 There are free electrons between layers of carbon atoms.
- 3 Graphite is a useful lubricant.
- 4 Graphite is a good conductor of electricity.

**A** 1 and 2      **B** 1, 3 and 4      **C** 2, 3 and 4      **D** 3 and 4 only

8 Aluminium oxide has the formula  $Al_2O_3$ .

Which statement about aluminium oxide is correct?

- A** 2g of aluminium atoms are combined with 3g of oxygen atoms.
- B** 2g of aluminium atoms are combined with 3g of oxygen molecules.
- C** Aluminium oxide has a relative formula mass of 102.
- D** Pure aluminium oxide contains a higher mass of oxygen than of aluminium.

9 Dilute sulfuric acid is electrolysed using carbon electrodes.

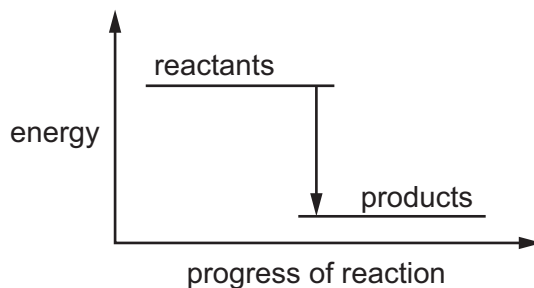
Which row shows the products formed at each electrode?

|          | anode    | cathode  |
|----------|----------|----------|
| <b>A</b> | hydrogen | oxygen   |
| <b>B</b> | hydrogen | sulfur   |
| <b>C</b> | oxygen   | hydrogen |
| <b>D</b> | oxygen   | sulfur   |

10 Which fuel does **not** rely on combustion to release energy?

- A** gasoline
- B** hydrogen
- C** methane
- D** uranium

- 11 The energy level diagram shows the energy of the reactants and products in a chemical reaction.



Which row correctly describes the energy change and the type of reaction shown?

|          | description of energy change             | type of reaction |
|----------|------------------------------------------|------------------|
| <b>A</b> | energy is given out to the surroundings  | endothermic      |
| <b>B</b> | energy is given out to the surroundings  | exothermic       |
| <b>C</b> | energy is taken in from the surroundings | endothermic      |
| <b>D</b> | energy is taken in from the surroundings | exothermic       |

- 12 Which list contains **only** chemical changes?

- A** melting, evaporating, dissolving
- B** rusting, freezing, subliming
- C** neutralisation, polymerisation, combustion
- D** boiling, condensing, distillation

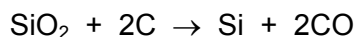
- 13 Which row shows the changes that **all** increase the rate of a chemical reaction?

|          | concentration of reactants | temperature | particle size |
|----------|----------------------------|-------------|---------------|
| <b>A</b> | decrease                   | decrease    | decrease      |
| <b>B</b> | decrease                   | increase    | increase      |
| <b>C</b> | increase                   | decrease    | increase      |
| <b>D</b> | increase                   | increase    | decrease      |

14 Which reaction is reversible?

- A an iron nail rusting when left in moist air
- B limestone reacting with an acid to form carbon dioxide gas
- C magnesium burning in air to produce a white ash
- D white anhydrous copper(II) sulfate turning blue when water is added

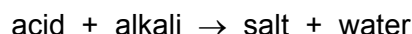
15 When heated strongly, silicon(IV) oxide reacts with carbon.



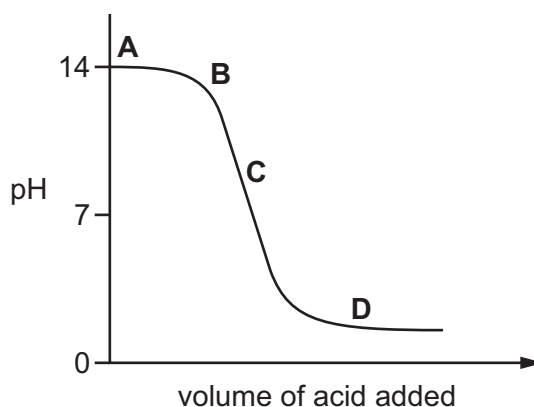
Which term describes what happens to silicon(IV) oxide?

- A thermal decomposition
- B neutralisation
- C oxidation
- D reduction

16 The graph shows how the pH of a solution changes as an acid is added to an alkali.



Which letter represents the area of the graph where both acid and salt are present?

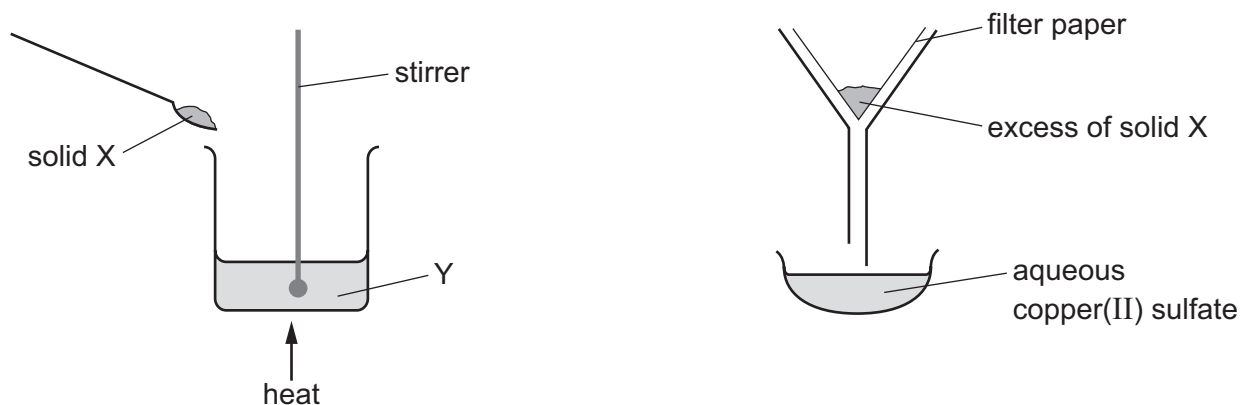


17 Element G is a metal in Group I of the Periodic Table and element H is a non-metal in Group VII. Both of these elements form oxides.

Which statement about their oxides is correct?

- A Both oxides are acidic.
- B Both oxides are basic.
- C The oxide of G is acidic and the oxide of H is basic.
- D The oxide of H is acidic and the oxide of G is basic.

18 The apparatus shown is used to prepare aqueous copper(II) sulfate.



What are X and Y?

|          | X                   | Y                           |
|----------|---------------------|-----------------------------|
| <b>A</b> | copper              | aqueous iron(II) sulfate    |
| <b>B</b> | copper(II) chloride | dilute sulfuric acid        |
| <b>C</b> | copper(II) oxide    | dilute sulfuric acid        |
| <b>D</b> | sulfur              | aqueous copper(II) chloride |

19 An aqueous solution of a compound M is tested.

The results are shown.

- It gave a lilac colour in a flame test.
- It produced a white precipitate when tested with acidified barium nitrate.

What is M?

- A** copper(II) chloride
- B** copper(II) sulfate
- C** potassium carbonate
- D** potassium sulfate

- 20** The character of the elements and charges on the ions of the elements change across the Periodic Table.

Which row describes the elements on the left of the table and the elements on the right?

|          | elements on the left |               | elements on the right |               |
|----------|----------------------|---------------|-----------------------|---------------|
|          | character            | charge on ion | character             | charge on ion |
| <b>A</b> | metallic             | positive      | non-metallic          | negative      |
| <b>B</b> | metallic             | negative      | non-metallic          | positive      |
| <b>C</b> | non-metallic         | positive      | metallic              | negative      |
| <b>D</b> | non-metallic         | negative      | metallic              | positive      |

- 21** Which statement about Group I and Group VII elements is correct?

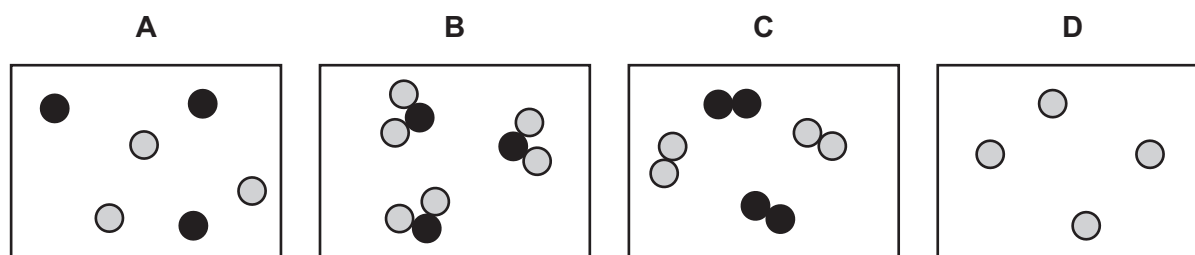
- A** Group VII elements are monoatomic non-metals.
- B** Lithium is more reactive with water than caesium.
- C** The melting points of Group I metals increase down the group.
- D** Potassium bromide reacts with chlorine to produce an orange solution.

- 22** The properties of the element titanium, Ti, can be predicted from its position in the Periodic Table.

Which row identifies the properties of titanium?

|          | can be used as a catalyst | conducts electricity when solid | has low density | forms coloured compounds |
|----------|---------------------------|---------------------------------|-----------------|--------------------------|
| <b>A</b> | ✓                         | ✓                               | ✓               | ✗                        |
| <b>B</b> | ✓                         | ✓                               | ✗               | ✓                        |
| <b>C</b> | ✓                         | ✗                               | ✓               | ✓                        |
| <b>D</b> | ✗                         | ✓                               | ✓               | ✓                        |

- 23** Which diagram shows a mixture of noble gases?



24 Which property is shown by **all** metals?

- A They are extracted from their ores by heating with carbon.
- B They conduct electricity.
- C They form acidic oxides.
- D They react with hydrochloric acid to form hydrogen.

25 The reactions of four metals, W, X, Y and Z, are shown.

| metal | observations                                                   |
|-------|----------------------------------------------------------------|
| W     | reacts with steam and hydrochloric acid but not cold water     |
| X     | reacts with hydrochloric acid but not with steam or cold water |
| Y     | reacts with hydrochloric acid and cold water                   |
| Z     | does not react with hydrochloric acid                          |

What is the order of reactivity for metals W, X, Y and Z?

|   | most<br>reactive | → | least<br>reactive |   |
|---|------------------|---|-------------------|---|
| A | Y                | W | X                 | Z |
| B | Y                | X | W                 | Z |
| C | Z                | W | X                 | Y |
| D | Z                | X | W                 | Y |

26 Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

What are the reasons for blowing in oxygen and adding calcium oxide?

|   | blowing in oxygen                         | adding calcium oxide                      |
|---|-------------------------------------------|-------------------------------------------|
| A | carbon is removed by reacting with oxygen | reacts with acidic impurities making slag |
| B | carbon is removed by reacting with oxygen | reacts with slag and so removes it        |
| C | iron reacts with the oxygen               | reacts with acidic impurities making slag |
| D | iron reacts with the oxygen               | reacts with slag and so removes it        |



27 Which statement about mild steel explains why it is a good choice for car bodies?

- A It is cheap and strong.
- B It is a good conductor.
- C It is low density.
- D It resists rusting.

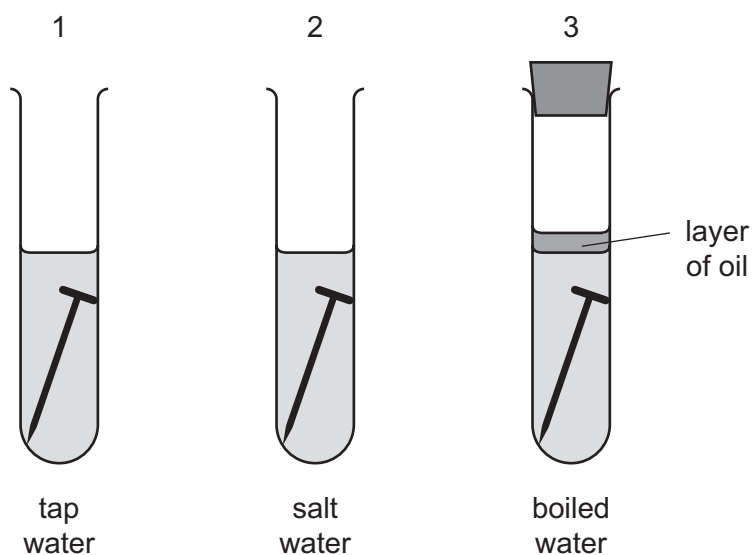
28 Which statement about pure water is **not** correct?

- A It condenses at 100 °C.
- B It freezes at 0 °C.
- C It turns cobalt(II) chloride paper blue.
- D It turns anhydrous copper(II) sulfate blue.

29 Which compounds **both** contribute to 'acid rain'?

- A carbon monoxide and lead compounds
- B carbon monoxide and oxides of nitrogen
- C oxides of nitrogen and sulfur dioxide
- D sulfur dioxide and lead compounds

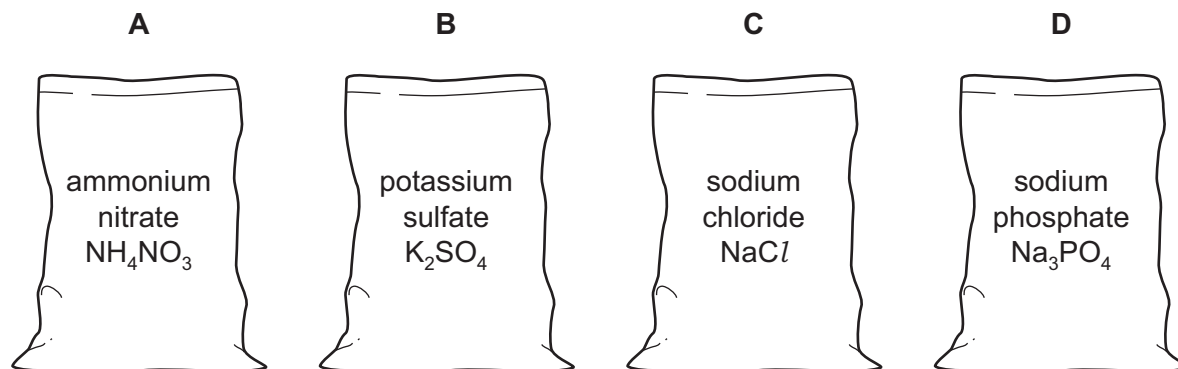
30 The diagrams show experiments to investigate rusting of iron nails.



In which test-tubes do the nails rust?

- A 1 only
- B 1 and 2 only
- C 1 and 3 only
- D 1, 2 and 3

31 Which bag is **not** suitable for use as a fertiliser?



32 X is a colourless greenhouse gas.

It is a waste product from digestion in animals.

It is generally unreactive, but it can be burnt.

What is X?

- A carbon dioxide
- B methane
- C nitrogen
- D sulfur dioxide

33 The list shows four methods that were suggested for the formation of carbon dioxide.

- 1 cracking methane using steam
- 2 action of heat on a carbonate
- 3 complete combustion of methane
- 4 reaction of a carbonate with oxygen

Which methods would result in the production of carbon dioxide?

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

34 A student suggests three uses of calcium carbonate (limestone).

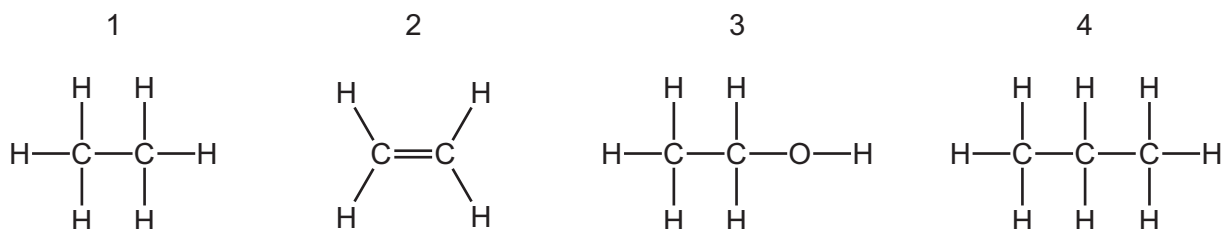
- 1 manufacture of cement
- 2 manufacture of iron
- 3 treating alkaline soils

Which suggestions are correct?

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

- 35 Which list shows the fractions obtained from distilling petroleum, in order of increasing boiling point?
- A** bitumen → diesel oil → fuel oil → lubricating oil
- B** diesel oil → gasoline → naphtha → kerosene
- C** gasoline → naphtha → kerosene → diesel oil
- D** kerosene → lubricating oil → naphtha → refinery gas

- 36 The structures of four compounds are shown.



Which compounds are members of the same homologous series?

- A** 1 and 2 only    **B** 1 and 4    **C** 1, 2 and 3    **D** 2 and 4
- 37 Increasing the number of atoms in one molecule of a hydrocarbon increases the amount of energy released when it burns.

What is the correct order?

|          | less energy released | →       | more energy released |
|----------|----------------------|---------|----------------------|
| <b>A</b> | ethene               | ethane  | methane              |
| <b>B</b> | ethene               | methane | ethane               |
| <b>C</b> | methane              | ethane  | ethene               |
| <b>D</b> | methane              | ethene  | ethane               |

- 38 Which statement about alcohols is correct?

- A** Alcohols and carboxylic acids have the same functional group.
- B** Ethanoic acid is produced from the reduction of ethanol.
- C** Ethanol is produced in an addition reaction between ethene and hydrogen.
- D** Water is produced from the combustion of alcohols.

**39** An organic compound, P, reacts with zinc to produce a gas, Q.

What are P and Q?

|          | P             | Q              |
|----------|---------------|----------------|
| <b>A</b> | ethanoic acid | carbon dioxide |
| <b>B</b> | ethanoic acid | hydrogen       |
| <b>C</b> | ethanol       | carbon dioxide |
| <b>D</b> | ethanol       | hydrogen       |

**40** Which substances are natural polymers?

- 1 proteins
- 2 carbohydrates
- 3 nylon
- 4 poly(ethene)

**A** 1 and 2

**B** 1 and 3

**C** 2 and 3

**D** 3 and 4

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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                            |                              |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|------------------------------|----------------------------|------------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                            | V                           | VI                           | VII                        | VIII                         |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12        | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19   | 10<br>Ne<br>neon<br>20       |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                            |                              |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73   | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79   | 35<br>Br<br>bromine<br>80  | 36<br>Kr<br>krypton<br>84    |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119        | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127   | 54<br>Xe<br>xenon<br>131     |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207       | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—    | 85<br>At<br>astatine<br>—  | 86<br>Rn<br>radon<br>—       |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— | 116<br>Lv<br>livermorium<br>— | 118<br>Og<br>oganesson<br>— | 119<br>Ts<br>tennessine<br>— | 120<br>Nh<br>nihonium<br>— | 121<br>Lr<br>lawrencium<br>— |

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/12**

Paper 1 Multiple Choice (Core)

**May/June 2020**

MARK SCHEME

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

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This document consists of **3** printed pages.



| Question | Answer | Marks |
|----------|--------|-------|
| 1        | D      | 1     |
| 2        | D      | 1     |
| 3        | B      | 1     |
| 4        | B      | 1     |
| 5        | B      | 1     |
| 6        | A      | 1     |
| 7        | C      | 1     |
| 8        | C      | 1     |
| 9        | C      | 1     |
| 10       | D      | 1     |
| 11       | B      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | D      | 1     |
| 15       | D      | 1     |
| 16       | D      | 1     |
| 17       | D      | 1     |
| 18       | C      | 1     |
| 19       | D      | 1     |
| 20       | A      | 1     |
| 21       | D      | 1     |
| 22       | B      | 1     |
| 23       | A      | 1     |
| 24       | B      | 1     |
| 25       | A      | 1     |
| 26       | A      | 1     |
| 27       | A      | 1     |
| 28       | C      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | B      | 1     |
| 31       | C      | 1     |
| 32       | B      | 1     |
| 33       | C      | 1     |
| 34       | A      | 1     |
| 35       | C      | 1     |
| 36       | B      | 1     |
| 37       | D      | 1     |
| 38       | D      | 1     |
| 39       | B      | 1     |
| 40       | A      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Blank pages are indicated.



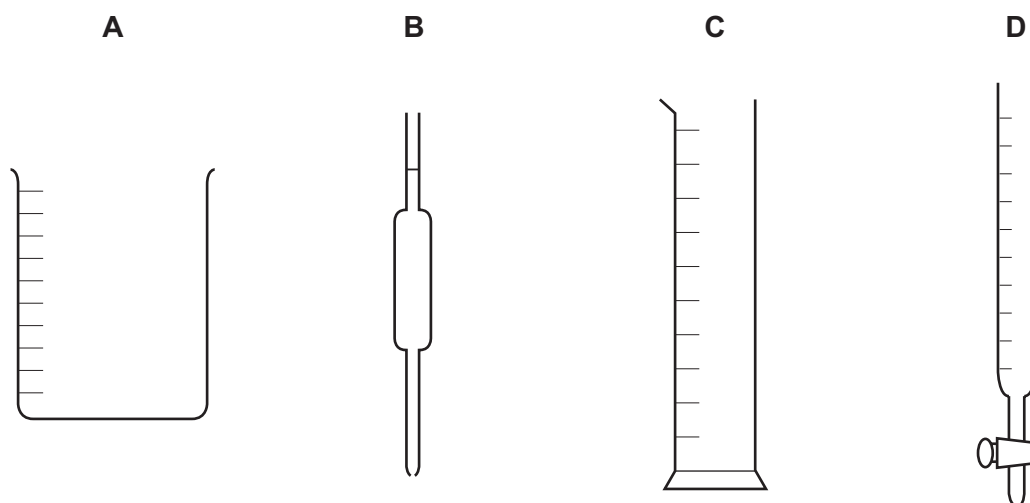
- 1 'The movement of a substance **very slowly** from an area of high concentration to an area of low concentration.'

Which process is being described?

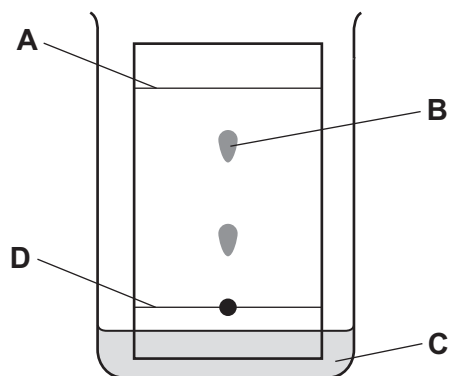
- A a liquid being frozen
  - B a solid melting
  - C a substance diffusing through a liquid
  - D a substance diffusing through the air
- 2 Oxygen melts at  $-219^{\circ}\text{C}$  and boils at  $-183^{\circ}\text{C}$ .

At which temperature is oxygen a liquid?

- A  $-225^{\circ}\text{C}$
  - B  $-189^{\circ}\text{C}$
  - C  $-175^{\circ}\text{C}$
  - D  $25^{\circ}\text{C}$
- 3 Which diagram shows a burette?



- 4 In the chromatography experiment shown, which label represents the solvent front?



- 5 Different methods of separation rely on substances having different properties.

Which property does distillation make use of?

- A boiling point
- B colour
- C particle size
- D solubility in different solvents

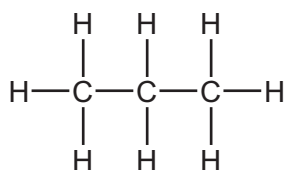
- 6 Which statement about atomic structure is correct?

- A Isotopes have a different nucleon number but the same proton number.
- B Metal atoms gain electrons to achieve a noble gas electronic structure.
- C The nucleon number is the total number of electrons and neutrons in the nucleus of the atom.
- D Protons and neutrons are oppositely charged particles.

- 7 Which element is a non-metal?

- A scandium
- B sodium
- C strontium
- D sulfur

- 8 The structure of propane,  $C_3H_8$ , is shown.



How many electrons are involved in the bonding of propane?

- A 8
- B 10
- C 16
- D 20

- 9 Rubidium is in Group I of the Periodic Table and bromine is in Group VII.

Rubidium reacts with bromine to form an ionic compound.

Which row shows the electron change taking place for rubidium and the correct formula of the rubidium ion?

|          | electron change | formula of ion formed |
|----------|-----------------|-----------------------|
| <b>A</b> | electron gained | $\text{Rb}^+$         |
| <b>B</b> | electron gained | $\text{Rb}^-$         |
| <b>C</b> | electron lost   | $\text{Rb}^+$         |
| <b>D</b> | electron lost   | $\text{Rb}^-$         |

- 10 Which statement explains why graphite is used as a lubricant?

- A** All bonds between the atoms are weak.
- B** It conducts electricity.
- C** It has a low melting point.
- D** Layers in the structure can slide over each other.

- 11 The formula of which compound contains the largest number of Group VII atoms?

- A**  $\text{C}_{13}\text{H}_{13}\text{IO}_8$       **B**  $\text{Cl}_2\text{O}_6$       **C**  $\text{Al}(\text{BrO}_3)_3$       **D**  $\text{NaFC}_2\text{H}_2\text{O}_2$

- 12 The relative atomic mass of chlorine is 35.5.

When calculating relative atomic mass, which particle is the mass of a chlorine atom compared to?

- A** a neutron
- B** a proton
- C** an atom of carbon-12
- D** an atom of hydrogen-1

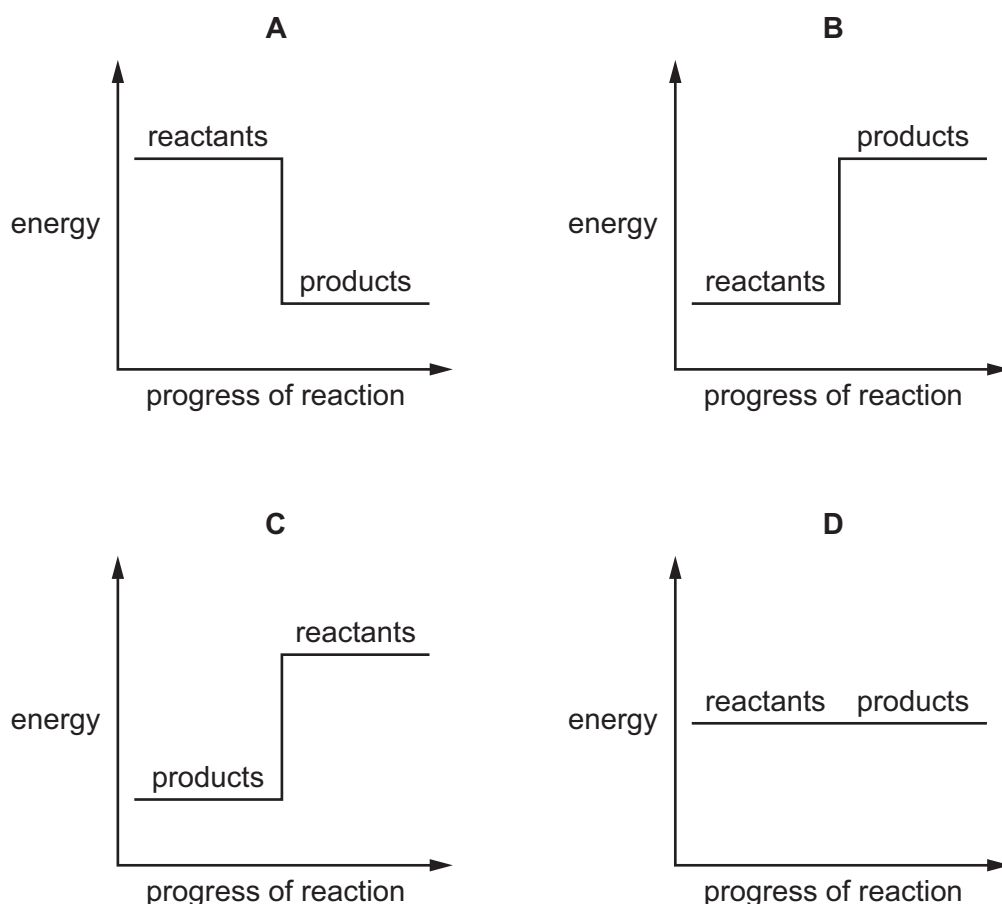
- 13 Universal indicator solution is added to a neutral solution of concentrated aqueous sodium chloride.

The solution, which contains  $\text{H}^+$  (hydrogen),  $\text{Na}^+$  (sodium),  $\text{Cl}^-$  (chloride) and  $\text{OH}^-$  (hydroxide) ions, is electrolysed.

The product at the cathode is hydrogen gas and the product at the anode is chlorine gas.

What happens to the colour of the indicator **in the solution** during electrolysis?

- A The colour changes from blue to green.  
B The colour changes from blue to red.  
C The colour changes from green to blue.  
D The colour changes from green to red.
- 14 Which energy level diagram represents an endothermic reaction?



- 15 Which process is a physical change?

- A burning a piece of magnesium  
B dissolving calcium carbonate in hydrochloric acid  
C melting an ice cube  
D the rusting of an iron nail

16 Which substance does **not** require oxygen in order to produce energy?

- A coal
- B hydrogen
- C natural gas
- D  $^{235}\text{U}$

17 Nitrogen,  $\text{N}_2$ , and hydrogen,  $\text{H}_2$ , can be converted into ammonia,  $\text{NH}_3$ , using a catalyst.

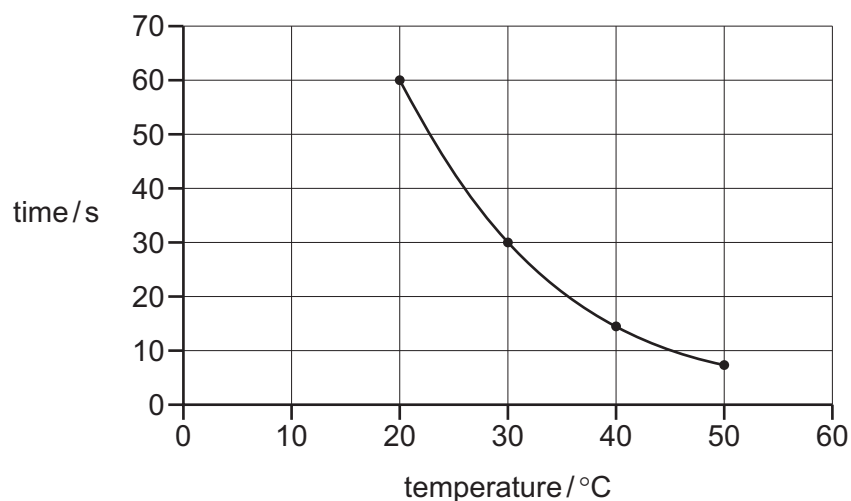
What is the purpose of the catalyst?

- A to increase the amount of ammonia produced
- B to increase the rate of reaction
- C to reduce the amount of reactants needed
- D to reduce the rate of reaction

18 A reaction is carried out at four different temperatures.

The time taken for the reaction to complete at each temperature is measured.

The results are shown.

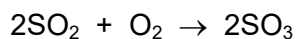


What is the relationship between temperature and rate of reaction?

- A The rate decreases as the temperature increases.
- B The rate increases as the temperature increases.
- C The rate is proportional to the temperature.
- D The rate is inversely proportional to the temperature.

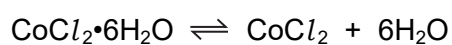


- 19 During the manufacture of sulfuric acid, sulfur dioxide is converted to sulfur trioxide.



Which type of reaction is this?

- A displacement
  - B neutralisation
  - C oxidation
  - D thermal decomposition
- 20 When pink crystals of cobalt(II) chloride are heated, steam is given off and the colour of the solid changes to blue.



What happens when water is added to the blue solid?

|   | colour          | temperature |
|---|-----------------|-------------|
| A | changes to pink | decreases   |
| B | changes to pink | increases   |
| C | remains blue    | decreases   |
| D | remains blue    | increases   |

- 21 Which statement describes a base?

- A It reacts with ammonium chloride to produce ammonia gas.
- B It reacts with calcium carbonate to produce carbon dioxide gas.
- C It reacts with copper to produce hydrogen gas.
- D It turns blue litmus red.

- 22 Which compound is an acidic oxide?

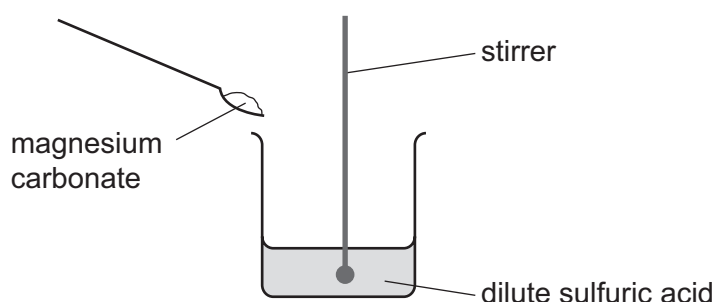
- A aluminium oxide
- B carbon dioxide
- C copper(II) oxide
- D magnesium oxide

23 Which statement describes how a flame test is done?

- A The tip of a clean wire is dipped into the substance and the wire is placed in a blue Bunsen burner flame.
- B The tip of a clean wire is dipped into the substance and the wire is placed in a yellow Bunsen burner flame.
- C A wooden splint is lit and is placed above a test-tube containing the gas being tested.
- D A wooden splint is lit, blown out and the glowing splint put into a test-tube of the gas being tested.

24 A student carries out an experiment to prepare pure magnesium sulfate crystals.

The diagram shows the first stage of the preparation.



He adds magnesium carbonate until no more reacts.

Which process should he use for the next stage?

- A crystallisation
- B evaporation
- C filtration
- D neutralisation

25 Which row about elements in the Periodic Table is correct?

|          | statement 1                                                      | statement 2                          |
|----------|------------------------------------------------------------------|--------------------------------------|
| <b>A</b> | two elements in the same group have similar chemical properties  | metals are on the left of the table  |
| <b>B</b> | two elements in the same group have similar chemical properties  | metals are on the right of the table |
| <b>C</b> | two elements in the same period have similar chemical properties | metals are on the left of the table  |
| <b>D</b> | two elements in the same period have similar chemical properties | metals are on the right of the table |

**26** Tennessine, Ts, is a newly discovered element.

The atomic number of tennessine is 117 and it is placed directly underneath astatine in Group VII of the Periodic Table.

The trends in properties of Group VII elements are shown.

| element  | boiling point<br>/°C | colour      | density<br>in g/cm <sup>3</sup> | reactivity     |
|----------|----------------------|-------------|---------------------------------|----------------|
| fluorine | −188                 | pale yellow | 0.002                           | extremely high |
| chlorine | −35                  | green       | 0.003                           | very high      |
| bromine  | 60                   | red-brown   | 3.103                           | high           |
| iodine   | 184                  | dark grey   | 4.933                           | low            |

Which statement about the properties of tennessine is likely to be correct?

- A** Tennessine has a higher reactivity than astatine.
- B** Tennessine has a lower boiling point than astatine.
- C** Tennessine is a lighter colour than astatine.
- D** Tennessine is more dense than astatine.

**27** A flammable gas needs to be removed from a tank at an industrial plant.

For safety reasons, an inert gas is used.

Which gas is suitable?

- A** argon
- B** hydrogen
- C** methane
- D** oxygen

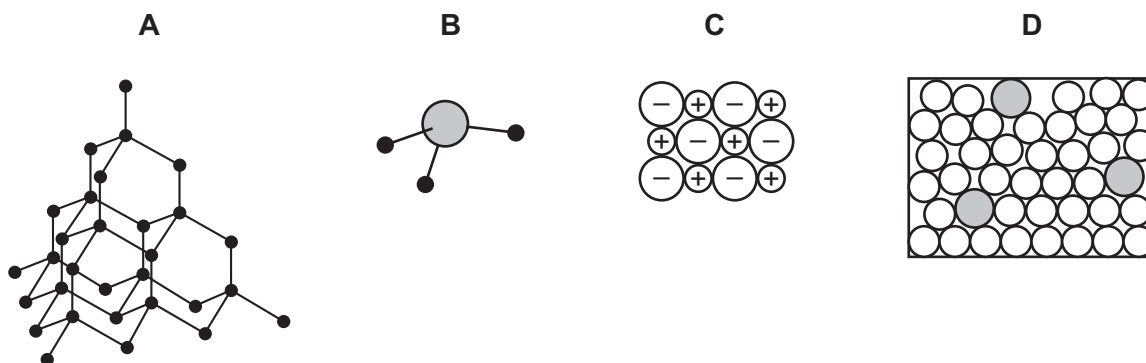
28 A substance, X, has the following properties.

- 1 It has a high melting point.
- 2 It conducts electricity in the solid and liquid states.
- 3 It is malleable.
- 4 It has a high density.

What is X?

- A a ceramic
- B copper
- C graphite
- D sodium chloride

29 Which diagram best represents the structure of a substance that is a good conductor of electricity at 25 °C?



30 Some properties of element Y are listed.

- It reacts with hydrochloric acid to make hydrogen gas.
- It reacts with steam but not with cold water.
- The oxide of Y cannot be reduced by carbon.

What is element Y?

- A copper
- B iron
- C magnesium
- D sodium

**31** Oxides of nitrogen are given out from car exhausts.

Which row best shows why oxides of nitrogen are unwanted in the atmosphere?

|          | acidic | toxic |
|----------|--------|-------|
| <b>A</b> | no     | no    |
| <b>B</b> | no     | yes   |
| <b>C</b> | yes    | no    |
| <b>D</b> | yes    | yes   |

**32** Water is purified using several processes.

Four of the processes are listed.

- 1 Chlorine is added to water to kill any bacteria.
- 2 Water is passed through coarse gravel to remove large pieces of dirt.
- 3 Water is passed through wire screens to remove large twigs.
- 4 Water is passed through fine sand to remove small particles.

In which order are the processes carried out?

- A** 1 → 2 → 3 → 4
- B** 2 → 1 → 4 → 3
- C** 3 → 2 → 4 → 1
- D** 4 → 3 → 2 → 1

**33** When solid S is heated strongly, it forms gas G.

G turns limewater cloudy.

What are S and G and which type of reaction does S undergo?

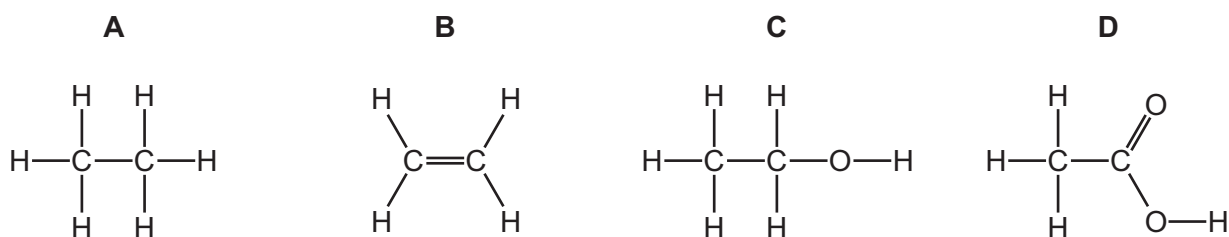
|          | S                 | G              | type of reaction      |
|----------|-------------------|----------------|-----------------------|
| <b>A</b> | calcium carbonate | carbon dioxide | combustion            |
| <b>B</b> | calcium carbonate | carbon dioxide | thermal decomposition |
| <b>C</b> | sodium carbonate  | oxygen         | combustion            |
| <b>D</b> | sodium carbonate  | oxygen         | thermal decomposition |

**34** The element sulfur is found in a number of different minerals.

Which mineral contains the greatest percentage by mass of sulfur?

- A** barite,  $\text{BaSO}_4$
- B** galena,  $\text{PbS}$
- C** gypsum,  $\text{CaSO}_4$
- D** pyrite,  $\text{FeS}_2$

**35** Which structure represents a molecule of ethanol?



**36** Petroleum is separated into fractions by fractional distillation.

Separation occurs in a fractionating column.

Some properties of three of these fractions are shown.

| fraction | boiling point range / °C | number of carbon atoms in the molecules |
|----------|--------------------------|-----------------------------------------|
| 1        |                          | 5–10                                    |
| 2        | 320–350                  | 16–24                                   |
| 3        | 120–210                  |                                         |

Which statement is correct?

- A** Fraction 1 has a higher boiling point range than fraction 2.
- B** Fraction 2 is removed from a higher point in the fractionating column than fraction 1.
- C** Molecules in fraction 3 have shorter chains than those in fraction 2.
- D** None of the fractions are liquid at room temperature.

37 Which statement describes methane?

- A It is an alcohol.
- B It is an unsaturated molecule.
- C It contains carbon, hydrogen and oxygen atoms only.
- D Each molecule contains four single covalent bonds.

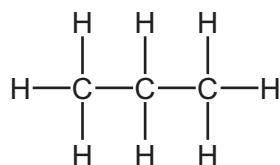
38 The flow chart shows the preparation of ethanol and some important chemistry of ethanol.



What are X, Y and Z?

|   | X       | Y              | Z       |
|---|---------|----------------|---------|
| A | yeast   | combustion     | oxygen  |
| B | glucose | combustion     | steam   |
| C | glucose | polymerisation | water   |
| D | yeast   | fermentation   | glucose |

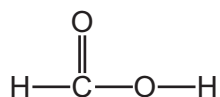
39 The structure of propane is shown.



Which statement about the atoms and the bonding in propane is correct?

- A All the bonds are single bonds.
- B Each carbon atom only bonds with two hydrogen atoms.
- C Propane is an unsaturated molecule.
- D There are three carbon-carbon bonds.

40 The structure of a compound X is shown.



X is in the same homologous series as ethanoic acid.

Which row describes some of the properties of an aqueous solution of X?

|          | reacts with $\text{CaCO}_3$<br>to produce a gas | neutralises<br>$\text{CuO}$ | turns methyl<br>orange red |
|----------|-------------------------------------------------|-----------------------------|----------------------------|
| <b>A</b> | no                                              | no                          | no                         |
| <b>B</b> | no                                              | yes                         | no                         |
| <b>C</b> | yes                                             | yes                         | yes                        |
| <b>D</b> | yes                                             | no                          | yes                        |



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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                             |                                 |                              |                              |                              |                            |                              |                                |                               |                               |                             |                             |                               |                             |                           |                              |                         |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|-----------------------------|---------------------------------|------------------------------|------------------------------|------------------------------|----------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-------------------------------|-----------------------------|---------------------------|------------------------------|-------------------------|
| I                          | II                          |                                                                                             |                             |                                 |                              |                              |                              |                            |                              |                                |                               | III                           | IV                          | V                           | VI                            | VII                         | VIII                      |                              |                         |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                             |                                 |                              |                              |                              |                            |                              |                                |                               | 1<br>H<br>hydrogen<br>1       |                             |                             |                               |                             |                           |                              |                         |
|                            |                             |                                                                                             |                             |                                 |                              |                              |                              |                            |                              |                                |                               |                               |                             |                             |                               |                             |                           |                              |                         |
|                            |                             |                                                                                             |                             |                                 |                              |                              |                              |                            |                              |                                |                               |                               |                             |                             |                               |                             |                           |                              |                         |
|                            |                             |                                                                                             |                             |                                 |                              |                              |                              |                            |                              |                                |                               | 11<br>Na<br>sodium<br>23      | 12<br>Mg<br>magnesium<br>24 | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28     | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32   | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40 |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48  | 23<br>V<br>vanadium<br>51       | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55  | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59   | 28<br>Ni<br>nickel<br>59     | 29<br>Cu<br>copper<br>64       | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70     | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79    | 35<br>Br<br>bromine<br>80   | 36<br>Kr<br>krypton<br>84 |                              |                         |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93       | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>—  | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108      | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115     | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127    | 54<br>Xe<br>xenon<br>131  |                              |                         |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        |                             | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181  | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186   | 76<br>Os<br>osmium<br>190  | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209    | 84<br>Po<br>polonium<br>—   | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>—       |                         |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         |                             | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—    | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—    | 108<br>Hs<br>hassium<br>—  | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— |                             | 116<br>Lv<br>livermorium<br>— |                             |                           |                              |                         |

lanthanoids

actinoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY**

**0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2020**

**MARK SCHEME**

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | B      | 1     |
| 3        | D      | 1     |
| 4        | A      | 1     |
| 5        | A      | 1     |
| 6        | A      | 1     |
| 7        | D      | 1     |
| 8        | D      | 1     |
| 9        | C      | 1     |
| 10       | D      | 1     |
| 11       | C      | 1     |
| 12       | C      | 1     |
| 13       | C      | 1     |
| 14       | B      | 1     |
| 15       | C      | 1     |
| 16       | D      | 1     |
| 17       | B      | 1     |
| 18       | B      | 1     |
| 19       | C      | 1     |
| 20       | B      | 1     |
| 21       | A      | 1     |
| 22       | B      | 1     |
| 23       | A      | 1     |
| 24       | C      | 1     |
| 25       | A      | 1     |
| 26       | D      | 1     |
| 27       | A      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | D      | 1     |
| 30       | C      | 1     |
| 31       | D      | 1     |
| 32       | C      | 1     |
| 33       | B      | 1     |
| 34       | D      | 1     |
| 35       | C      | 1     |
| 36       | C      | 1     |
| 37       | D      | 1     |
| 38       | B      | 1     |
| 39       | A      | 1     |
| 40       | C      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/11

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

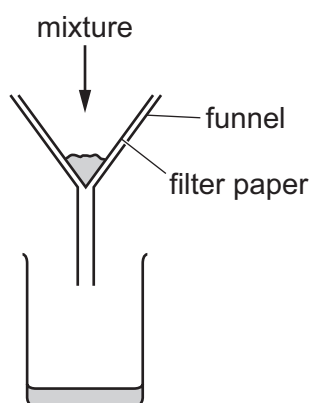
This document has **16** pages. Any blank pages are indicated.



1 Which row describes the arrangement and movement of particles in a liquid?

|          | arrangement of particles | movement of particles    |
|----------|--------------------------|--------------------------|
| <b>A</b> | touching and regular     | vibrating                |
| <b>B</b> | touching and random      | moving around each other |
| <b>C</b> | touching and regular     | moving around each other |
| <b>D</b> | touching and random      | moving very fast         |

2 A mixture is separated using the apparatus shown.



What is the mixture?

- A** aqueous copper(II) sulfate and aqueous sodium chloride
  - B** aqueous copper(II) sulfate and copper
  - C** copper and sulfur
  - D** ethanol and ethanoic acid
- 3 Which statement about paper chromatography is correct?
- A** A solvent is needed to dissolve the paper.
  - B** Paper chromatography separates mixtures of solvents.
  - C** The solvent should cover the baseline.
  - D** The baseline should be drawn in pencil.

- 4 Element X has 7 protons.

Element Y has 8 more protons than X.

Which statement about element Y is correct?

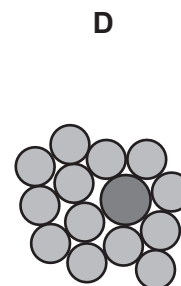
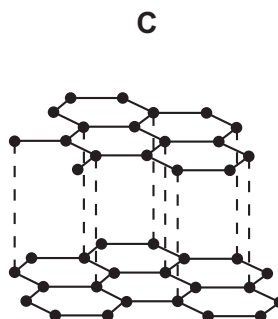
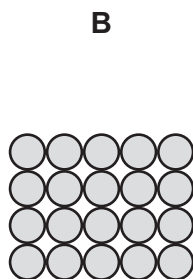
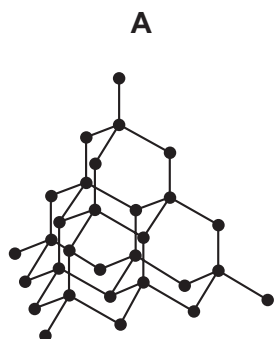
- A Y has more electron shells than X.
  - B Y has more electrons in its outer shell than X.
  - C Y is in a different group of the Periodic Table from X.
  - D Y is in the same period of the Periodic Table as X.
- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A ammonia,  $\text{NH}_3$
  - B chlorine,  $\text{Cl}_2$
  - C methane,  $\text{CH}_4$
  - D water,  $\text{H}_2\text{O}$
- 6 Which row describes how an ionic bond forms between a sodium atom and a chlorine atom?

|   | sodium atom              | chlorine atom            |
|---|--------------------------|--------------------------|
| A | two electrons are lost   | two electrons are gained |
| B | one electron is gained   | one electron is lost     |
| C | two electrons are gained | two electrons are lost   |
| D | one electron is lost     | one electron is gained   |

- 7 Which diagram shows the structure of an alloy?





- 8 Methane burns in oxygen to produce carbon dioxide and water.

What is the balanced equation for this reaction?

- A  $\text{CH}_4 + 2\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$   
B  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$   
C  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$   
D  $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

- 9 What is the relative formula mass of magnesium nitrate,  $\text{Mg}(\text{NO}_3)_2$ ?

- A 74                      B 86                      C 134                      D 148

- 10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.

What would happen in **both** experiments?

- A A halogen would be formed at the anode.  
B A metal would be formed at the cathode.  
C Hydrogen would be formed at the anode.  
D Hydrogen would be formed at the cathode.

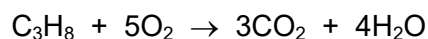
- 11 Steel core aluminium cables are used for overhead electricity cables.

Which statement explains why these cables are used?

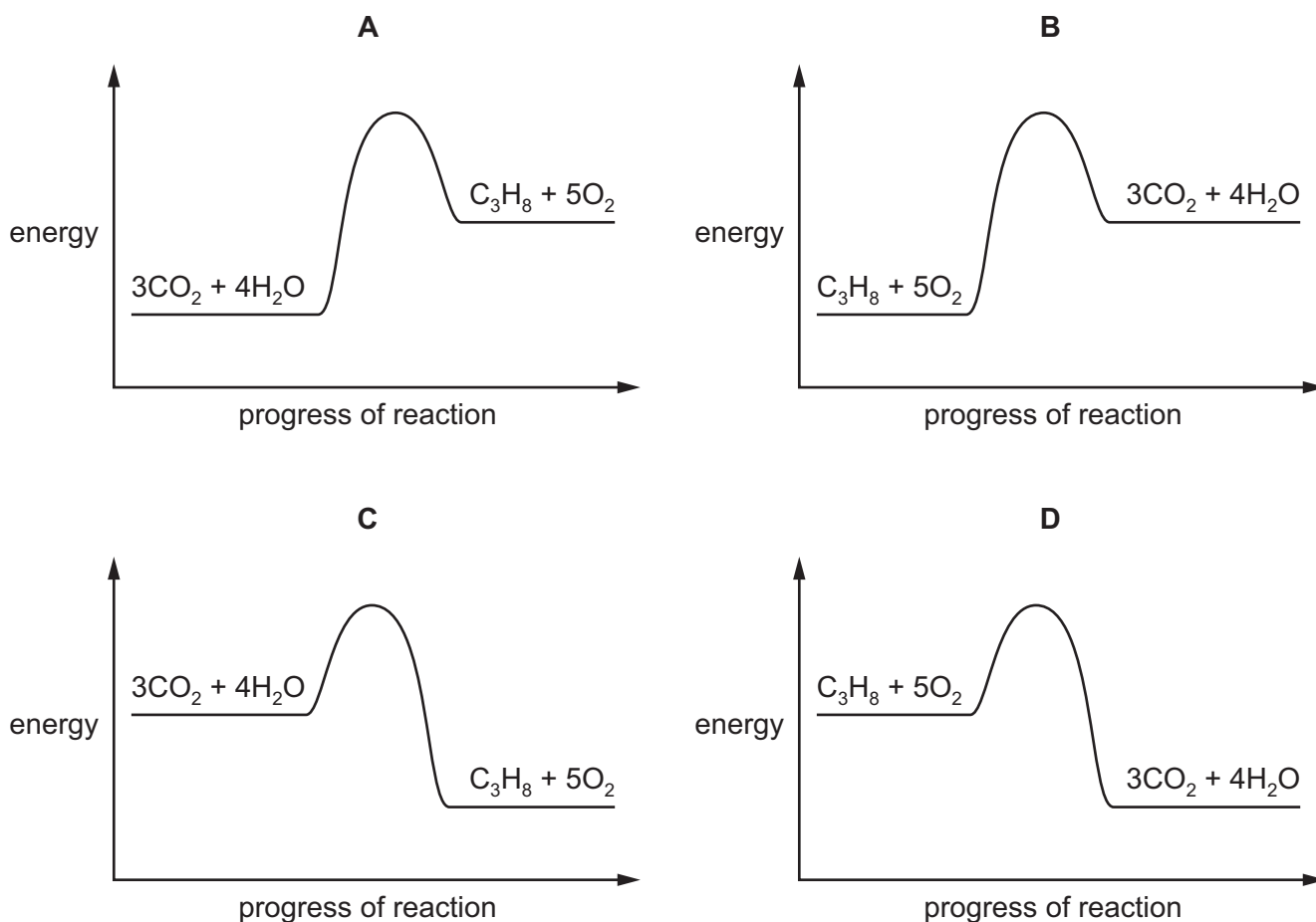
- A Aluminium conducts electricity only when it surrounds a steel core.  
B Aluminium conducts electricity and the steel core makes the cable stronger.  
C Steel conducts electricity and is surrounded by aluminium because aluminium is an insulator.  
D Steel conducts electricity and is surrounded by aluminium to stop the steel from corroding.

12 The complete combustion of propane is exothermic.

The equation for this reaction is shown.



Which energy level diagram represents the complete combustion of propane?



13 Which changes occur when hydrogen is burned in oxygen?

|          | energy change | product                                |
|----------|---------------|----------------------------------------|
| <b>A</b> | endothermic   | $\text{H}_2\text{O}$ only              |
| <b>B</b> | endothermic   | $\text{H}_2\text{O}$ and $\text{CO}_2$ |
| <b>C</b> | exothermic    | $\text{H}_2\text{O}$ only              |
| <b>D</b> | exothermic    | $\text{H}_2\text{O}$ and $\text{CO}_2$ |

- 14** When sulfur is heated it undergoes a .....1..... change as it melts.

Further heating causes the sulfur to undergo a .....2..... change and form sulfur dioxide.

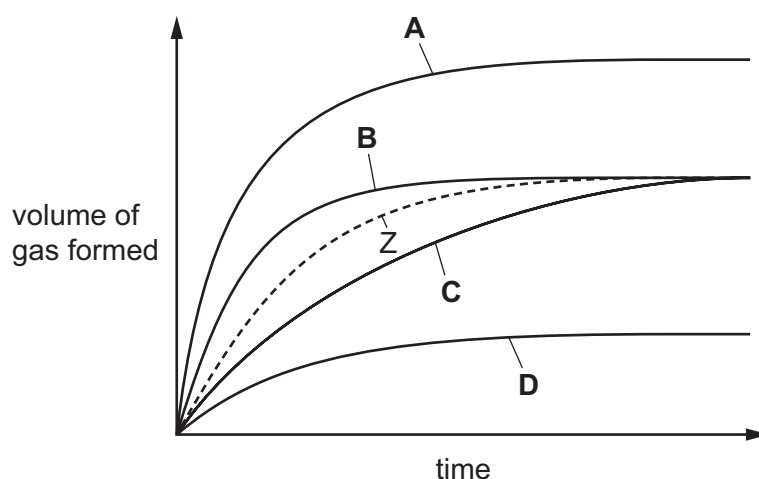
Which words complete gaps 1 and 2?

|          | 1        | 2        |
|----------|----------|----------|
| <b>A</b> | chemical | chemical |
| <b>B</b> | chemical | physical |
| <b>C</b> | physical | chemical |
| <b>D</b> | physical | physical |

- 15** Zinc reacts with an acid to form a gas. The volume of gas produced is measured at intervals. The results are shown as curve Z.

The reaction is repeated in the presence of a catalyst.

Which curve shows the results for the catalysed reaction?



- 16** Which statement is correct?

- A** When anhydrous copper(II) sulfate is heated its colour changes to a deeper blue.
- B** When hydrated copper(II) sulfate is heated its colour changes to a deeper blue.
- C** When water is added to blue cobalt(II) chloride paper it turns pink.
- D** When water is added to pink cobalt(II) chloride paper it turns blue.

**17** Three separate experiments are carried out on an aqueous solution of S.

The results are shown.

- 1 Magnesium does not react with the solution.
- 2 A gas is given off when ammonium sulfate is heated with the solution.
- 3 Methyl orange turns yellow when added to the solution.

What is S?

- A** hydrochloric acid
- B** sodium hydroxide
- C** sodium chloride
- D** sulfur dioxide

**18** Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

|          | element X | nature of oxide, XO |
|----------|-----------|---------------------|
| <b>A</b> | metal     | acidic              |
| <b>B</b> | metal     | basic               |
| <b>C</b> | non-metal | acidic              |
| <b>D</b> | non-metal | basic               |

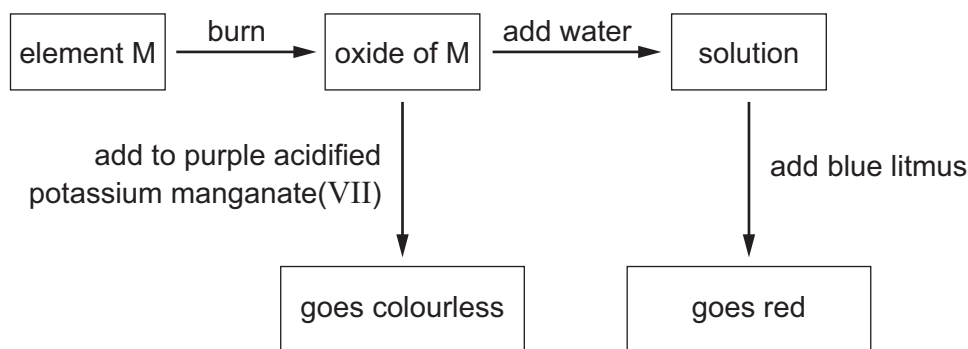
**19** Copper(II) sulfate is prepared by adding excess copper(II) oxide to warm dilute sulfuric acid.

Which purification methods are used to obtain pure solid copper(II) sulfate from the reaction mixture?

- 1 crystallisation
- 2 filtration
- 3 chromatography
- 4 distillation

- A** 1 and 4
- B** 1 and 2
- C** 2 and 3
- D** 3 and 4

20 Some reactions of element M are shown.



What is element M?

- A carbon
- B iron
- C magnesium
- D sulfur

21 Element X is in Group II of the Periodic Table.

Which statements about X are correct?

- 1 X is a metal.
- 2 X has two electrons in its outer shell.
- 3 X is a liquid at room temperature.

- A 1 and 2 only    B 1 and 3 only    C 2 and 3 only    D 1, 2 and 3

22 Why is helium used to fill balloons?

- A Helium is monoatomic.
- B Helium is in Group VIII of the Periodic Table.
- C Helium has a full outer electron shell.
- D Helium is less dense than air.

**23** Which row describes the trend in properties of the elements in Group I as the group is descended?

|          | melting point | reactivity with water |
|----------|---------------|-----------------------|
| <b>A</b> | decreases     | decreases             |
| <b>B</b> | decreases     | increases             |
| <b>C</b> | increases     | decreases             |
| <b>D</b> | increases     | increases             |

**24** An element melts at 1455 °C, has a density of 8.90 g/cm<sup>3</sup> and forms a green chloride.

Where in the Periodic Table is this element found?

A 10x10 grid representing a periodic table. The grid is divided into several sections. The top row has a single cell labeled 'A' at the far right. The second row has a single cell labeled 'B' at the far left. The third row has a single cell labeled 'C' in the middle. The fourth row has a single cell labeled 'D' at the far right. The grid is otherwise empty.

**25** Some properties of metal J are listed.

- J does not react with cold water.
- J reacts with dilute hydrochloric acid.
- No reaction occurs when the oxide of J is heated with carbon.

## What is J?

- A** copper
- B** iron
- C** magnesium
- D** sodium

**26** Iron from a blast furnace is treated with oxygen and with calcium oxide to make steel.

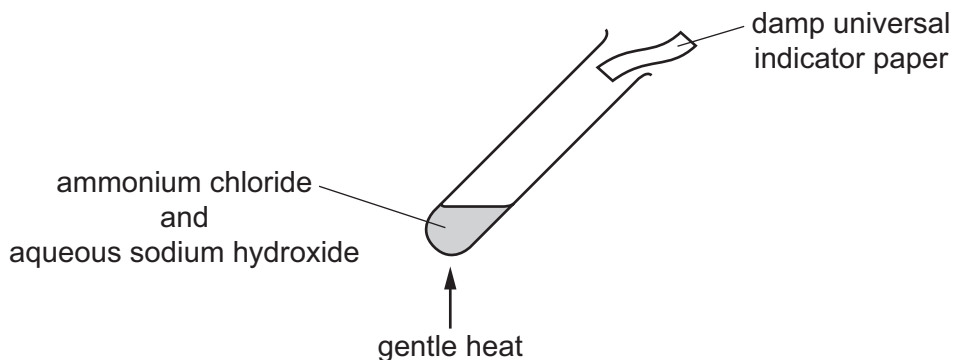
Which substances in the iron are removed?

|          | oxygen removes | calcium oxide removes |
|----------|----------------|-----------------------|
| <b>A</b> | carbon         | acidic oxides         |
| <b>B</b> | carbon         | basic oxides          |
| <b>C</b> | iron           | acidic oxides         |
| <b>D</b> | iron           | basic oxides          |

**27** Which row describes a use of the metal and explains why it is used?

|          | metal     | use                | reason                        |
|----------|-----------|--------------------|-------------------------------|
| <b>A</b> | aluminium | food containers    | good conductor of electricity |
| <b>B</b> | aluminium | aircraft wings     | high density                  |
| <b>C</b> | copper    | cooking utensils   | good conductor of heat        |
| <b>D</b> | copper    | electricity cables | good electrical insulator     |

**28** Ammonium chloride is heated with aqueous sodium hydroxide.



A gas is produced which turns damp universal indicator paper blue.

Which gas has been produced?

- A** ammonia
- B** hydrogen
- C** oxygen
- D** sulfur dioxide

29 Which two gases make up approximately 99% of clean, dry air?

- A carbon dioxide and nitrogen
- B carbon dioxide and oxygen
- C nitrogen and oxygen
- D argon and nitrogen

30 A student writes three statements about potassium nitrate,  $\text{KNO}_3$ .

- 1 The relative formula mass of  $\text{KNO}_3$  is 101.
- 2 Potassium nitrate contains the three essential elements for plant growth.
- 3 Potassium nitrate could be used as a fertiliser.

Which statements are correct?

- A 1 and 2 only    B 1 and 3 only    C 2 and 3 only    D 1, 2 and 3

31 Which row describes the uses of sulfur and sulfur dioxide?

|   | sulfur                       | sulfur dioxide        |
|---|------------------------------|-----------------------|
| A | extraction of aluminium      | food preservative     |
| B | extraction of aluminium      | manufacture of cement |
| C | manufacture of sulfuric acid | food preservative     |
| D | manufacture of sulfuric acid | manufacture of cement |

32 A white solid Z reacts with dilute hydrochloric acid to produce a gas.

The same gas is produced when compound Z is heated strongly.

What is Z?

- A calcium
- B calcium carbonate
- C calcium hydroxide
- D calcium oxide



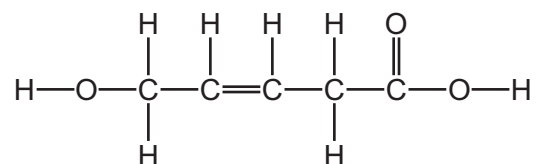
33 Some information about compound L is listed.

- 1 L is an organic compound which contains four hydrogen atoms.
- 2 L is soluble in water.
- 3 An aqueous solution of L reacts with copper(II) carbonate to produce a gas.

What is L?

- A methane
- B ethene
- C ethanoic acid
- D ethanol

34 The structure of an organic molecule is shown.



Which functional groups does this molecule contain?

|   | alcohol | alkene | carboxylic acid |
|---|---------|--------|-----------------|
| A | no      | no     | no              |
| B | no      | yes    | yes             |
| C | yes     | no     | yes             |
| D | yes     | yes    | yes             |

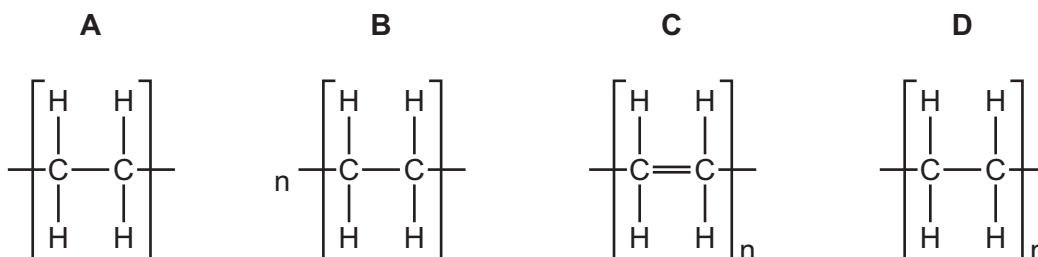
35 Which compounds belong to the same homologous series?

- A ethane and propane
- B ethanoic acid and ethanol
- C methane and ethene
- D propene and ethanoic acid

36 Which statement about alkanes is correct?

- A They burn in oxygen.
- B They contain carbon, hydrogen and oxygen atoms.
- C They contain double bonds.
- D They contain ionic bonds.

37 Which structure represents poly(ethene)?



38 P, Q, R and S are four organic compounds.

P is an unsaturated hydrocarbon.

Q burns but otherwise is unreactive.

R contains a C–C single bond and a C=C double bond.

S undergoes addition polymerisation.

Which compounds are alkenes?

- A** P and R only    **B** P, R and S    **C** P, Q and S    **D** Q, R and S

39 Which statement about petroleum fractions is correct?

- A All petroleum fractions are used as fuels.
- B Gas oil is used to make bottled gas for heating.
- C Hydrocarbons in diesel have higher boiling points than hydrocarbons in gasoline.
- D Molecules in kerosene are larger than molecules in fuel oil.

40 Which substance is a natural polymer?

- A ethene
- B Terylene
- C nylon
- D protein

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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                           |                          |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|---------------------------|--------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                          | V                           | VI                            | VII                       | VIII                     |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12      | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16        | 9<br>F<br>fluorine<br>19  | 10<br>Ne<br>neon<br>20   |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                           |                          |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        |                             |                             |                             |                               |                           |                          |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127  | 54<br>Xe<br>xenon<br>131 |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—     | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>—   |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— |                             | 114<br>Fl<br>flerovium<br>— |                             | 116<br>Lv<br>livermorium<br>— |                           |                          |

lanthanoids

actinoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2021**

**MARK SCHEME**

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | B      | 1     |
| 2        | B      | 1     |
| 3        | D      | 1     |
| 4        | A      | 1     |
| 5        | A      | 1     |
| 6        | D      | 1     |
| 7        | D      | 1     |
| 8        | B      | 1     |
| 9        | D      | 1     |
| 10       | A      | 1     |
| 11       | B      | 1     |
| 12       | D      | 1     |
| 13       | C      | 1     |
| 14       | C      | 1     |
| 15       | B      | 1     |
| 16       | C      | 1     |
| 17       | B      | 1     |
| 18       | B      | 1     |
| 19       | B      | 1     |
| 20       | D      | 1     |
| 21       | A      | 1     |
| 22       | D      | 1     |
| 23       | B      | 1     |
| 24       | C      | 1     |
| 25       | C      | 1     |
| 26       | A      | 1     |
| 27       | C      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | B      | 1     |
| 31       | C      | 1     |
| 32       | B      | 1     |
| 33       | C      | 1     |
| 34       | D      | 1     |
| 35       | A      | 1     |
| 36       | A      | 1     |
| 37       | D      | 1     |
| 38       | B      | 1     |
| 39       | C      | 1     |
| 40       | D      | 1     |





# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Which row describes what happens to the particles in solid iodine when it is heated and turned into a gas?

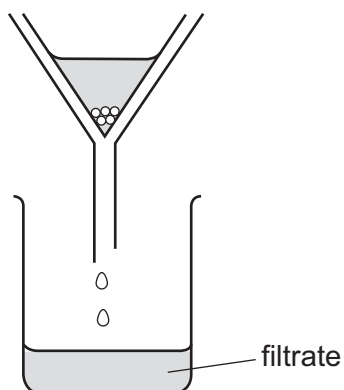
|          | separation of particles | speed of particles |
|----------|-------------------------|--------------------|
| <b>A</b> | closer together         | faster             |
| <b>B</b> | closer together         | slower             |
| <b>C</b> | further apart           | faster             |
| <b>D</b> | further apart           | slower             |

- 2 A student put exactly  $25.00\text{ cm}^3$  of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A** balance, measuring cylinder, thermometer  
**B** balance, pipette, stopwatch  
**C** balance, pipette, thermometer  
**D** burette, pipette, thermometer
- 3 A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.



What is the filtrate?

- A** broken glass only  
**B** broken glass and sugar solution  
**C** pure water  
**D** sugar solution

- 4 How many protons, neutrons and electrons are there in one atom of the isotope  $^{27}_{13}\text{Al}$ ?

|          | protons | neutrons | electrons |
|----------|---------|----------|-----------|
| <b>A</b> | 13      | 13       | 13        |
| <b>B</b> | 13      | 14       | 13        |
| <b>C</b> | 14      | 13       | 13        |
| <b>D</b> | 14      | 14       | 13        |

- 5 Which description of brass is correct?

- A** alloy  
**B** compound  
**C** element  
**D** non-metal

- 6 Rubidium is in Group I and iodine is in Group VII of the Periodic Table.

Which row describes what happens when rubidium and iodine react together to form rubidium iodide?

|          | rubidium                                      | iodine                                        |
|----------|-----------------------------------------------|-----------------------------------------------|
| <b>A</b> | each atom gains one electron                  | each atom loses one electron                  |
| <b>B</b> | each atom loses one electron                  | each atom gains one electron                  |
| <b>C</b> | each atom loses more than one electron        | each atom gains more than one electron        |
| <b>D</b> | each atom neither gains nor loses an electron | each atom neither gains nor loses an electron |

- 7 Which row shows the properties for an ionic compound?

|          | volatility | electrical conductivity when solid |
|----------|------------|------------------------------------|
| <b>A</b> | high       | good                               |
| <b>B</b> | high       | poor                               |
| <b>C</b> | low        | good                               |
| <b>D</b> | low        | poor                               |

8 Which substance is described as a macromolecule?

- A ammonia
- B graphite
- C iron
- D sodium chloride

9 The formula of sodium chlorate(V) is  $\text{NaClO}_3$ .

What is the relative formula mass of sodium chlorate(V),  $\text{NaClO}_3$ ?

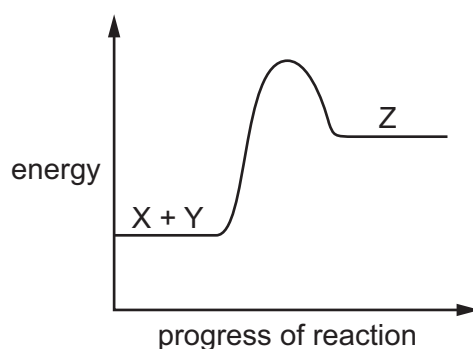
- A 52.0                      B 74.5                      C 106.5                      D 223.5

10 Iron can be electroplated with zinc to make it resistant to corrosion.

Which row about electroplating iron with zinc is correct?

|   | positive electrode<br>(anode) | negative electrode<br>(cathode) | electrolyte  |
|---|-------------------------------|---------------------------------|--------------|
| A | iron                          | zinc                            | iron nitrate |
| B | iron                          | zinc                            | zinc nitrate |
| C | zinc                          | iron                            | iron nitrate |
| D | zinc                          | iron                            | zinc nitrate |

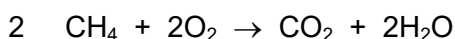
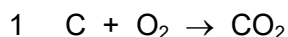
11 An energy level diagram for the reaction between substance X and substance Y to form substance Z is shown.



Which statement is correct?

- A Energy is released as substance Z is formed.
- B Substance Z has more energy than substance X and substance Y.
- C The reaction is exothermic.
- D When substance X and substance Y react, the temperature increases.

12 Which reactions are exothermic?



- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

13 Solid copper(II) carbonate reacts with dilute sulfuric acid.



The rate of the reaction can be changed by varying the conditions.

Which changes always increase the rate of this chemical reaction?

- 1 increasing the concentration of sulfuric acid
- 2 increasing the size of the pieces of copper(II) carbonate
- 3 increasing the temperature
- 4 increasing the volume of sulfuric acid

- A** 1, 3 and 4      **B** 1 and 3 only      **C** 2 and 3      **D** 3 and 4 only

14 When a piece of marble is added to hydrochloric acid, bubbles of carbon dioxide gas are given off.

Which method is used to find the rate of the reaction?

- A** counting the number of gas bubbles formed
- B** measuring the diameter of the gas bubbles
- C** measuring the speed at which the gas bubbles rise upwards through the acid
- D** measuring the time taken for  $10\text{ cm}^3$  of gas to be collected

15 Solid X is heated strongly.

The colour of the solid changes from blue to white.

What is solid X?

- A** anhydrous cobalt(II) chloride
- B** calcium carbonate
- C** hydrated copper(II) sulfate
- D** lead(II) bromide

**16** What happens to a chemical substance when it is reduced?

- A** It burns.
- B** It decomposes.
- C** It loses oxygen.
- D** It gains mass.

**17** Which statements about acids and bases are correct?

- 1 An acid reacts with a metal to give off hydrogen.
- 2 A base reacts with an ammonium salt to give off ammonia.
- 3 An acid reacts with a carbonate to give off carbon dioxide.
- 4 Alkaline solutions are orange in methyl orange.

- A** 1, 2 and 3      **B** 1, 2 and 4      **C** 1, 3 and 4      **D** 2, 3 and 4

**18** Oxide 1 is a solid that reacts with dilute hydrochloric acid.

Oxide 2 is a gas that reacts with sodium hydroxide solution.

What are the formulae of the oxides?

|          | oxide 1         | oxide 2         |
|----------|-----------------|-----------------|
| <b>A</b> | CaO             | MgO             |
| <b>B</b> | MgO             | NO <sub>2</sub> |
| <b>C</b> | NO <sub>2</sub> | SO <sub>2</sub> |
| <b>D</b> | SO <sub>2</sub> | CaO             |

**19** In the preparation of zinc sulfate crystals, excess zinc oxide is added to dilute sulfuric acid.

Why is an excess of zinc oxide added?

- A** to make sure crystals are formed and not powder
- B** to avoid filtering the mixture
- C** to use up all of the sulfuric acid
- D** to use up all of the zinc oxide

**20** Which statement about aqueous sodium hydroxide is correct?

- A** When it is added to a solution containing sulfate ions, a white precipitate is formed.
- B** When it is added to a solution of copper(II) ions, a blue precipitate is formed which dissolves in excess to give deep blue solution.
- C** When it is added to a solution of iron(II) ions, a green precipitate is formed which does not dissolve in excess.
- D** When it is added to ammonium chloride, a gas is produced which turns blue litmus red.

**21** A period of the Periodic Table is shown.

|         |   |    |     |    |   |    |     |      |
|---------|---|----|-----|----|---|----|-----|------|
| group   | I | II | III | IV | V | VI | VII | VIII |
| element | R | S  | T   | V  | W | X  | Y   | Z    |

The letters are not their chemical symbols.

Which statement is correct?

- A** Element R does not conduct electricity.
- B** Elements R and Y react together to form an ionic compound.
- C** Element Z exists as a diatomic molecule.
- D** Element Z reacts with element T.

**22** Which statement about the elements in Group VII of the Periodic Table is correct?

- A** Chlorine can displace bromine from bromides.
- B** Group VII elements are all solids at room temperature.
- C** Group VII elements occur as monoatomic covalent molecules.
- D** Reactivity increases down Group VII.

**23** Part of the Periodic Table is shown.

Which element is a transition element?

**24** The noble gases are in Group VIII of the Periodic Table.

Which statement explains why noble gases are unreactive?

- A** They all have eight electrons in their outer shells.
- B** They all have full outer shells.
- C** They are all gases.
- D** They are all monoatomic.

**25** Which statement is correct for **all** metals?

- A** They conduct electricity when molten.
- B** They gain electrons when they form ions.
- C** They have a low density.
- D** They have a low melting point.

**26** Which row describes the method of extraction of aluminium and iron from their ores?

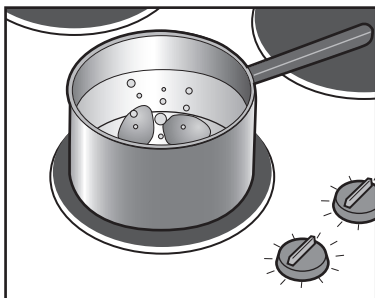
|          | aluminium             | iron                  |
|----------|-----------------------|-----------------------|
| <b>A</b> | electrolysis          | electrolysis          |
| <b>B</b> | electrolysis          | reduction with carbon |
| <b>C</b> | reduction with carbon | electrolysis          |
| <b>D</b> | reduction with carbon | reduction with carbon |

**27** Which statement about metals and their uses is correct?

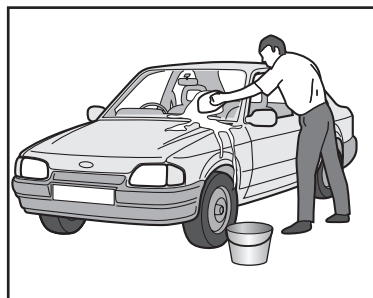
- A** Aluminium is used to make food containers because it is resistant to corrosion.
- B** Aluminium is used to make aircraft wings because it is strong and has a high density.
- C** Iron is used to make electrical wires because it is a good insulator of electricity.
- D** Iron is used to make cooking utensils because it is easily recycled.



28 The diagrams show some uses of water in the home.



1



2



3

For which uses is it important for the water to have been treated?

- A** 1 only      **B** 2 only      **C** 3 only      **D** 1, 2 and 3

29 Four different test-tubes containing water and an iron nail are left for two weeks.

Which nail showed the least amount of rusting?

**A**

tap water

**B**boiled  
tap water**C**boiled  
tap water**D**

tap water

30 Which process does **not** produce a greenhouse gas?

- A** acid rain on limestone buildings  
**B** combustion of wood  
**C** digestion in cows  
**D** zinc reacting with sulfuric acid

- 31 Sulfur burns to make sulfur dioxide.

Which row describes a source of sulfur and a use of sulfur dioxide?

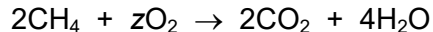
|          | source of sulfur     | use of sulfur dioxide |
|----------|----------------------|-----------------------|
| <b>A</b> | the air              | food preservative     |
| <b>B</b> | the air              | treating acidic soils |
| <b>C</b> | underground deposits | food preservative     |
| <b>D</b> | underground deposits | treating acidic soils |

- 32 Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A** ammonia
- B** sodium chloride
- C** sodium hydroxide
- D** sulfuric acid

- 33 A chemical equation for the complete combustion of methane is shown.



What is the value of  $z$ ?

- A** 2                      **B** 3                      **C** 4                      **D** 6

- 34 Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.

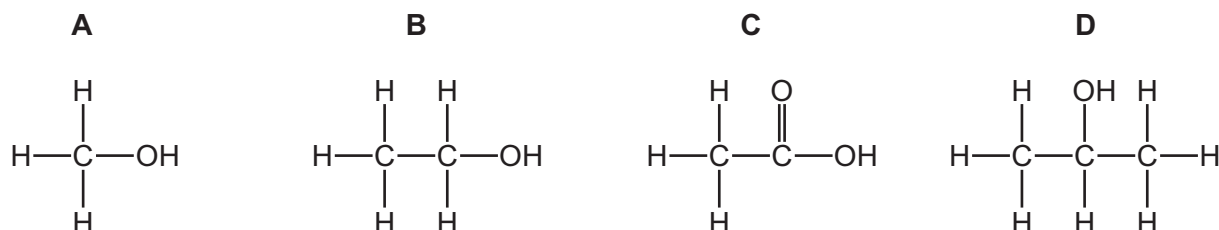
What could X and Y be?

|          | X               | Y                              |
|----------|-----------------|--------------------------------|
| <b>A</b> | C               | H <sub>2</sub>                 |
| <b>B</b> | C               | C <sub>8</sub> H <sub>18</sub> |
| <b>C</b> | CH <sub>4</sub> | H <sub>2</sub>                 |
| <b>D</b> | CH <sub>4</sub> | C <sub>8</sub> H <sub>18</sub> |

35 Which substance is **not** a fossil fuel?

- A ethanol
- B gasoline
- C kerosene
- D methane

36 Which compound belongs to a different homologous series to the others?



37 What is a property of aqueous ethanoic acid?

- A It changes red litmus blue.
- B It has a deep purple colour.
- C It has a pH of less than 7.
- D It reacts with a metal oxide to form carbon dioxide.

38 Which statements about unsaturated hydrocarbons are correct?

- 1 They contain both single and double bonds.
- 2 They turn aqueous bromine from colourless to brown.
- 3 They can be manufactured by cracking.

- A 1 and 2 only    B 1 and 3 only    C 2 and 3 only    D 1, 2 and 3

39 Which substance is used to produce alcohol by fermentation?

- A phosphoric acid
- B platinum
- C iron
- D yeast

**40** Which statements are correct?

- 1 Polymers are large molecules built up from monomers.
- 2 Proteins are natural polymers.
- 3 Proteins and carbohydrates are constituents of food.

**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                              |                              |                            |                             |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|------------------------------|------------------------------|----------------------------|-----------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                            | V                            | VI                           | VII                        | VIII                        |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12        | 7<br>N<br>nitrogen<br>14     | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19   | 10<br>Ne<br>neon<br>20      |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 12<br>Mg<br>magnesium<br>24 | 13<br>Al<br>aluminium<br>27   | 14<br>Si<br>silicon<br>28    | 15<br>P<br>phosphorus<br>31  | 16<br>S<br>sulfur<br>32    |                             |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73   | 33<br>As<br>arsenic<br>75    | 34<br>Se<br>selenium<br>79   | 35<br>Br<br>bromine<br>80  | 36<br>Kr<br>krypton<br>84   |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119        | 51<br>Sb<br>antimony<br>122  | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127   | 54<br>Xe<br>xenon<br>131    |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207       | 83<br>Bi<br>bismuth<br>209   | 84<br>Po<br>polonium<br>—    | 85<br>At<br>astatine<br>—  | 86<br>Rn<br>radon<br>—      |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesson<br>—  | 119<br>Nh<br>nihonium<br>— | 120<br>Fl<br>flerovium<br>— |

lanthanoids

actinoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).





## Cambridge IGCSE™ (9–1)

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**CHEMISTRY**

**0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2021**

**MARK SCHEME**

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | C      | 1     |
| 3        | D      | 1     |
| 4        | B      | 1     |
| 5        | A      | 1     |
| 6        | B      | 1     |
| 7        | D      | 1     |
| 8        | B      | 1     |
| 9        | C      | 1     |
| 10       | D      | 1     |
| 11       | B      | 1     |
| 12       | A      | 1     |
| 13       | B      | 1     |
| 14       | D      | 1     |
| 15       | C      | 1     |
| 16       | C      | 1     |
| 17       | A      | 1     |
| 18       | B      | 1     |
| 19       | C      | 1     |
| 20       | C      | 1     |
| 21       | B      | 1     |
| 22       | A      | 1     |
| 23       | C      | 1     |
| 24       | B      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | A      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | B      | 1     |
| 30       | D      | 1     |
| 31       | C      | 1     |
| 32       | D      | 1     |
| 33       | C      | 1     |
| 34       | D      | 1     |
| 35       | A      | 1     |
| 36       | C      | 1     |
| 37       | C      | 1     |
| 38       | B      | 1     |
| 39       | D      | 1     |
| 40       | D      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/11

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** write on any bar codes.
- You may use a calculator.

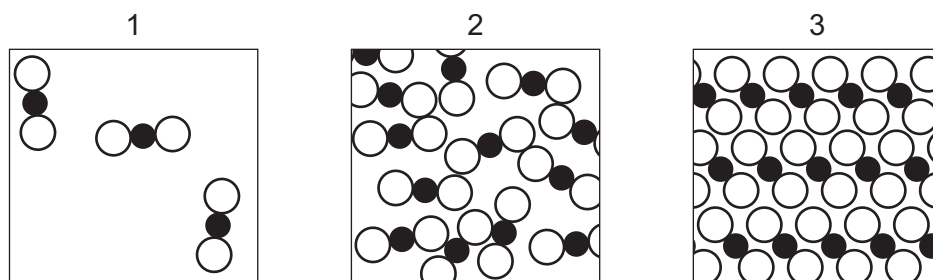
## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Diagrams of the three states of matter for carbon dioxide are shown.



Which two diagrams show the states of matter before and after the sublimation of carbon dioxide?

- A** 2 to 1      **B** 2 to 3      **C** 3 to 1      **D** 3 to 2
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm<sup>3</sup> of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
- 2 measuring cylinder
- 3 thermometer
- 4 balance

- A** 1, 2 and 4      **B** 1 and 2 only      **C** 1 and 4 only      **D** 2, 3 and 4
- 3 Which method is used to separate a mixture of the following liquids?

| liquid      | boiling point/°C |
|-------------|------------------|
| methanol    | 64.5             |
| ethanol     | 78.5             |
| propan-1-ol | 97.2             |
| butan-1-ol  | 117.0            |

- A** crystallisation
- B** evaporation
- C** filtration
- D** fractional distillation

4 Which substance should be pure for the intended use?

- A a drug for curing disease
- B limestone for iron extraction
- C petroleum for fractional distillation
- D water for washing a car

5 Which row identifies an alloy, a pure metal and a non-metal?

|          | alloy  | pure metal | non-metal |
|----------|--------|------------|-----------|
| <b>A</b> | brass  | carbon     | copper    |
| <b>B</b> | brass  | copper     | carbon    |
| <b>C</b> | copper | brass      | carbon    |
| <b>D</b> | copper | carbon     | brass     |

6 Information about the structures of three atoms, X, Y and Z, is shown.

| atom | proton number | nucleon number |
|------|---------------|----------------|
| X    | 1             | 1              |
| Y    | 1             | 2              |
| Z    | 1             | 3              |

Which statements about atoms X, Y and Z are correct?

- 1 They are isotopes of the same element.
- 2 They contain the same number of electrons.
- 3 They contain the same number of neutrons.
- 4 They contain one occupied electron shell.

**A** 1, 2 and 4      **B** 1 and 2 only      **C** 3 and 4      **D** 3 only

7 What happens to an atom when it becomes an ion with a charge of +1?

- A It gains an electron.
- B It gains a proton.
- C It loses an electron.
- D It loses a proton.

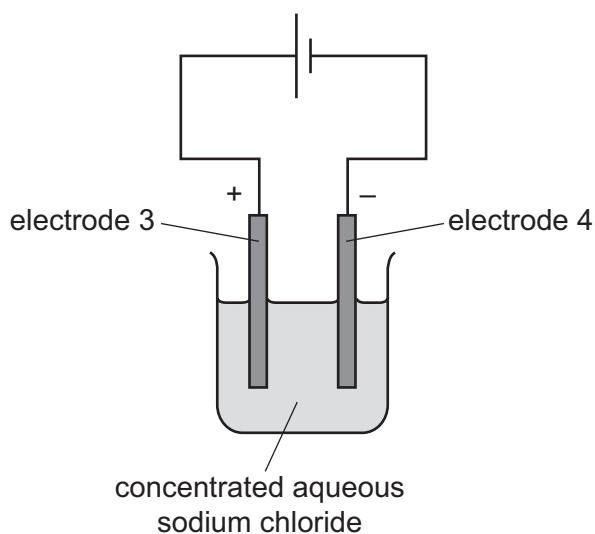
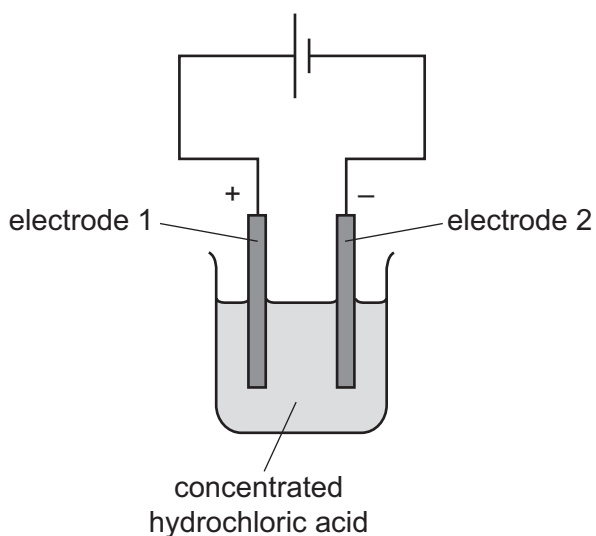
- 8 The relative atomic mass,  $A_r$ , of an element is determined by comparing the mass of one atom of the element with the mass of one atom of element Q.

What is Q?

- A carbon
  - B chlorine
  - C hydrogen
  - D oxygen
- 9 Which equation for the decomposition of calcium nitrate is correct?

- A  $\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + \text{NO}_2 + \text{O}_2$
- B  $\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \text{O}_2$
- C  $2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 2\text{NO}_2 + \text{O}_2$
- D  $2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$

- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
- B electrodes 1 and 3
- C electrode 2 only
- D electrodes 2 and 4

- 11** Overhead power cables made from (steel-cored) aluminium are used to carry electricity over long distances.

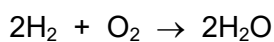
Which property of (steel-cored) aluminium makes it suitable for use in power cables?

- A** Aluminium has a low density.
- B** Aluminium has low strength.
- C** Steel is a good conductor of heat.
- D** Steel is resistant to corrosion.

- 12** Which row identifies a chemical change and a physical change?

|          | chemical change             | physical change             |
|----------|-----------------------------|-----------------------------|
| <b>A</b> | boiling ethanol             | burning ethanol             |
| <b>B</b> | burning ethanol             | evaporating ethanol         |
| <b>C</b> | dissolving ethanol in water | burning ethanol             |
| <b>D</b> | evaporating ethanol         | dissolving ethanol in water |

- 13** The equation for the reaction when hydrogen is used as a fuel is shown.



Which statement about this reaction is correct?

- A** Energy is given out so the temperature of the surroundings decreases.
- B** Energy is taken in so the temperature of the surroundings increases.
- C** The reaction is endothermic so the temperature of the surroundings decreases.
- D** The reaction is exothermic so the temperature of the surroundings increases.

- 14** Which fuels release carbon dioxide when burned?

- 1 gasoline
- 2 hydrogen
- 3 methane

- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 3 only

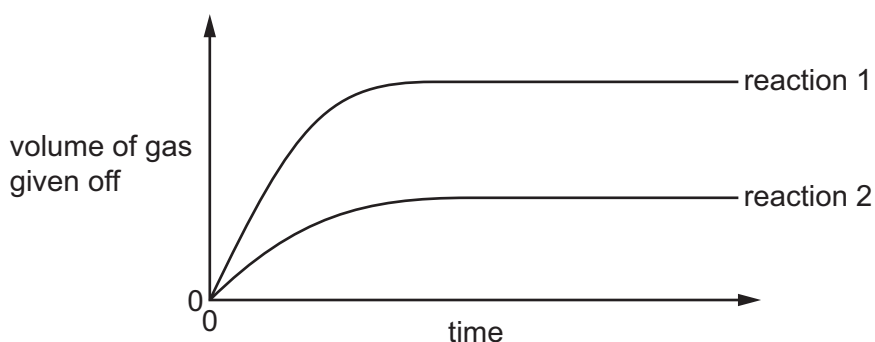


- 15** Water is added to anhydrous copper(II) sulfate.

What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
  - B** The copper(II) sulfate turns blue and the solution formed gets hotter.
  - C** The copper(II) sulfate turns white and the solution formed gets colder.
  - D** The copper(II) sulfate turns white and the solution formed gets hotter.
- 16** Excess magnesium ribbon is reacted with a fixed volume of hydrochloric acid and the volume of gas given off over time is measured.

The results of two different experiments are shown.



Which statement explains the differences between the results of the two experiments?

- A** Reaction 1 uses a catalyst.
  - B** The acid used is twice as concentrated in reaction 1.
  - C** The magnesium ribbon is in smaller pieces in reaction 2.
  - D** The temperature is higher in reaction 2.
- 17** Which products are formed when magnesium hydroxide reacts with hydrochloric acid?
- A** magnesium chloride, carbon dioxide and water
  - B** magnesium chloride, hydrogen and water
  - C** magnesium chloride and hydrogen only
  - D** magnesium chloride and water only

- 18** The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

| oxide tested | pH of solution |
|--------------|----------------|
| X            | 1              |
| Y            | 13             |

Which information about X and Y is correct?

|          | oxide is acidic | oxide is basic | metal | non-metal |
|----------|-----------------|----------------|-------|-----------|
| <b>A</b> | X               | Y              | X     | Y         |
| <b>B</b> | X               | Y              | Y     | X         |
| <b>C</b> | Y               | X              | X     | Y         |
| <b>D</b> | Y               | X              | Y     | X         |

- 19** An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration  
**B** evaporation → crystallisation → filtration  
**C** filtration → crystallisation → evaporation  
**D** filtration → evaporation → crystallisation

- 20** Three separate samples of an aqueous compound T are tested.

The results of the tests are shown.

| test                                                                | observation                          |
|---------------------------------------------------------------------|--------------------------------------|
| acidify with dilute nitric acid,<br>then add aqueous barium nitrate | white precipitate                    |
| add aqueous ammonia                                                 | white precipitate, soluble in excess |
| add aqueous sodium hydroxide                                        | white precipitate, soluble in excess |

## What is T?

- A** aluminium chloride  
**B** aluminium sulfate  
**C** zinc chloride  
**D** zinc sulfate

- 21** Part of the Periodic Table is shown.

Which element is a metal?

A simplified periodic table grid is shown, with elements A, B, C, and D marked. The grid consists of 18 columns and 6 rows. The layout is as follows:

- Row 1: Columns 1-2 are filled. Column 18 is filled.
- Row 2: Columns 1-2 are filled. Columns 13-18 are filled.
- Row 3: Columns 3-12 are filled. Column 13 is labeled 'C'. Column 15 is labeled 'B'. Column 18 is filled.
- Row 4: Columns 3-18 are filled. Column 17 is labeled 'D'.
- Row 5: Columns 3-18 are filled.
- Row 6: Columns 3-18 are filled.

Element A is located in the box above the gap between columns 2 and 3 in row 1. Element B is in row 3, column 15. Element C is in row 3, column 13. Element D is in row 4, column 17.

- 22** The elements sodium to argon form Period 3 of the Periodic Table.

Which row describes the trend across Period 3 from left to right?

|          | number of<br>outer-shell electrons | metallic<br>character | group<br>number |
|----------|------------------------------------|-----------------------|-----------------|
| <b>A</b> | decreases                          | decreases             | decreases       |
| <b>B</b> | decreases                          | increases             | decreases       |
| <b>C</b> | increases                          | decreases             | increases       |
| <b>D</b> | increases                          | increases             | increases       |

**23** Lithium, sodium and potassium are elements in Group I of the Periodic Table.

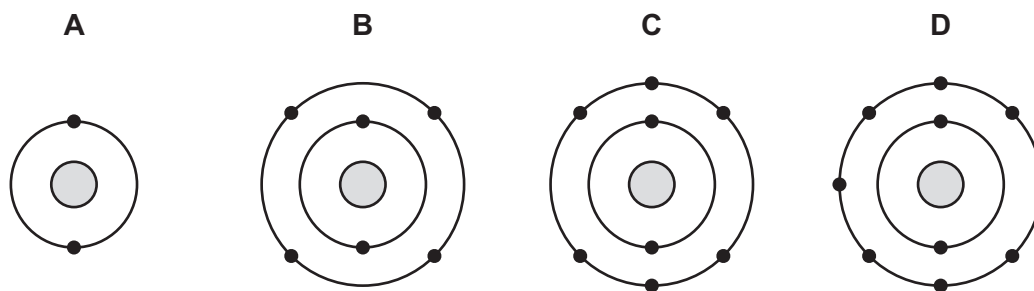
Which statement about these elements is correct?

- A** Lithium has the highest melting point and the lowest density.
- B** Lithium has the highest density and the most violent reaction with water.
- C** Potassium has the highest melting point and the highest density.
- D** Potassium has the lowest melting point and the least violent reaction with water.

**24** Which statement describes a transition element?

- A** It can act as a catalyst and some of its compounds can also act as catalysts.
- B** It forms white compounds with sulfur, oxygen, chlorine and bromine.
- C** It has a low density and a piece of it will float on water.
- D** It is a very poor conductor of electricity.

**25** Which diagram represents the arrangement of the outer-shell electrons of a noble gas?



**26** Which statements about the general properties of metals are correct?

- 1 They are good conductors of heat and electricity.
- 2 They have low melting points.
- 3 They react with dilute acids to form a salt and water.
- 4 They react with oxygen to form basic oxides.

- A** 1 and 2              **B** 1 and 4              **C** 2 and 3              **D** 3 and 4

27 Reactions of three metals and their oxides are shown.

| metal | add dilute hydrochloric acid to metal | heat metal oxide with carbon |
|-------|---------------------------------------|------------------------------|
| 1     | ✓                                     | ✓                            |
| 2     | ✓                                     | ✗                            |
| 3     | ✗                                     | ✓                            |

key

✓ = reacts

✗ = does not react

What is the order of reactivity of these metals, from most reactive to least reactive?

- A** 1 → 2 → 3      **B** 1 → 3 → 2      **C** 2 → 1 → 3      **D** 2 → 3 → 1

28 Which uses of the metals shown are correct?

|          | aluminium       | stainless steel |
|----------|-----------------|-----------------|
| <b>A</b> | aircraft bodies | car bodies      |
| <b>B</b> | car bodies      | aircraft bodies |
| <b>C</b> | chemical plant  | food containers |
| <b>D</b> | food containers | cutlery         |

29 Which statement about the reactions in the blast furnace is correct?

- A** Carbon reacts with oxygen and heats the furnace.  
**B** Carbon monoxide removes the silicon dioxide impurity forming slag.  
**C** Iron(III) oxide is oxidised to iron.  
**D** Limestone reduces iron(III) oxide to iron.

30 Iron rusts when exposed to air.

Which two substances in air cause iron to rust?

- A** carbon dioxide and oxygen  
**B** nitrogen and oxygen  
**C** oxygen and water  
**D** carbon dioxide and water

**31** Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_2\text{SO}_4$
- B**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_3\text{PO}_4$
- C**  $\text{KNO}_3$  and  $(\text{NH}_4)_2\text{SO}_4$
- D**  $\text{KNO}_3$  and  $(\text{NH}_4)_3\text{PO}_4$

**32** Which process produces methane?

- A** combustion of hydrocarbons
- B** decomposition of vegetation
- C** respiration
- D** reaction between hydrochloric acid and calcium carbonate

**33** Which statements about sulfur dioxide are correct?

- 1 Sulfur dioxide decolourises acidified potassium manganate(VII).
- 2 Sulfur dioxide forms when acids react with carbonates.
- 3 Sulfur dioxide is used as a bleach.
- 4 Sulfur dioxide is used to treat acidic soil.

- A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

**34** What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
- B** calcium hydroxide and carbon monoxide
- C** calcium oxide and carbon dioxide
- D** calcium oxide and carbon monoxide

**35** In which lists are the compounds in the same homologous series?

- 1  $\text{CH}_4$ ,  $\text{C}_2\text{H}_4$ ,  $\text{C}_3\text{H}_8$
- 2  $\text{CH}_3\text{OH}$ ,  $\text{C}_2\text{H}_5\text{OH}$ ,  $\text{C}_3\text{H}_7\text{OH}$
- 3  $\text{CH}_3\text{CO}_2\text{H}$ ,  $\text{CH}_3\text{CH}_2\text{OH}$ ,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- A** 1 and 2      **B** 1 and 3      **C** 2 only      **D** 3 only

36 Which row about aqueous ethanoic acid and dilute hydrochloric acid is correct?

|          | both contain carbon | both contain hydrogen | both react with carbonates |
|----------|---------------------|-----------------------|----------------------------|
| <b>A</b> | ✓                   | ✗                     | ✓                          |
| <b>B</b> | ✓                   | ✓                     | ✗                          |
| <b>C</b> | ✗                   | ✓                     | ✓                          |
| <b>D</b> | ✗                   | ✗                     | ✗                          |

key

✓ = yes

✗ = no

37 Some properties of colourless liquid L are listed.

- It boils at 65 °C.
- When added to water, two layers form which do not mix.
- It does not react with sodium carbonate.
- It has no effect on bromine water.

What is L?

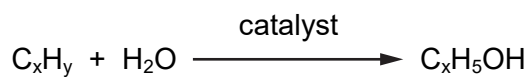
- A** ethanol
- B** hexane
- C** hexene
- D** ethanoic acid

38 A molecule of compound P contains two carbon atoms and four hydrogen atoms.

Which row represents P?

|          | name of compound | $M_r$ | reacts with aqueous bromine |
|----------|------------------|-------|-----------------------------|
| <b>A</b> | ethane           | 30    | ✗                           |
| <b>B</b> | ethene           | 16    | ✓                           |
| <b>C</b> | ethene           | 28    | ✓                           |
| <b>D</b> | ethene           | 28    | ✗                           |

- 39 The equation representing the reaction of a hydrocarbon with water is shown.



What are the values of x and y?

|          | x | y |
|----------|---|---|
| <b>A</b> | 1 | 4 |
| <b>B</b> | 1 | 6 |
| <b>C</b> | 2 | 4 |
| <b>D</b> | 2 | 6 |

- 40 Many molecules of J join together in reaction R to form a long chain molecule K.

K is the only product.

Which row describes molecule J, reaction R and molecule K?

|          | molecule J | reaction R | molecule K |
|----------|------------|------------|------------|
| <b>A</b> | polymer    | addition   | monomer    |
| <b>B</b> | monomer    | addition   | polymer    |
| <b>C</b> | polymer    | cracking   | monomer    |
| <b>D</b> | monomer    | cracking   | polymer    |



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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                             |                           |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|------------------------------|-----------------------------|---------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                            | V                           | VI                           | VII                         | VIII                      |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12        | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19    | 10<br>Ne<br>neon<br>20    |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             | 12<br>Mg<br>magnesium<br>24   | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28    | 15<br>P<br>phosphorus<br>31 |                           |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73   | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79   | 35<br>Br<br>bromine<br>80   | 36<br>Kr<br>krypton<br>84 |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119        | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127    | 54<br>Xe<br>xenon<br>131  |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207       | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—    | 85<br>At<br>astatine<br>—   | 86<br>Rn<br>radon<br>—    |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— | 116<br>Lv<br>livermorium<br>— | —                           | —                            | —                           | —                         |

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2022**

MARK SCHEME

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | A      | 1     |
| 3        | D      | 1     |
| 4        | A      | 1     |
| 5        | B      | 1     |
| 6        | A      | 1     |
| 7        | C      | 1     |
| 8        | A      | 1     |
| 9        | D      | 1     |
| 10       | D      | 1     |
| 11       | A      | 1     |
| 12       | B      | 1     |
| 13       | D      | 1     |
| 14       | B      | 1     |
| 15       | B      | 1     |
| 16       | B      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | D      | 1     |
| 20       | D      | 1     |
| 21       | C      | 1     |
| 22       | C      | 1     |
| 23       | A      | 1     |
| 24       | A      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | D      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | A             | <b>1</b>     |
| 30              | C             | <b>1</b>     |
| 31              | D             | <b>1</b>     |
| 32              | B             | <b>1</b>     |
| 33              | A             | <b>1</b>     |
| 34              | C             | <b>1</b>     |
| 35              | C             | <b>1</b>     |
| 36              | C             | <b>1</b>     |
| 37              | B             | <b>1</b>     |
| 38              | C             | <b>1</b>     |
| 39              | C             | <b>1</b>     |
| 40              | B             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/12

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Substances change state when their temperature is changed.

Which changes of state take place when the temperature of a substance is lowered?

- 1 boiling
- 2 condensation
- 3 freezing
- 4 melting

**A** 1 and 4      **B** 2, 3 and 4      **C** 2 and 3 only      **D** 3 only

- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm<sup>3</sup> of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
- 2 measuring cylinder
- 3 thermometer
- 4 balance

**A** 1, 2 and 4      **B** 1 and 2 only      **C** 1 and 4 only      **D** 2, 3 and 4

- 3 Which method is used to separate a mixture of the following liquids?

| liquid      | boiling point/°C |
|-------------|------------------|
| methanol    | 64.5             |
| ethanol     | 78.5             |
| propan-1-ol | 97.2             |
| butan-1-ol  | 117.0            |

- A** crystallisation
- B** evaporation
- C** filtration
- D** fractional distillation



- 4 X and Y are two different elements.

X and Y have the same number of nucleons.

Which statement about X and Y is correct?

- A They have the same physical properties.
- B Their atoms have the same number of electrons.
- C They are in different groups of the Periodic Table.
- D They have different relative masses.

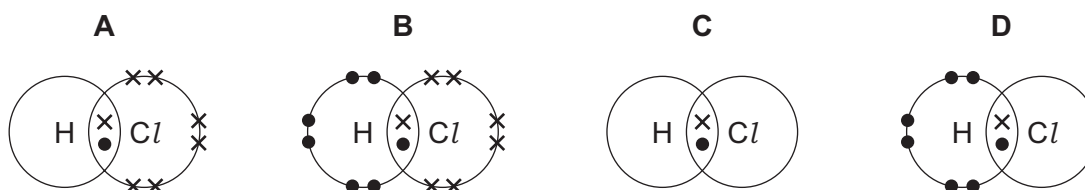
- 5 Which row identifies an alloy, a pure metal and a non-metal?

|          | alloy  | pure metal | non-metal |
|----------|--------|------------|-----------|
| <b>A</b> | brass  | carbon     | copper    |
| <b>B</b> | brass  | copper     | carbon    |
| <b>C</b> | copper | brass      | carbon    |
| <b>D</b> | copper | carbon     | brass     |

- 6 Which statement about ions and ionic bonding is correct?

- A Caesium atoms gain electrons to form negatively charged caesium ions.
- B Ionic bonding involves sharing of pairs of electrons.
- C Potassium ions and chloride ions have the same number of outer-shell electrons.
- D Sodium ions have an equal number of protons and electrons.

- 7 Which dot-and-cross diagram shows the arrangement of outer shell electrons in a molecule of hydrogen chloride?



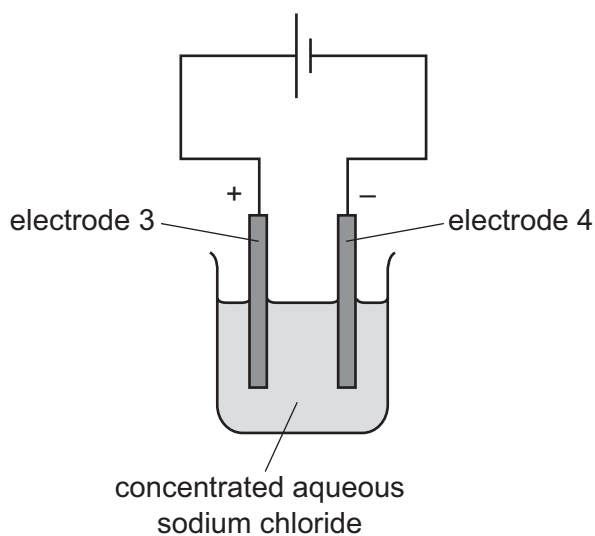
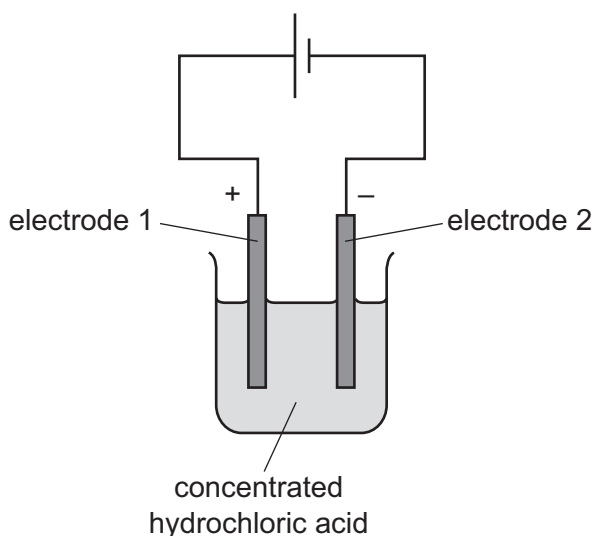
- 8 Which statement explains why graphite can be used as a lubricant?

- A All of the atoms in graphite are carbon.
- B Each carbon atom forms three bonds.
- C Graphite has a macromolecular structure.
- D The layers in graphite can slide over each other.

- 9 A compound of element X has the formula  $X_2O$  and a relative formula mass of 144.

What is element X?

- A copper, Cu
  - B gadolinium, Gd
  - C sulfur, S
  - D tellurium, Te
- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
  - B electrodes 1 and 3
  - C electrode 2 only
  - D electrodes 2 and 4
- 11 Which type of reaction occurs when calcium carbonate is heated at a high temperature to produce calcium oxide and carbon dioxide?
- A combustion
  - B endothermic
  - C oxidation
  - D reduction

12 Which row identifies a chemical change and a physical change?

|          | chemical change             | physical change             |
|----------|-----------------------------|-----------------------------|
| <b>A</b> | boiling ethanol             | burning ethanol             |
| <b>B</b> | burning ethanol             | evaporating ethanol         |
| <b>C</b> | dissolving ethanol in water | burning ethanol             |
| <b>D</b> | evaporating ethanol         | dissolving ethanol in water |

13 Which statement about rate of reaction is correct?

- A** Catalysts increase the time for the reaction to be completed.
- B** Decreasing particle size increases the rate of reaction.
- C** Decreasing temperature increases the rate of reaction.
- D** Rate of reaction decreases as the concentration increases.

14 Some common household substances are tested with litmus and methyl orange.

| household substance | colour of litmus | colour of methyl orange |
|---------------------|------------------|-------------------------|
| bicarbonate of soda | blue             | yellow                  |
| lemonade            | red              | red                     |
| milk                | red              | red                     |
| milk of magnesia    | blue             | yellow                  |
| washing powder      | blue             | yellow                  |
| vinegar             | red              | red                     |

Which statement is correct?

- A** Lemonade, milk and bicarbonate of soda are all acidic.
- B** Milk of magnesia can neutralise washing powder.
- C** Milk of magnesia, washing powder and vinegar are all bases.
- D** Vinegar can neutralise bicarbonate of soda.

**15** Water is added to anhydrous copper(II) sulfate.

What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

**16** In which equation is carbon both oxidised and reduced?

- A**  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- B**  $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
- C**  $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 3\text{CO}_2 + 2\text{Fe}$
- D**  $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$

**17** Aqueous solutions containing copper(II) ions can be identified using flame tests and by adding aqueous sodium hydroxide.

Which row describes what is observed in these tests?

|          | flame test       | aqueous sodium hydroxide |
|----------|------------------|--------------------------|
| <b>A</b> | blue-green flame | light blue precipitate   |
| <b>B</b> | blue-green flame | green precipitate        |
| <b>C</b> | lilac flame      | light blue precipitate   |
| <b>D</b> | lilac flame      | green precipitate        |

- 18** The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

| oxide tested | pH of solution |
|--------------|----------------|
| X            | 1              |
| Y            | 13             |

Which information about X and Y is correct?

|          | oxide is acidic | oxide is basic | metal | non-metal |
|----------|-----------------|----------------|-------|-----------|
| <b>A</b> | X               | Y              | X     | Y         |
| <b>B</b> | X               | Y              | Y     | X         |
| <b>C</b> | Y               | X              | X     | Y         |
| <b>D</b> | Y               | X              | Y     | X         |

- 19** An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration  
**B** evaporation → crystallisation → filtration  
**C** filtration → crystallisation → evaporation  
**D** filtration → evaporation → crystallisation

- 20** Some statements about gas G are listed.

G is monoatomic.

G is found in clean, dry air.

G is used in lamps.

Which element is G?

- A** argon  
**B** helium  
**C** nitrogen  
**D** oxygen

**21** Part of the Periodic Table is shown.

Which element is a metal?

The diagram shows a simplified periodic table grid. The elements are positioned as follows:

- A** is located in the first period, group 18 (top right).
- B** is located in the second period, group 16.
- C** is located in the third period, group 17.
- D** is located in the fourth period, group 18.

**22** The elements sodium to argon form Period 3 of the Periodic Table.

Which row describes the trend across Period 3 from left to right?

|          | number of<br>outer-shell electrons | metallic<br>character | group<br>number |
|----------|------------------------------------|-----------------------|-----------------|
| <b>A</b> | decreases                          | decreases             | decreases       |
| <b>B</b> | decreases                          | increases             | decreases       |
| <b>C</b> | increases                          | decreases             | increases       |
| <b>D</b> | increases                          | increases             | increases       |

**23** Some properties of element E are listed.

It has a high density.

It has a high melting point.

## What is E?

- A** aluminium  
**B** bromine  
**C** iron  
**D** lithium

**24** Lithium, sodium and potassium are elements in Group I of the Periodic Table.

Chlorine, bromine and iodine are elements in Group VII of the Periodic Table.

Which row identifies the **least** dense of these elements in each group?

|          | Group I   | Group VII |
|----------|-----------|-----------|
| <b>A</b> | lithium   | chlorine  |
| <b>B</b> | lithium   | iodine    |
| <b>C</b> | potassium | chlorine  |
| <b>D</b> | potassium | iodine    |

**25** The reactions of metals P, Q, R and S are shown.

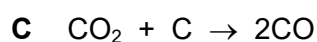
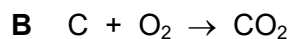
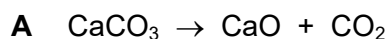
| metal | reaction with water | reaction with hydrochloric acid | reduction of the metal oxide with carbon |
|-------|---------------------|---------------------------------|------------------------------------------|
| P     | no reaction         | no reaction                     | reduced                                  |
| Q     | slow                | vigorous                        | no reaction                              |
| R     | vigorous            | vigorous                        | no reaction                              |
| S     | very slow           | vigorous                        | reduced                                  |

What is the order of reactivity of the metals?

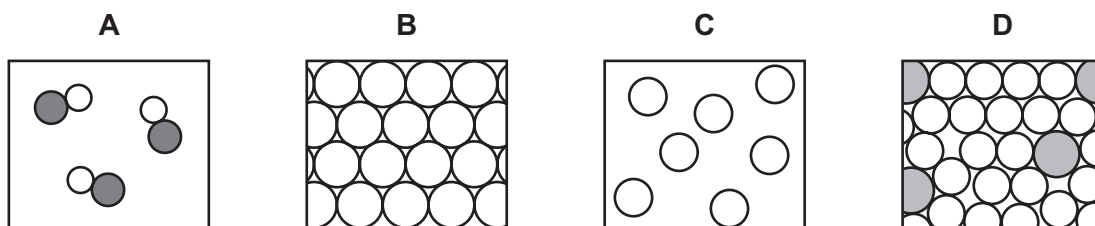
|          | <div style="display: flex; align-items: center; justify-content: space-between;"> <span>least reactive</span> <span>—————→</span> <span>most reactive</span> </div> |   |   |   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| <b>A</b> | P                                                                                                                                                                   | S | Q | R |
| <b>B</b> | P                                                                                                                                                                   | Q | S | R |
| <b>C</b> | R                                                                                                                                                                   | S | Q | P |
| <b>D</b> | R                                                                                                                                                                   | Q | S | P |

**26** Iron is extracted from hematite in the blast furnace at a temperature of about 1550 °C.

Which equation shows the main reaction that increases the temperature in the furnace?



27 Which diagram represents the arrangement of atoms in an alloy?



28 Which uses of the metals shown are correct?

|          | aluminium       | stainless steel |
|----------|-----------------|-----------------|
| <b>A</b> | aircraft bodies | car bodies      |
| <b>B</b> | car bodies      | aircraft bodies |
| <b>C</b> | chemical plant  | food containers |
| <b>D</b> | food containers | cutlery         |

29 Which row identifies a substance present in clean air and a substance that is a pollutant in air?

|          | present in clean air | pollutant in air |
|----------|----------------------|------------------|
| <b>A</b> | oxides of nitrogen   | nitrogen         |
| <b>B</b> | carbon dioxide       | sulfur dioxide   |
| <b>C</b> | carbon monoxide      | lead compounds   |
| <b>D</b> | nitrogen             | argon            |

30 Which property of sulfur dioxide explains why it is used as a food preservative?

- A** acidic oxide
- B** bleach
- C** kills bacteria
- D** pungent smell



**31** Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_2\text{SO}_4$
- B**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_3\text{PO}_4$
- C**  $\text{KNO}_3$  and  $(\text{NH}_4)_2\text{SO}_4$
- D**  $\text{KNO}_3$  and  $(\text{NH}_4)_3\text{PO}_4$

**32** Compound J is an unsaturated carboxylic acid.

Which bonds are present in a molecule of J?

|          | C=C | C=O | O-H |                          |
|----------|-----|-----|-----|--------------------------|
| <b>A</b> | ✓   | ✓   | ✓   | key<br>✓ = yes<br>✗ = no |
| <b>B</b> | ✗   | ✓   | ✓   |                          |
| <b>C</b> | ✓   | ✗   | ✗   |                          |
| <b>D</b> | ✗   | ✓   | ✗   |                          |

**33** Petroleum is separated into useful fractions by fractional distillation.

Which fraction is used as a fuel for jet aeroplanes?

- A** fuel oil
- B** gasoline
- C** naphtha
- D** kerosene / paraffin

**34** What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
- B** calcium hydroxide and carbon monoxide
- C** calcium oxide and carbon dioxide
- D** calcium oxide and carbon monoxide

**35** Ethene reacts with substance X to form ethanol.

What is X?

- A** ethanoic acid
- B** glucose
- C** hydrogen
- D** steam

**36** What is the equation for the complete combustion of methane?

- A**  $\text{CH}_4 + 4\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- B**  $2\text{CH}_4 + 3\text{O}_2 \rightarrow 2\text{CO} + 4\text{H}_2\text{O}$
- C**  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- D**  $\text{C}_2\text{H}_6 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$

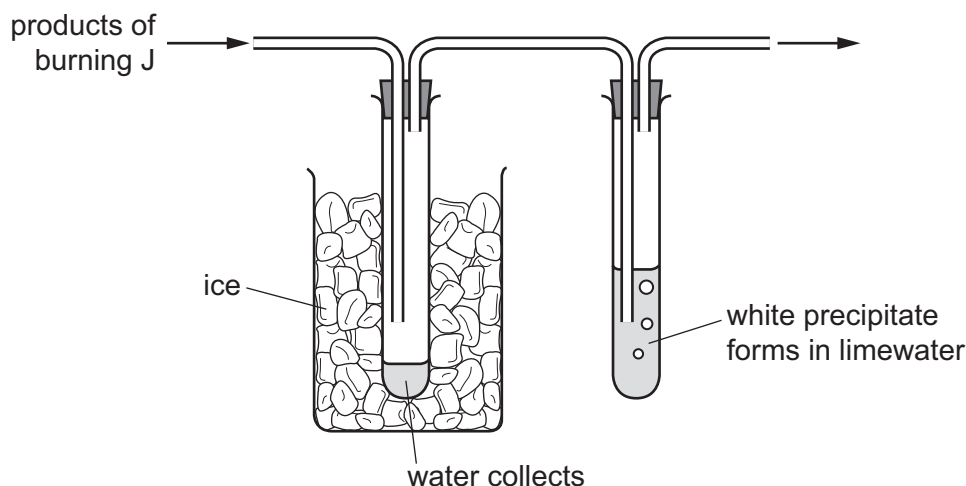
**37** Alkenes can be produced by cracking large hydrocarbon molecules to form smaller hydrocarbon molecules.

Which equations represent possible reactions when tetradecane,  $\text{C}_{14}\text{H}_{30}$ , is cracked?

- 1  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + \text{C}_3\text{H}_6 + \text{C}_4\text{H}_8 + \text{C}_5\text{H}_{10}$
- 2  $\text{C}_{14}\text{H}_{30} \rightarrow \text{H}_2 + \text{C}_2\text{H}_4 + \text{C}_3\text{H}_6 + \text{C}_4\text{H}_8 + \text{C}_5\text{H}_{10}$
- 3  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + 4\text{C}_3\text{H}_6$
- 4  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + \text{C}_3\text{H}_8 + \text{C}_9\text{H}_{18}$

- A** 1 only                      **B** 1 and 4                      **C** 1, 2 and 3                      **D** 3 and 4

38 The products formed by burning substance J are passed through the apparatus shown.



What is substance J?

- A carbon monoxide
- B ethanol
- C hydrogen
- D sulfur

39 Which statements about ethanoic acid are correct?

- 1 Aqueous ethanoic acid reacts with magnesium to form magnesium ethanoate.
- 2 Carbon dioxide is formed when aqueous ethanoic acid reacts with sodium carbonate.
- 3 Hydrogen is formed when aqueous ethanoic acid reacts with sodium hydroxide.
- 4 Ethanoic acid turns red litmus paper blue.

- A 1 and 2      B 1 and 3      C 2 and 3      D 2 and 4

40 Which statement about polymerisation is correct?

- A Large monomer molecules join to form small polymer molecules.
- B Large polymer molecules join to form small monomer molecules.
- C Small monomer molecules join to form large polymer molecules.
- D Small polymer molecules join to form large monomer molecules.

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# The Periodic Table of Elements

lanthanoids

|    |    |              |     |
|----|----|--------------|-----|
| 57 | La | lanthanum    | 139 |
| 58 | Ce | cerium       | 140 |
| 59 | Pr | praseodymium | 141 |
| 60 | Nd | neodymium    | 144 |
| 61 | Pm | promethium   | —   |
| 62 | Sm | samarium     | 150 |
| 63 | Eu | euroium      | 152 |
| 64 | Gd | gadolinium   | 157 |
| 65 | Tb | terbium      | 159 |
| 66 | Dy | dysprosium   | 163 |
| 67 | Ho | holmium      | 165 |
| 68 | Er | erbium       | 167 |
| 69 | Tm | thulium      | 169 |
| 70 | Yb | ytterbium    | 173 |
| 71 | Lu | lutetium     | 175 |

actinoids

|     |    |              |     |
|-----|----|--------------|-----|
| 89  | Ac | actinium     | —   |
| 90  | Th | thorium      | 232 |
| 91  | Pa | protactinium | 231 |
| 92  | U  | uranium      | 238 |
| 93  | Np | neptunium    | —   |
| 94  | Pu | plutonium    | —   |
| 95  | Am | americium    | —   |
| 96  | Cm | curium       | —   |
| 97  | Bk | berkelium    | —   |
| 98  | Cf | californium  | —   |
| 99  | Es | einsteinium  | —   |
| 100 | Fm | fermium      | —   |
| 101 | Md | mendelevium  | —   |
| 102 | No | nobelium     | —   |
| 103 | Lr | lawrencium   | —   |

The volume of one mole of any gas is  $24\text{ dm}^3$  at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/12**

Paper 1 Multiple Choice (Core)

**May/June 2022**

MARK SCHEME

Maximum Mark: 40

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|                         |
|-------------------------|
| <p><b>Published</b></p> |
|-------------------------|

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | A      | 1     |
| 3        | D      | 1     |
| 4        | C      | 1     |
| 5        | B      | 1     |
| 6        | C      | 1     |
| 7        | A      | 1     |
| 8        | D      | 1     |
| 9        | A      | 1     |
| 10       | D      | 1     |
| 11       | B      | 1     |
| 12       | B      | 1     |
| 13       | B      | 1     |
| 14       | D      | 1     |
| 15       | B      | 1     |
| 16       | B      | 1     |
| 17       | A      | 1     |
| 18       | B      | 1     |
| 19       | D      | 1     |
| 20       | A      | 1     |
| 21       | C      | 1     |
| 22       | C      | 1     |
| 23       | C      | 1     |
| 24       | A      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | D      | 1     |
| 28       | D      | 1     |



| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | B             | <b>1</b>     |
| 30              | C             | <b>1</b>     |
| 31              | D             | <b>1</b>     |
| 32              | A             | <b>1</b>     |
| 33              | D             | <b>1</b>     |
| 34              | C             | <b>1</b>     |
| 35              | D             | <b>1</b>     |
| 36              | C             | <b>1</b>     |
| 37              | C             | <b>1</b>     |
| 38              | B             | <b>1</b>     |
| 39              | A             | <b>1</b>     |
| 40              | C             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

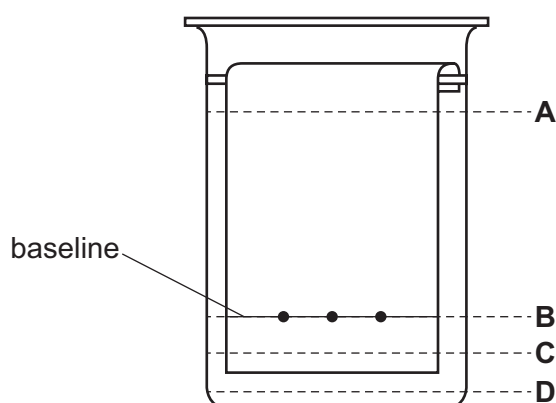
## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

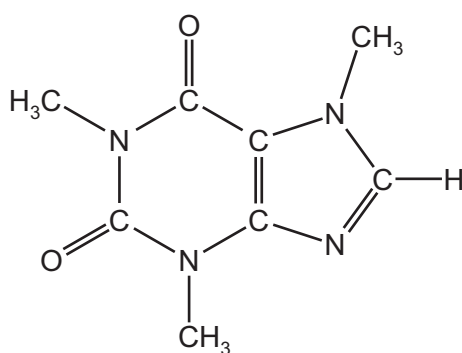


- 1 Which statement describes the particles in a liquid?
- A** They are close together but have no regular arrangement.  
**B** They are densely packed in a regular order.  
**C** They move freely at high speed and are widely spaced.  
**D** They vibrate but do not move from a fixed position.
- 2 The apparatus used in a chromatography experiment is shown.  
Which line shows the starting depth of the solvent in the beaker?



- 3 Filtration is used to separate mixtures.  
Which type of mixture is separated by filtration?
- A** an insoluble solid from a liquid  
**B** a liquid solvent from a solution  
**C** a dissolved solid from a solution  
**D** a liquid from a mixture of liquids
- 4 How many neutrons are present in one atom of  $^{35}_{17}\text{Cl}$ ?
- A** 17                      **B** 18                      **C** 35                      **D** 52
- 5 Which statement about an alloy is correct?
- A** It is a compound made of two or more elements, one of which is a metal.  
**B** It is a layer of a metal plated onto another metal.  
**C** It is a mixture of a metal with one or more other elements.  
**D** It is a single element.

- 6 Which statement about compounds is correct?
- A** Covalent compounds are less volatile than ionic compounds.
- B** Covalent compounds conduct electricity when they are solid.
- C** Ionic compounds conduct electricity when molten.
- D** Ionic compounds are insoluble in water.
- 7 Which statement explains why diamond is used in cutting tools?
- A** It has no free electrons.
- B** It has a high melting point.
- C** It is colourless.
- D** It is hard.
- 8 Caffeine is a stimulant found in coffee.

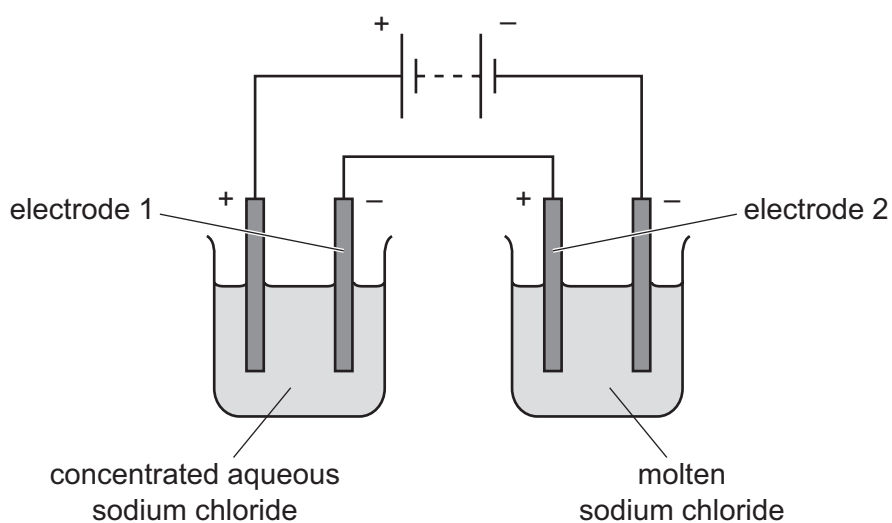


caffeine

Which formula represents caffeine?

- A**  $C_7H_{10}N_4O_2$       **B**  $C_8H_{10}N_3O_2$       **C**  $C_8H_{10}N_4O_2$       **D**  $C_8H_{11}N_4O_2$
- 9 What is the relative formula mass of ammonium sulfate,  $(NH_4)_2SO_4$ ?
- A** 63                      **B** 114                      **C** 118                      **D** 132

10 The electrolysis of concentrated aqueous sodium chloride and molten sodium chloride is shown.



What are the products at electrodes 1 and 2?

|          | electrode 1 | electrode 2 |
|----------|-------------|-------------|
| <b>A</b> | chlorine    | chlorine    |
| <b>B</b> | hydrogen    | chlorine    |
| <b>C</b> | hydrogen    | sodium      |
| <b>D</b> | sodium      | sodium      |

11 When an acid is added to an alkali, the temperature of the reaction mixture rises.

Which words describe this reaction?

- A** decomposition and endothermic
- B** decomposition and exothermic
- C** neutralisation and endothermic
- D** neutralisation and exothermic

12 Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

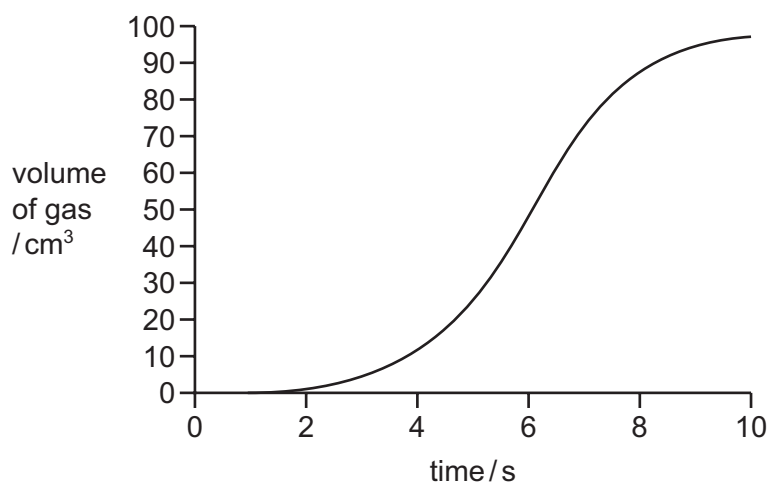
|          | fuel     | formula                         | melting point / °C | boiling point / °C |
|----------|----------|---------------------------------|--------------------|--------------------|
| <b>A</b> | hydrogen | H <sub>2</sub>                  | -259               | -253               |
| <b>B</b> | methane  | CH <sub>4</sub>                 | -182               | -164               |
| <b>C</b> | octane   | C <sub>8</sub> H <sub>18</sub>  | -57                | 126                |
| <b>D</b> | wax      | C <sub>31</sub> H <sub>64</sub> | 60                 | 400                |

13 Which process is a physical change?

- A** burning wood
- B** cooking an egg
- C** melting an ice cube
- D** rusting iron

14 The volume of gas given off in a chemical reaction is measured over time.

The results are shown.



At which time is the rate of reaction greatest?

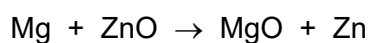
- A** 0 s
- B** 4 s
- C** 6 s
- D** 10 s

15 Which row describes the colours of the named salts?

|          | hydrated<br>copper(II) sulfate | hydrated<br>cobalt(II) chloride | anhydrous<br>copper(II) sulfate | anhydrous<br>cobalt(II) chloride |
|----------|--------------------------------|---------------------------------|---------------------------------|----------------------------------|
| <b>A</b> | blue                           | blue                            | white                           | pink                             |
| <b>B</b> | blue                           | pink                            | white                           | blue                             |
| <b>C</b> | white                          | blue                            | blue                            | pink                             |
| <b>D</b> | white                          | pink                            | blue                            | white                            |

16 When magnesium is heated with zinc oxide a reaction occurs.

The equation is shown.



Which substance is oxidised?

- A** magnesium
- B** magnesium oxide
- C** zinc
- D** zinc oxide

17 X and Y are oxides of two different elements.

- X reacts with water to produce aqueous solution Z.
- Z turns universal indicator paper blue.
- An aqueous solution of Y reacts with sodium carbonate to produce carbon dioxide gas.

Which statement is correct?

- A** X and Y are both the oxides of metals.
- B** X and Y are both the oxides of non-metals.
- C** X is the oxide of a metal and Y is the oxide of a non-metal.
- D** X is the oxide of a non-metal and Y is the oxide of a metal.

**18** Copper(II) sulfate is made by reacting excess insoluble solid M and solution N.

Which row identifies M and N and the method used to extract crystals of copper(II) sulfate from the mixture?

|          | M                    | N              | method                                                              |
|----------|----------------------|----------------|---------------------------------------------------------------------|
| <b>A</b> | copper               | sodium sulfate | crystals are filtered out from the mixture                          |
| <b>B</b> | copper               | sulfuric acid  | mixture is filtered and the filtrate evaporated until crystals form |
| <b>C</b> | copper(II) carbonate | sulfuric acid  | mixture is filtered and the filtrate evaporated until crystals form |
| <b>D</b> | copper(II) oxide     | sulfuric acid  | mixture is filtered and the residue dried                           |

**19** Which row shows the observation when a few drops of aqueous P is added to concentrated aqueous Q?

|          | P                                  | Q                   | observation       |
|----------|------------------------------------|---------------------|-------------------|
| <b>A</b> | acidified potassium manganate(VII) | sodium sulfite      | purple solution   |
| <b>B</b> | sodium hydroxide                   | zinc chloride       | white precipitate |
| <b>C</b> | ammonia                            | potassium carbonate | fizzing           |
| <b>D</b> | barium chloride                    | iron(III) sulfate   | brown precipitate |

**20** Which statement about the Periodic Table is correct?

- A** Elements in the same group have the same number of electron shells.
- B** Elements are arranged in order of increasing proton number.
- C** Metals are on the right and non-metals are on the left.
- D** The most reactive elements are at the bottom of every group.



- 21** Elements J and K are in the same period in the Periodic Table.

J reacts with acids to produce a salt and hydrogen.

K reacts with sodium to form an ionic compound.

Which statement about J and K is correct?

- A** An atom of J has more electrons than an atom of K.
- B** J and K are both metals.
- C** J and K are both non-metals.
- D** J is to the left of K in the Periodic Table.

- 22** Part of the Periodic Table is shown.

Which element has a high density, a high melting point and forms a brown oxide?

A simplified periodic table with 18 columns and 4 rows. The first two columns are labeled 'A' in the top-left cell. The next six columns are labeled 'B' in the third row, seventh column. The next six columns are labeled 'D' in the second row, thirteenth column. The last two columns are labeled 'C' in the bottom row, first and second columns. There are also empty boxes at the top center and top right.

- 23** Gas G has 10 electrons. Gas H has eight more electrons than gas G. Both gases are monoatomic.

Which statement about G and H is correct?

- A** Both gases are in the same group of the Periodic Table.
- B** Both gases are in the same period of the Periodic Table.
- C** Both gases are very reactive.
- D** Gas G has a higher atomic mass than gas H.

- 24** Which property is correct for all metals?

- A** They are good conductors of electricity.
- B** They are hard.
- C** They have high melting points.
- D** They react with dilute acids.

**25** Silver is below copper in the reactivity series.

Which row describes the reactions of silver?

|          | reaction with steam            | reaction with dilute hydrochloric acid |
|----------|--------------------------------|----------------------------------------|
| <b>A</b> | no reaction                    | no reaction                            |
| <b>B</b> | no reaction                    | reacts to produce hydrogen gas         |
| <b>C</b> | reacts to produce hydrogen gas | no reaction                            |
| <b>D</b> | reacts to produce hydrogen gas | reacts to produce hydrogen gas         |

**26** Which types of reaction do hematite and limestone undergo in the blast furnace?

|          | hematite              | limestone             |
|----------|-----------------------|-----------------------|
| <b>A</b> | reduction             | reduction             |
| <b>B</b> | reduction             | thermal decomposition |
| <b>C</b> | thermal decomposition | reduction             |
| <b>D</b> | thermal decomposition | thermal decomposition |

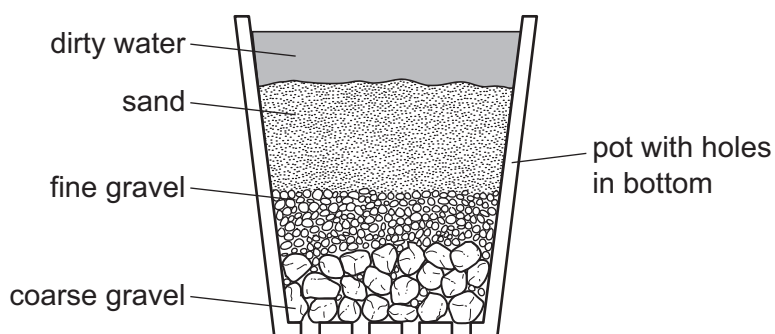
**27** Some properties and uses of different metals are shown.

|   | metal           | property                      | use               |
|---|-----------------|-------------------------------|-------------------|
| 1 | aluminium       | low density                   | aircraft          |
| 2 | copper          | good conductor of electricity | electrical wiring |
| 3 | copper          | poor conductor of heat        | cooking utensils  |
| 4 | stainless steel | corrodes easily               | cutlery           |

Which rows link a use of the metal to its stated property?

**A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

28 The diagram shows a stage in the purification of dirty water.



Which process does this apparatus show?

- A chlorination
  - B condensation
  - C distillation
  - D filtration
- 29 Which substance in polluted air damages stonework and kills trees?
- A carbon dioxide
  - B carbon monoxide
  - C lead compounds
  - D sulfur dioxide
- 30 Ammonium nitrate,  $\text{NH}_4\text{NO}_3$ , is a fertiliser and is added to fields to help crops grow.

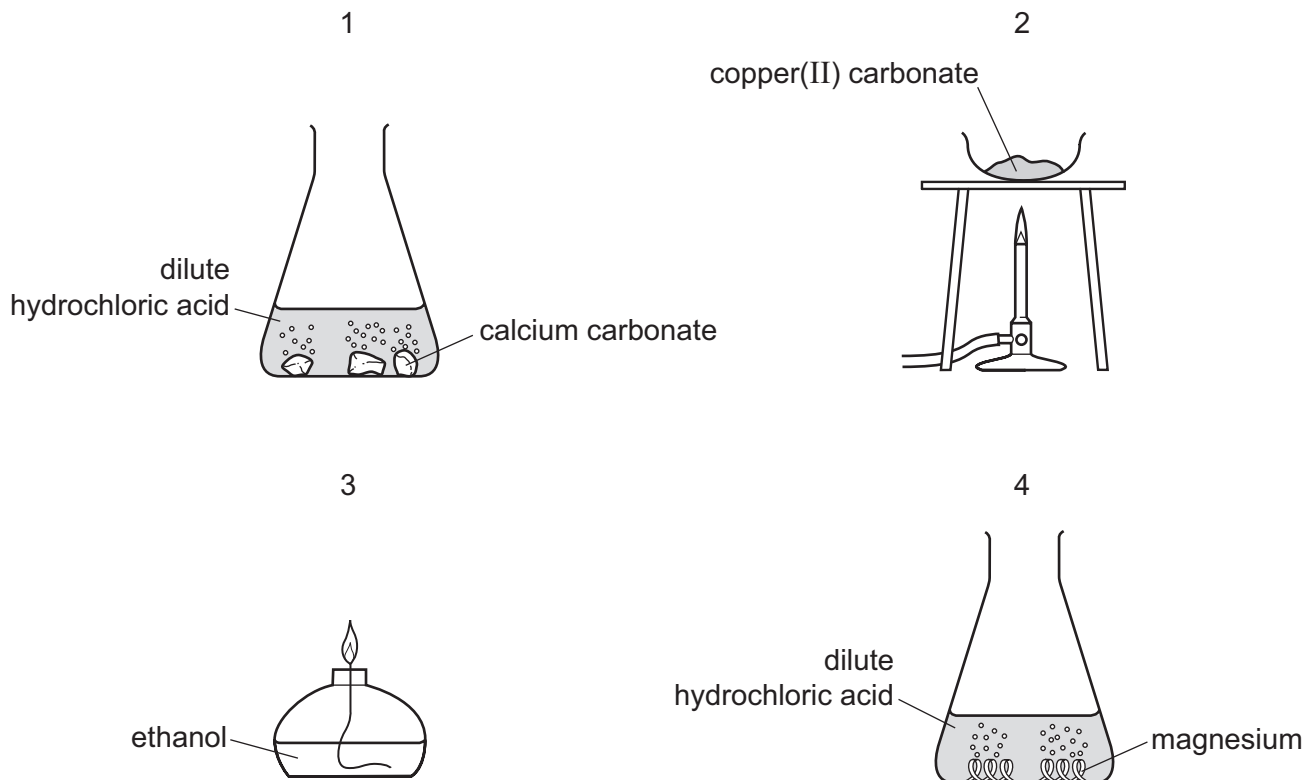
Slaked lime,  $\text{Ca}(\text{OH})_2$ , is an alkali and is added to fields to reduce the acidity of the soil.

Ammonium nitrate and slaked lime should not be added to a field at the same time because they react with each other to form a gas, Z.

What is Z?

- A ammonia
- B hydrogen
- C nitrogen
- D oxygen

31 Four reactions are shown.



Which reactions produce water?

- A** 1 and 2      **B** 1 and 3      **C** 2, 3 and 4      **D** 3 and 4 only

32 Which element has an oxide that is used as a food preservative?

- A** helium  
**B** hydrogen  
**C** iron  
**D** sulfur

33 Which substance gives off carbon dioxide on heating?

- A** lime  
**B** limestone  
**C** limewater  
**D** slaked lime

**34** Which statement about both ethane and ethanol is correct?

- A** They are hydrocarbons.
- B** They contain oxygen.
- C** They contain the same number of atoms.
- D** They produce water when burned.

**35** Fuel oil and naphtha are two fractions obtained from petroleum.

What are the major uses of these fractions?

|          | fuel oil  | naphtha          |
|----------|-----------|------------------|
| <b>A</b> | jet fuel  | making chemicals |
| <b>B</b> | jet fuel  | making roads     |
| <b>C</b> | ship fuel | making chemicals |
| <b>D</b> | ship fuel | making roads     |

**36** Which homologous series of compounds reacts to form an addition polymer?

- A** alcohols
- B** alkanes
- C** alkenes
- D** carboxylic acids

**37** What is the total number of shared electrons in ethane,  $C_2H_6$ ?

- A** 6                      **B** 7                      **C** 12                      **D** 14

**38** Which process produces ethanol from glucose?

- A** catalytic addition
- B** cracking
- C** fermentation
- D** polymerisation

39 Which statement about unsaturated hydrocarbons is correct?

- A  $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$  is an unsaturated hydrocarbon.
- B Ethene has more hydrogen atoms per molecule than ethane.
- C Unsaturated hydrocarbons have double bonds between carbon and hydrogen atoms.
- D Unsaturated hydrocarbons turn aqueous bromine from colourless to brown.

40 An organic compound X contains two carbon atoms in each molecule.

X reacts with aqueous sodium carbonate to give carbon dioxide.

What is compound X?

- A ethanol
- B ethane
- C  $\text{CH}_2=\text{CH}_2$
- D  $\text{CH}_3\text{COOH}$



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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                           |                          |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|------------------------------|---------------------------|--------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                                                     |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                            | V                           | VI                           | VII                       | VIII                     |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12        | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19  | 10<br>Ne<br>neon<br>20   |
|                            | 11<br>Na<br>sodium<br>23    |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                               |                             |                              |                           |                          |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        |                             |                               |                             |                              |                           |                          |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119        | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127  | 54<br>Xe<br>xenon<br>131 |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207       | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—    | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>—   |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 114<br>Fl<br>flerovium<br>— | 116<br>Lv<br>livermorium<br>— |                             |                              |                           |                          |

lanthanoids

actinoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## **Cambridge IGCSE™ (9–1)**

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**CHEMISTRY**

**0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2022**

**MARK SCHEME**

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | A      | 1     |
| 2        | C      | 1     |
| 3        | A      | 1     |
| 4        | B      | 1     |
| 5        | C      | 1     |
| 6        | C      | 1     |
| 7        | D      | 1     |
| 8        | C      | 1     |
| 9        | D      | 1     |
| 10       | B      | 1     |
| 11       | D      | 1     |
| 12       | B      | 1     |
| 13       | C      | 1     |
| 14       | C      | 1     |
| 15       | B      | 1     |
| 16       | A      | 1     |
| 17       | C      | 1     |
| 18       | C      | 1     |
| 19       | B      | 1     |
| 20       | B      | 1     |
| 21       | D      | 1     |
| 22       | B      | 1     |
| 23       | A      | 1     |
| 24       | A      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | A      | 1     |
| 28       | D      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | D             | <b>1</b>     |
| 30              | A             | <b>1</b>     |
| 31              | B             | <b>1</b>     |
| 32              | D             | <b>1</b>     |
| 33              | B             | <b>1</b>     |
| 34              | D             | <b>1</b>     |
| 35              | C             | <b>1</b>     |
| 36              | C             | <b>1</b>     |
| 37              | D             | <b>1</b>     |
| 38              | C             | <b>1</b>     |
| 39              | A             | <b>1</b>     |
| 40              | D             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/11

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows the result of dropping a purple crystal into water.



Which processes take place in this experiment?

|          | chemical reaction | diffusing | dissolving |
|----------|-------------------|-----------|------------|
| <b>A</b> | ✓                 | ✓         | ✗          |
| <b>B</b> | ✓                 | ✗         | ✗          |
| <b>C</b> | ✗                 | ✗         | ✓          |
| <b>D</b> | ✗                 | ✓         | ✓          |

- 2 Which row about elements, mixtures and compounds is correct?

|          | metallic element | non-metallic element | mixture | compound |
|----------|------------------|----------------------|---------|----------|
| <b>A</b> | copper           | methane              | brass   | sulfur   |
| <b>B</b> | brass            | sulfur               | copper  | methane  |
| <b>C</b> | copper           | sulfur               | brass   | methane  |
| <b>D</b> | brass            | methane              | copper  | sulfur   |

- 3 What are the relative charge and relative mass of an electron?

|          | relative charge | relative mass    |
|----------|-----------------|------------------|
| <b>A</b> | 0               | 1                |
| <b>B</b> | 0               | $\frac{1}{2000}$ |
| <b>C</b> | -1              | 1                |
| <b>D</b> | -1              | $\frac{1}{2000}$ |

- 4 The atomic structures of four particles, W, X, Y and Z, are shown.

|   | electrons | neutrons | protons |
|---|-----------|----------|---------|
| W | 2         | 2        | 2       |
| X | 2         | 2        | 3       |
| Y | 2         | 3        | 2       |
| Z | 3         | 2        | 3       |

Which particles are isotopes of the same element?

- A** W and X      **B** W and Y      **C** X and Y      **D** X and Z

- 5 Which row shows the properties of an ionic compound?

|          | electrical conductivity of solid | melting point / °C |
|----------|----------------------------------|--------------------|
| <b>A</b> | good                             | 98                 |
| <b>B</b> | good                             | 3652               |
| <b>C</b> | poor                             | 78                 |
| <b>D</b> | poor                             | 801                |

- 6 Which row describes the formation of single covalent bonds in methane?

|          |                                                    |                                                                   |
|----------|----------------------------------------------------|-------------------------------------------------------------------|
| <b>A</b> | atoms share a pair of electrons                    | both atoms gain a noble gas electronic structure                  |
| <b>B</b> | atoms share a pair of electrons                    | both atoms have the same number of electrons in their outer shell |
| <b>C</b> | electrons are transferred from one atom to another | both atoms gain a noble gas electronic structure                  |
| <b>D</b> | electrons are transferred from one atom to another | both atoms have the same number of electrons in their outer shell |

- 7 Which equation represents the neutralisation of nitric acid using sodium hydroxide?

- A**  $\text{NaOH(aq)} + \text{HNO}_3\text{(aq)} \rightarrow \text{NaNO}_3\text{(aq)} + \text{H}_2\text{O(l)}$   
**B**  $\text{NaOH(aq)} + \text{HNO}_3\text{(aq)} \rightarrow \text{NaNO}_3\text{(l)} + \text{H}_2\text{O(l)}$   
**C**  $\text{NaOH(l)} + \text{HNO}_3\text{(l)} \rightarrow \text{NaNO}_3\text{(l)} + \text{H}_2\text{O(aq)}$   
**D**  $\text{NaOH(l)} + \text{HNO}_3\text{(l)} \rightarrow \text{NaNO}_3\text{(l)} + \text{H}_2\text{O(l)}$

8 What is the relative formula mass of ammonium nitrate,  $\text{NH}_4\text{NO}_3$ ?

A 80

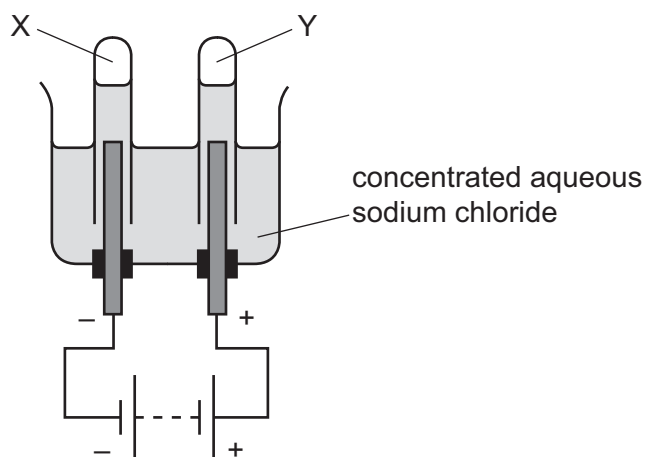
B 108

C 122

D 150

9 Concentrated aqueous sodium chloride is electrolysed using inert electrodes.

Gases X and Y are produced at the electrodes shown.



What are X and Y?

|   | X        | Y        |
|---|----------|----------|
| A | chlorine | hydrogen |
| B | hydrogen | chlorine |
| C | hydrogen | oxygen   |
| D | oxygen   | hydrogen |

10 Which statement about hydrogen fuel cells is correct?

A Hydrogen fuel cells do not produce carbon dioxide.

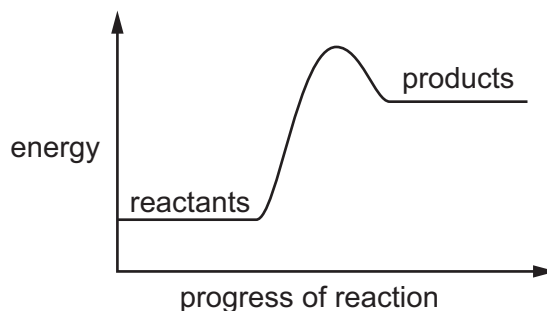
B Hydrogen fuel cells do not need oxygen.

C The waste from a hydrogen fuel cell is an acidic gas.

D The reaction in a fuel cell is endothermic.



11 A reaction pathway diagram is shown.



Which statement about this reaction is correct?

- A The reaction rate increases during the reaction.
- B The reaction is endothermic.
- C The reaction transfers thermal energy to the surroundings.
- D The temperature of the surroundings increases.

12 Lumps of calcium carbonate react with dilute hydrochloric acid as shown.



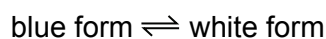
Which change in conditions decreases the rate of the reaction?

- A increasing the concentration of the acid
- B increasing the volume of the acid
- C increasing the size of the lumps of calcium carbonate
- D increasing the temperature

13 Solid copper(II) sulfate exists in two different forms, anhydrous and hydrated.

One of these forms is blue and the other is white.

The change between these two forms is reversible.



What is the blue form and how is the change from the blue form to the white form brought about?

|   | blue form | change to white form |
|---|-----------|----------------------|
| A | anhydrous | add water            |
| B | anhydrous | heat                 |
| C | hydrated  | add water            |
| D | hydrated  | heat                 |

14 Four redox equations and statements about the equations are shown.

|   | reaction                                                                   | statement                   |
|---|----------------------------------------------------------------------------|-----------------------------|
| 1 | $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$                            | carbon is oxidised          |
| 2 | $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$                            | carbon dioxide is oxidised  |
| 3 | $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$                            | carbon is oxidised          |
| 4 | $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ | iron(III) oxide is oxidised |

Which statements about the equations are correct?

- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

15 Sodium hydroxide forms an alkaline solution with a pH of 14.

Which indicator turns yellow when added to this solution?

- A** litmus  
**B** methyl orange  
**C** thymolphthalein  
**D** universal indicator

16 Which row identifies an acidic oxide and a basic oxide?

|          | acidic oxide    | basic oxide     |
|----------|-----------------|-----------------|
| <b>A</b> | CaO             | CuO             |
| <b>B</b> | CaO             | SO <sub>2</sub> |
| <b>C</b> | CO <sub>2</sub> | CuO             |
| <b>D</b> | CO <sub>2</sub> | SO <sub>2</sub> |

17 A student makes aqueous copper(II) chloride by adding excess copper(II) carbonate to dilute hydrochloric acid.

What is the next step in the method in the formation of solid copper(II) chloride?

- A** crystallisation  
**B** evaporation  
**C** filtration  
**D** titration

18 Which statements about the trends across a period of the Periodic Table are correct?

- 1 Aluminium is more metallic than sodium.
- 2 Beryllium is more metallic than carbon.
- 3 Boron is more metallic than lithium.
- 4 Magnesium is more metallic than silicon.

**A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

19 Which row shows the trend in melting point, density and reactivity as Group I is descended?

|          | melting point | density   | reactivity |
|----------|---------------|-----------|------------|
| <b>A</b> | increases     | decreases | decreases  |
| <b>B</b> | decreases     | increases | increases  |
| <b>C</b> | increases     | decreases | increases  |
| <b>D</b> | decreases     | increases | decreases  |

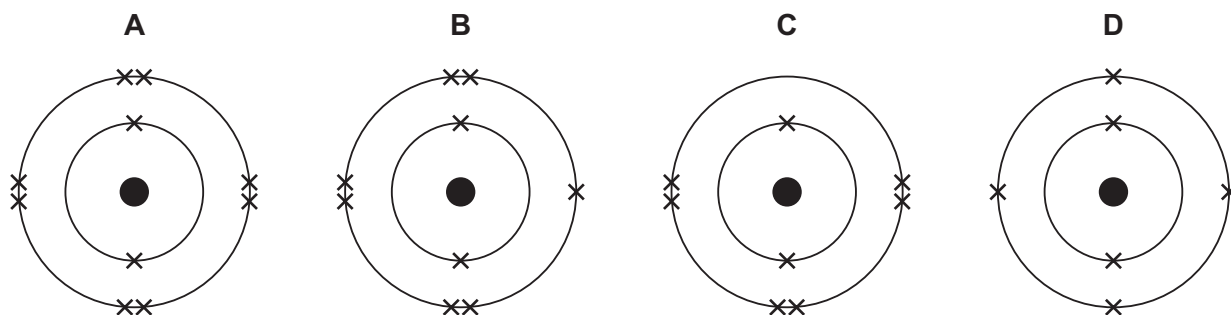
20 Which row describes a similarity and a difference between chlorine and bromine?

|          | similarity                                       | difference                                           |
|----------|--------------------------------------------------|------------------------------------------------------|
| <b>A</b> | both are gases at room temperature and pressure  | chlorine and bromine have different colours          |
| <b>B</b> | both exist as diatomic molecules                 | chlorine is more dense than bromine                  |
| <b>C</b> | both have atoms with seven outer-shell electrons | only bromine will react with aqueous sodium chloride |
| <b>D</b> | both react with aqueous potassium iodide         | chlorine is more reactive than bromine               |

21 Which statement describes transition elements?

- A** They have high densities and high melting points.
- B** They have high densities and low melting points.
- C** They have low densities and high melting points.
- D** They have low densities and low melting points.

22 Which diagram shows the electronic structure of a noble gas?



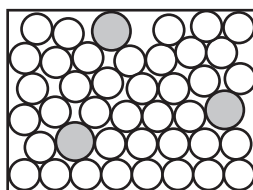
23 Which gas is made when powdered zinc is added to dilute hydrochloric acid?

- A** carbon dioxide
- B** chlorine
- C** hydrogen
- D** oxygen

24 Which metal is used in aircraft manufacture because it has a low density?

- A** aluminium
- B** copper
- C** iron
- D** potassium

25 The diagram represents the structure of a solid.



Which solids does the diagram represent?

|          | brass | graphite | sodium chloride |
|----------|-------|----------|-----------------|
| <b>A</b> | ✓     | ✓        | x               |
| <b>B</b> | ✓     | x        | x               |
| <b>C</b> | x     | ✓        | ✓               |
| <b>D</b> | x     | x        | ✓               |

- 26 Three students, X, Y and Z, are told that solid P reacts with dilute acids and also conducts electricity.

The table shows the students' suggestions about the identity of P.

|        |      |          |
|--------|------|----------|
| X      | Y    | Z        |
| copper | iron | graphite |

Which students are correct?

- A** X, Y and Z      **B** X only      **C** Y only      **D** Z only
- 27 Which substances in the air are needed for iron to rust?
- A** oxygen and water  
**B** oxygen only  
**C** water and carbon dioxide  
**D** water only
- 28 Part of the reactivity series of metals is shown.

|         |            |
|---------|------------|
| metal P | decreasing |
| carbon  | reactivity |
| metal Q | ↓          |
| metal R |            |

Which row shows how each metal is extracted from its ore?

|          | metal P                    | metal Q                    | metal R                    |
|----------|----------------------------|----------------------------|----------------------------|
| <b>A</b> | electrolysis of molten ore | electrolysis of molten ore | heating with carbon        |
| <b>B</b> | heating with carbon        | electrolysis of molten ore | electrolysis of molten ore |
| <b>C</b> | heating with carbon        | heating with carbon        | electrolysis of molten ore |
| <b>D</b> | electrolysis of molten ore | heating with carbon        | heating with carbon        |

**29** Several processes are used to treat domestic water.

Which row identifies a reason for the given process?

|          | process       | reason                   |
|----------|---------------|--------------------------|
| <b>A</b> | chlorination  | removes impurities       |
| <b>B</b> | filtration    | removes insoluble solids |
| <b>C</b> | sedimentation | removes soluble solids   |
| <b>D</b> | use of carbon | kills bacteria           |

**30** Which pair of compounds make an NPK fertiliser?

- A** ammonium sulfate and potassium phosphate
- B** calcium hydroxide and ammonium nitrate
- C** calcium phosphate and potassium chloride
- D** potassium nitrate and ammonium sulfate

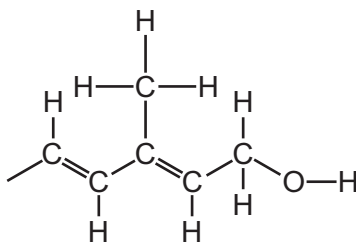
**31** Some information about gas X is listed.

- It is not present in clean, dry air.
- It is not a cause of respiratory problems.
- It is responsible for global warming.

What is X?

- A** carbon dioxide
- B** carbon monoxide
- C** methane
- D** nitrogen dioxide

32 Part of the structure of a molecule of vitamin A is shown.



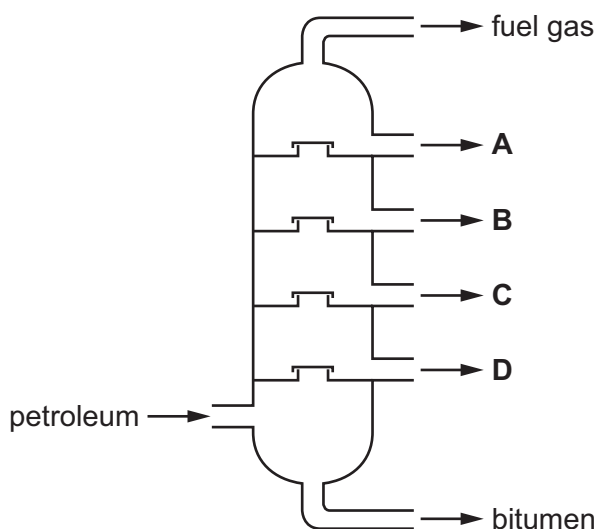
Which statements about this part of the structure are correct?

- 1 It is saturated.
- 2 There are two alkene groups.
- 3 The structure shows a carboxylic acid.

**A** 1 and 3      **B** 1 only      **C** 2 and 3      **D** 2 only

33 The fractional distillation of petroleum is shown.

Which fraction contains hydrocarbons with the longest chain length?



34 Which equation represents the cracking of an alkane?

- A**  $3\text{C}_2\text{H}_4 \rightarrow \text{C}_6\text{H}_{12}$
- B**  $\text{C}_6\text{H}_{12} + \text{H}_2 \rightarrow \text{C}_6\text{H}_{14}$
- C**  $\text{C}_6\text{H}_{14} \rightarrow 6\text{C} + 7\text{H}_2$
- D**  $\text{C}_6\text{H}_{14} \rightarrow \text{C}_2\text{H}_4 + \text{C}_4\text{H}_{10}$

35 Which statements about ethanol are correct?

- 1 Ethanol is made by reacting steam with ethene at 300 °C.
- 2 Ethanol is made by fermentation at 55 °C.
- 3 Ethanol burns to produce carbon dioxide and water.
- 4 Ethanol contains a carbon–carbon double bond.

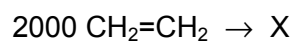
**A** 1 and 2      **B** 1 and 3      **C** 2 and 3      **D** 3 and 4

36 Which substances react with aqueous ethanoic acid to form a gas?

- 1 magnesium
- 2 magnesium carbonate
- 3 magnesium oxide

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

37 In reaction R, 2000 molecules of  $\text{CH}_2=\text{CH}_2$  react to form a single molecule X only.



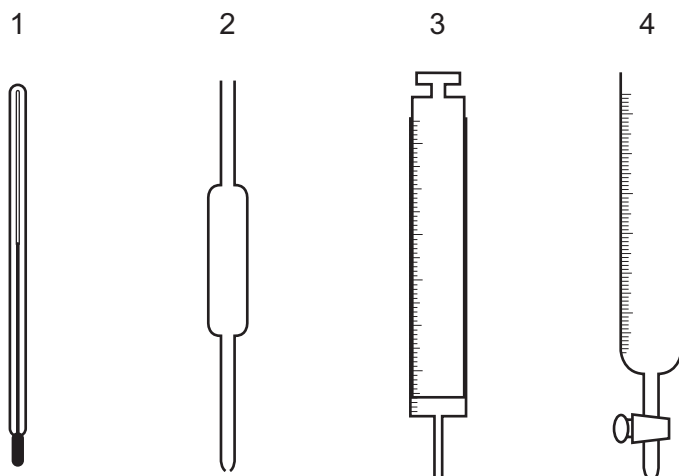
Which terms describe reaction R,  $\text{CH}_2=\text{CH}_2$  and X?

|          | reaction R   | $\text{CH}_2=\text{CH}_2$ | X       |
|----------|--------------|---------------------------|---------|
| <b>A</b> | addition     | monomer                   | polymer |
| <b>B</b> | addition     | polymer                   | monomer |
| <b>C</b> | substitution | monomer                   | polymer |
| <b>D</b> | substitution | polymer                   | monomer |



**38** The concentration of acids and alkalis can be determined by titration.

Which pieces of equipment are needed to perform a titration?



- A** 1 and 2      **B** 1 and 3      **C** 2 and 3      **D** 2 and 4

**39** Which process is used to produce drinking water from sea water?

- A** crystallisation  
**B** distillation  
**C** filtration  
**D** chlorination

**40** The results of two separate tests on a white solid X are shown.

| test                                  | result                                                 |
|---------------------------------------|--------------------------------------------------------|
| add dilute nitric acid                | effervescence                                          |
| add aqueous sodium hydroxide and warm | a gas is formed which turns damp red litmus paper blue |

What is X?

- A** aluminium carbonate  
**B** aluminium nitrate  
**C** ammonium carbonate  
**D** ammonium nitrate



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The Periodic Table of Elements

| Group                                                                                                             |                         |                         |                         |                             |                          |                          |                          |                        |                          |                            |                           |                           |                         |                         |                          |                           |                          |                        |
|-------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|--------------------------|--------------------------|--------------------------|------------------------|--------------------------|----------------------------|---------------------------|---------------------------|-------------------------|-------------------------|--------------------------|---------------------------|--------------------------|------------------------|
| I                                                                                                                 | II                      |                         |                         |                             |                          |                          |                          |                        |                          |                            |                           | III                       | IV                      | V                       | VI                       | VII                       | VIII                     |                        |
| <div>1Hhydrogen1</div> <div>Key<div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div></div> |                         |                         |                         |                             |                          |                          |                          |                        |                          |                            |                           |                           |                         |                         |                          |                           |                          |                        |
| 3Li<br>lithium<br>7                                                                                               | 4Be<br>beryllium<br>9   |                         |                         |                             |                          |                          |                          |                        |                          |                            |                           |                           |                         |                         |                          |                           |                          |                        |
| 11Na<br>sodium<br>23                                                                                              | 12Mg<br>magnesium<br>24 |                         |                         |                             |                          |                          |                          |                        |                          |                            |                           |                           |                         |                         |                          |                           |                          |                        |
| 13Al<br>aluminium<br>27                                                                                           | 14Si<br>silicon<br>28   | 15P<br>phosphorus<br>31 | 16S<br>sulfur<br>32     | 17Cl<br>chlorine<br>35.5    | 18Ar<br>argon<br>40      |                          |                          |                        |                          |                            |                           |                           |                         |                         |                          |                           |                          |                        |
| 19K<br>potassium<br>39                                                                                            | 20Ca<br>calcium<br>40   | 21Sc<br>scandium<br>45  | 22Ti<br>titanium<br>48  | 23V<br>vanadium<br>51       | 24Cr<br>chromium<br>52   | 25Mn<br>manganese<br>55  | 26Fe<br>iron<br>56       | 27Co<br>cobalt<br>59   | 28Ni<br>nickel<br>59     | 29Cu<br>copper<br>64       | 30Zn<br>zinc<br>65        | 31Ga<br>gallium<br>70     | 32Ge<br>germanium<br>73 | 33As<br>arsenic<br>75   | 34Se<br>selenium<br>79   | 35Br<br>bromine<br>80     | 36Kr<br>krypton<br>84    |                        |
| 37Rb<br>rubidium<br>85                                                                                            | 38Sr<br>strontium<br>88 | 39Y<br>yttrium<br>89    | 40Zr<br>zirconium<br>91 | 41Nb<br>niobium<br>93       | 42Mo<br>molybdenum<br>96 | 43Tc<br>technetium<br>—  | 44Ru<br>ruthenium<br>101 | 45Rh<br>rhodium<br>103 | 46Pd<br>palladium<br>106 | 47Ag<br>silver<br>108      | 48Cd<br>cadmium<br>112    | 49In<br>indium<br>115     | 50Sn<br>tin<br>119      | 51Sb<br>antimony<br>122 | 52Te<br>tellurium<br>128 | 53I<br>iodine<br>127      | 54Xe<br>xenon<br>131     |                        |
| 55Cs<br>caesium<br>133                                                                                            | 56Ba<br>barium<br>137   | 57–71<br>lanthanoids    |                         | 72Hf<br>hafnium<br>178      | 73Ta<br>tantalum<br>181  | 74W<br>tungsten<br>184   | 75Re<br>rhenium<br>186   | 76Os<br>osmium<br>190  | 77Ir<br>iridium<br>192   | 78Pt<br>platinum<br>195    | 79Au<br>gold<br>197       | 80Hg<br>mercury<br>201    | 81Tl<br>thallium<br>204 | 82Pb<br>lead<br>207     | 83Bi<br>bismuth<br>209   | 84Po<br>polonium<br>—     | 85At<br>astatine<br>—    | 86Rn<br>radon<br>—     |
| 87Fr<br>francium<br>—                                                                                             | 88Ra<br>radium<br>—     | 89–103<br>actinoids     |                         | 104Rf<br>rutherfordium<br>— | 105Db<br>dubnium<br>—    | 106Sg<br>seaborgium<br>— | 107Bh<br>bohrium<br>—    | 108Hs<br>hassium<br>—  | 109Mt<br>meitnerium<br>— | 110Ds<br>darmstadtium<br>— | 111Rg<br>roentgenium<br>— | 112Cn<br>copernicium<br>— | 113Nh<br>nihonium<br>—  | 114Fl<br>flerovium<br>— | 115Mc<br>moscovium<br>—  | 116Lv<br>livermorium<br>— | 117Ts<br>tennessine<br>— | 118Og<br>oganeson<br>— |

|             |                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|-------------|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| lanthanoids | 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
|             | 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).



## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2023**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | D      | 1     |
| 2        | C      | 1     |
| 3        | D      | 1     |
| 4        | B      | 1     |
| 5        | D      | 1     |
| 6        | A      | 1     |
| 7        | A      | 1     |
| 8        | A      | 1     |
| 9        | B      | 1     |
| 10       | A      | 1     |
| 11       | B      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | B      | 1     |
| 15       | B      | 1     |
| 16       | C      | 1     |
| 17       | C      | 1     |
| 18       | C      | 1     |
| 19       | B      | 1     |
| 20       | D      | 1     |
| 21       | A      | 1     |
| 22       | A      | 1     |
| 23       | C      | 1     |
| 24       | A      | 1     |
| 25       | B      | 1     |
| 26       | C      | 1     |
| 27       | A      | 1     |
| 28       | D      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | B             | <b>1</b>     |
| 30              | A             | <b>1</b>     |
| 31              | C             | <b>1</b>     |
| 32              | D             | <b>1</b>     |
| 33              | D             | <b>1</b>     |
| 34              | D             | <b>1</b>     |
| 35              | B             | <b>1</b>     |
| 36              | B             | <b>1</b>     |
| 37              | A             | <b>1</b>     |
| 38              | D             | <b>1</b>     |
| 39              | B             | <b>1</b>     |
| 40              | C             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/12

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.





1 Four physical changes of ethanol are listed.

- 1 condensation
- 2 evaporation
- 3 freezing
- 4 boiling

In which changes do the particles move further apart?

- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

2 Which statement explains why water is a compound?

- A** The hydrogen and oxygen atoms in a molecule of water can only be separated by chemical means.
- B** The hydrogen and oxygen atoms in a molecule of water can be separated using physical means.
- C** The number of hydrogen and oxygen atoms in a molecule of water is variable.
- D** Water has the same chemical properties as both hydrogen and oxygen.

3 An atom of element X contains:

- 5 protons
- 6 neutrons
- 5 electrons.

Which statements about element X are correct?

- 1 X has an atomic number of 6.
- 2 X has a nucleon number of 11.
- 3 X is in Group II of the Periodic Table.
- 4 X is in the second period of the Periodic Table.

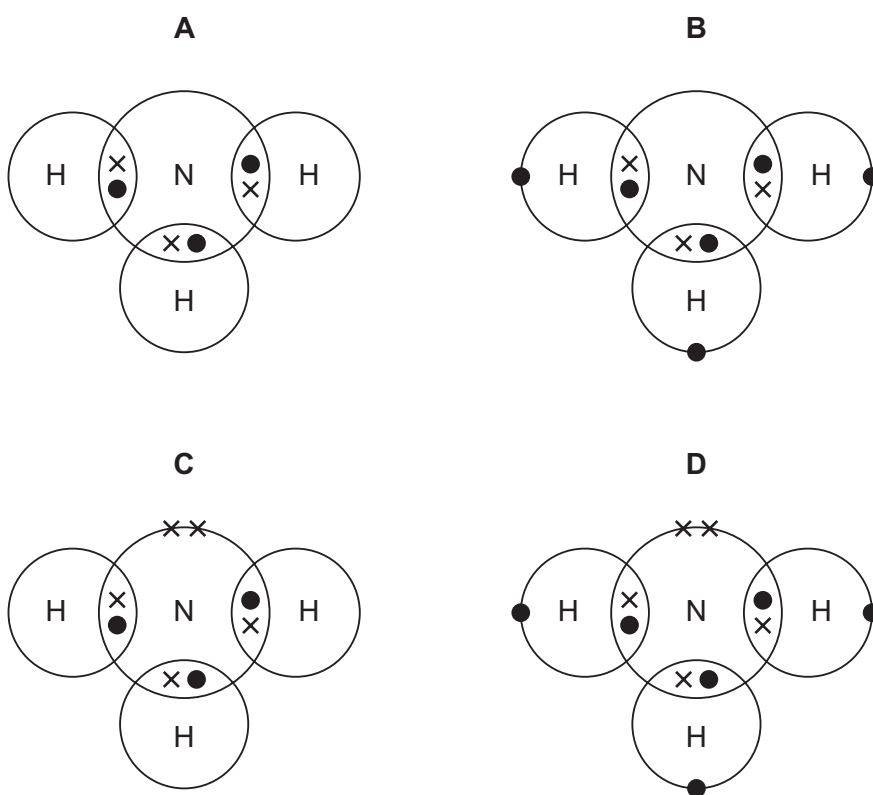
- A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

4 Which row describes properties of lithium fluoride?

|          | electrical conductivity<br>when solid | electrical conductivity<br>when molten | melting point |
|----------|---------------------------------------|----------------------------------------|---------------|
| <b>A</b> | does not conduct                      | conducts                               | high          |
| <b>B</b> | does not conduct                      | does not conduct                       | low           |
| <b>C</b> | conducts                              | conducts                               | high          |
| <b>D</b> | conducts                              | does not conduct                       | low           |

5 Ammonia,  $\text{NH}_3$ , is a covalent molecule.

Which diagram shows the outer-shell electron arrangement in a molecule of ammonia?



6 Which substance has a giant covalent structure?

- A** ethanol
- B** graphite
- C** methane
- D** sodium chloride

- 7 Sodium burns in oxygen to form sodium oxide.

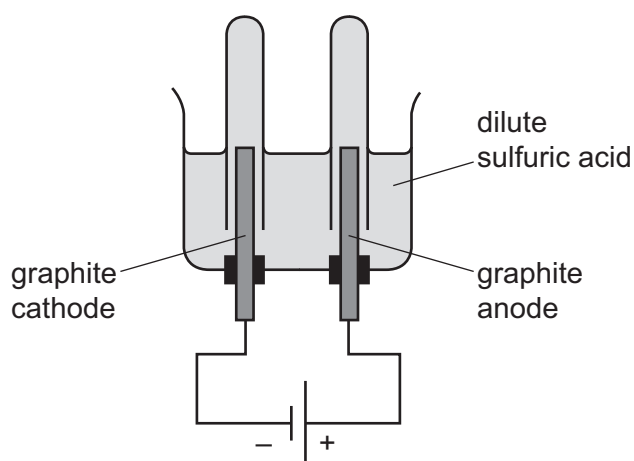
What is the balanced equation for the reaction?

- A**  $4\text{Na} + 2\text{O} \rightarrow 2\text{Na}_2\text{O}$   
**B**  $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$   
**C**  $2\text{Na}_2 + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$   
**D**  $2\text{Na}_2 + 2\text{O} \rightarrow 2\text{Na}_2\text{O}$

- 8 What is the relative formula mass of  $\text{Mg}(\text{OH})_2$ ?

- A** 21                      **B** 30                      **C** 42                      **D** 58

- 9 Dilute sulfuric acid is electrolysed using inert electrodes. The apparatus is set up as shown.



$30\text{ cm}^3$  of a gas is collected at the cathode. A different gas is collected at the anode.

Which row is correct?

|          | gas at cathode | gas at anode | volume of gas collected at anode / $\text{cm}^3$ |
|----------|----------------|--------------|--------------------------------------------------|
| <b>A</b> | hydrogen       | oxygen       | 15                                               |
| <b>B</b> | hydrogen       | oxygen       | 30                                               |
| <b>C</b> | oxygen         | hydrogen     | 15                                               |
| <b>D</b> | oxygen         | hydrogen     | 30                                               |

**10** Four statements about hydrogen fuel cells are listed.

- 1 The fuel cell converts chemical energy into electrical energy.
- 2 In the fuel cell, hydrogen combines with oxygen.
- 3 Carbon dioxide and water are produced in the fuel cell.
- 4 The hydrogen fuel is extracted from the air.

Which statements are correct?

- A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

**11** 5g of four different fuels are set alight and placed under a beaker containing 50 cm<sup>3</sup> of water.

The temperature of the water is taken at the start and after five minutes.

Which fuel releases the most energy?

|          | temperature<br>at start<br>/°C | temperature<br>after five minutes<br>/°C |
|----------|--------------------------------|------------------------------------------|
| <b>A</b> | 15                             | 23                                       |
| <b>B</b> | 21                             | 31                                       |
| <b>C</b> | 28                             | 47                                       |
| <b>D</b> | 30                             | 48                                       |

**12** Which changes increase the rate of reaction?

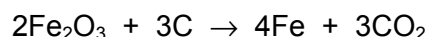
- 1 increasing the concentration of the reactants
- 2 increasing the particle size of a solid reactant
- 3 increasing the temperature

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**13** Which reaction is reversible?

- A** an iron nail rusting when left in moist air
- B** limestone reacting with an acid to form carbon dioxide gas
- C** magnesium burning in air to produce a white ash
- D** white anhydrous copper(II) sulfate turning blue when water is added

- 14 The equation for the reaction between iron(III) oxide and carbon is shown.



Which type of reaction does iron(III) oxide undergo?

- A reduction
  - B precipitation
  - C oxidation
  - D combustion
- 15 Copper(II) chloride is made when copper(II) carbonate reacts with dilute hydrochloric acid.

What are the other products in this reaction?

- A water and carbon dioxide
  - B carbon dioxide only
  - C water and hydrogen
  - D hydrogen only
- 16 Rubidium is in Group I and strontium is in Group II of the Periodic Table.

Which row describes the nature of rubidium oxide,  $\text{Rb}_2\text{O}$ , and strontium oxide,  $\text{SrO}$ ?

|   | $\text{Rb}_2\text{O}$ | $\text{SrO}$ |
|---|-----------------------|--------------|
| A | acidic                | acidic       |
| B | acidic                | basic        |
| C | basic                 | acidic       |
| D | basic                 | basic        |

- 17 Magnesium sulfate is a soluble solid which is formed when insoluble magnesium oxide reacts with dilute sulfuric acid.

Which method is used to prepare solid magnesium sulfate?

- A Excess sulfuric acid is reacted with magnesium oxide. The mixture is evaporated to dryness.
- B Excess sulfuric acid is reacted with magnesium oxide. The precipitate is filtered, washed and dried.
- C Sulfuric acid is reacted with excess magnesium oxide. The mixture is filtered and the filtrate is evaporated to dryness.
- D Sulfuric acid is reacted with excess magnesium oxide. The precipitate is filtered, washed and dried.

**18** Q and R are elements in the same period of the Periodic Table.

Q has 7 electrons in its outer shell and R has 2 electrons in its outer shell.

Which statement about Q and R is correct?

- A** Q is a metal and R is a non-metal.
- B** Q and R have different numbers of electron shells.
- C** R is found to the right of Q in the Periodic Table.
- D** The proton number of R is less than the proton number of Q.

**19** Which statement about alkali metals is correct?

- A** Lithium is more dense than sodium.
- B** Sodium is more reactive than potassium.
- C** Sodium has a higher melting point than potassium.
- D** They are in Group II of the Periodic Table.

**20** Aqueous bromine is added to aqueous sodium iodide.



What are the products of this reaction?

|          | 1      | 2              |
|----------|--------|----------------|
| <b>A</b> | iodide | sodium bromide |
| <b>B</b> | iodide | sodium bromine |
| <b>C</b> | iodine | sodium bromide |
| <b>D</b> | iodine | sodium bromine |

**21** Which row describes the properties of a transition element?

|          | melting point | density | forms coloured compounds |
|----------|---------------|---------|--------------------------|
| <b>A</b> | high          | low     | no                       |
| <b>B</b> | high          | high    | yes                      |
| <b>C</b> | low           | low     | no                       |
| <b>D</b> | low           | low     | yes                      |

22 Which row describes the properties of argon?

|          | property 1 | property 2 |
|----------|------------|------------|
| <b>A</b> | inert      | diatomic   |
| <b>B</b> | inert      | monatomic  |
| <b>C</b> | reactive   | diatomic   |
| <b>D</b> | reactive   | monatomic  |

23 Which row identifies the properties of zinc?

|          | thermal conductivity | reacts with dilute acid |
|----------|----------------------|-------------------------|
| <b>A</b> | good                 | yes                     |
| <b>B</b> | good                 | no                      |
| <b>C</b> | poor                 | yes                     |
| <b>D</b> | poor                 | no                      |

24 Uses of metals depend on their properties.

Which property is necessary for the use given?

|          | use of the metal           | property of the metal |
|----------|----------------------------|-----------------------|
| <b>A</b> | car bodies                 | ductile               |
| <b>B</b> | cutlery                    | conducts heat         |
| <b>C</b> | food containers            | resists corrosion     |
| <b>D</b> | overhead electrical cables | high density          |

25 Which compounds **both** contribute to acid rain?

- A** carbon monoxide and carbon dioxide
- B** carbon monoxide and oxides of nitrogen
- C** oxides of nitrogen and sulfur dioxide
- D** sulfur dioxide and carbon dioxide

**26** P, Q, R and S are metals.

P reacts with dilute hydrochloric acid, forming hydrogen.

Q reacts violently with water.

R reacts with water to give hydrogen.

S is formed by heating its oxide with carbon.

Which row identifies the metals?

|          | P      | Q         | R         | S         |
|----------|--------|-----------|-----------|-----------|
| <b>A</b> | copper | sodium    | potassium | iron      |
| <b>B</b> | zinc   | magnesium | calcium   | iron      |
| <b>C</b> | zinc   | sodium    | calcium   | magnesium |
| <b>D</b> | iron   | potassium | sodium    | zinc      |

**27** Which compound is formed when iron rusts?

- A** anhydrous iron(II) oxide
- B** anhydrous iron(III) oxide
- C** hydrated iron(III) hydroxide
- D** hydrated iron(III) oxide

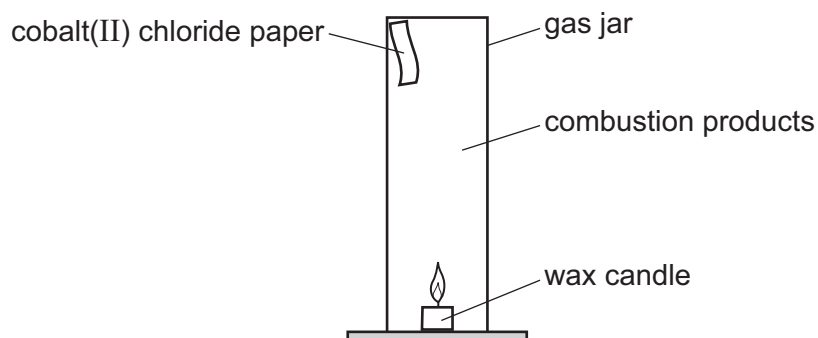
**28** Which reaction in the blast furnace releases heat energy?

- A**  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- B**  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- C**  $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
- D**  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$



- 29** A wax candle is made from a mixture of hydrocarbons.

The candle is lit and placed in a gas jar along with a strip of cobalt(II) chloride test paper as shown.



After a short time, the oxygen in the jar is used up and the candle flame goes out.

Which substance does the cobalt(II) chloride paper identify?

- A** carbon dioxide
  - B** carbon monoxide
  - C** sulfur dioxide
  - D** water
- 30** Urea,  $\text{CO}(\text{NH}_2)_2$ , is used as a fertiliser.

Which element that plants need for improved plant growth is provided by urea?

- A** carbon
- B** hydrogen
- C** nitrogen
- D** oxygen

- 31 The percentage composition of gases on Neptune is shown.

| gas         | percentage composition / % |
|-------------|----------------------------|
| hydrogen    | 80                         |
| helium      | 18                         |
| methane     | 1.5                        |
| other gases | 0.5                        |

Which statement about the atmospheres on Neptune and on the Earth is correct?

- A** There is more helium on Neptune than oxygen on the Earth.  
**B** There is less methane on Neptune than carbon dioxide on the Earth.  
**C** There is less hydrogen on the Earth than on Neptune.  
**D** There is more helium on the Earth than on Neptune.
- 32 Which row shows the general formula for alkenes and for alcohols?

|          | alkenes       | alcohols          |
|----------|---------------|-------------------|
| <b>A</b> | $C_nH_{2n}$   | $C_nH_{2n+1}COOH$ |
| <b>B</b> | $C_nH_{2n}$   | $C_nH_{2n+1}OH$   |
| <b>C</b> | $C_nH_{2n+2}$ | $C_nH_{2n+1}COOH$ |
| <b>D</b> | $C_nH_{2n+2}$ | $C_nH_{2n+1}OH$   |

- 33 A molecule has the formula  $C_2H_5Cl$ .

What is its chemical name?

- A** chloroethane  
**B** chloroethanol  
**C** chloroethene  
**D** chloromethanol
- 34 Which compound rapidly decolourises aqueous bromine?
- A** ethane  
**B** ethanoic acid  
**C** ethanol  
**D** ethene

**35** Compound Z has the molecular formula  $\text{C}_2\text{H}_6\text{O}$ .

Which statement about compound Z is correct?

- A** Z is unsaturated.
- B** Z is a carboxylic acid.
- C** Z is formed by the reaction of ethane with steam.
- D** Z is used as a fuel.

**36** What is the formula of the salt formed when aqueous ethanoic acid reacts with calcium carbonate?

- A**  $\text{Ca}(\text{CH}_3\text{COOH})_2$
- B**  $\text{Ca}(\text{CH}_3\text{COO})_2$
- C**  $\text{Ca}_2\text{CH}_3\text{COOH}$
- D**  $\text{Ca}_2\text{CH}_3\text{COO}$

**37** Rock salt is a mixture of salt and sand.

The method used to separate the sand from the salt is listed.

step 1 Crush the rock salt, add to warm water and stir.

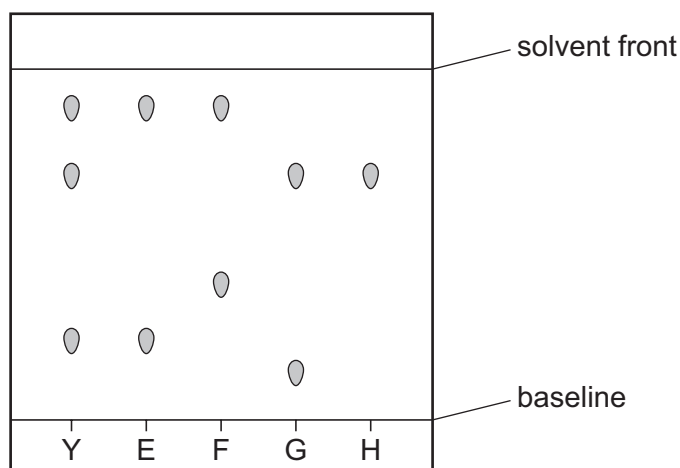
step 2 Pour the mixture through a filter paper held in a funnel.

step 3 Evaporate the water to crystallise the salt.

Which statement about the method is correct?

- A** The filtrate in step 2 is pure water.
- B** The residue in step 2 is pure crystals of salt.
- C** The solute is salt.
- D** The solvent is a mixture of salt and water.

- 38 Chromatography is carried out on mixture Y and dyes E, F, G and H. The chromatogram is shown.



Which dyes are present in mixture Y?

- A** E and G      **B** E and H      **C** F and G      **D** F and H
- 39 A fractionating column is used to separate the hydrocarbon fractions in petroleum by fractional distillation.

Which row describes the properties of the fractions that condense at the top of the fractionating column?

|          | size of molecule | boiling point |
|----------|------------------|---------------|
| <b>A</b> | large            | high          |
| <b>B</b> | large            | low           |
| <b>C</b> | small            | high          |
| <b>D</b> | small            | low           |

- 40 When acid is added to salt X, a gas is produced which turns limewater milky.

When sodium hydroxide is added to salt X, a gas is produced which turns litmus paper blue.

What is X?

- A**  $\text{CaCO}_3$       **B**  $(\text{NH}_4)_2\text{CO}_3$       **C**  $\text{NH}_4\text{NO}_3$       **D**  $\text{ZnCO}_3$

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## Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)**

**0971/12**

Paper 1 Multiple Choice (Core)

**May/June 2023**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | A      | 1     |
| 3        | D      | 1     |
| 4        | A      | 1     |
| 5        | C      | 1     |
| 6        | B      | 1     |
| 7        | B      | 1     |
| 8        | D      | 1     |
| 9        | A      | 1     |
| 10       | A      | 1     |
| 11       | C      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | A      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | C      | 1     |
| 18       | D      | 1     |
| 19       | C      | 1     |
| 20       | C      | 1     |
| 21       | B      | 1     |
| 22       | B      | 1     |
| 23       | A      | 1     |
| 24       | C      | 1     |
| 25       | C      | 1     |
| 26       | D      | 1     |
| 27       | D      | 1     |
| 28       | A      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | D             | <b>1</b>     |
| 30              | C             | <b>1</b>     |
| 31              | C             | <b>1</b>     |
| 32              | B             | <b>1</b>     |
| 33              | A             | <b>1</b>     |
| 34              | D             | <b>1</b>     |
| 35              | D             | <b>1</b>     |
| 36              | B             | <b>1</b>     |
| 37              | C             | <b>1</b>     |
| 38              | B             | <b>1</b>     |
| 39              | D             | <b>1</b>     |
| 40              | B             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The melting points and boiling points of four elements are shown.

| element | melting point/ $^{\circ}\text{C}$ | boiling point/ $^{\circ}\text{C}$ |
|---------|-----------------------------------|-----------------------------------|
| W       | -7                                | 60                                |
| X       | -101                              | -34                               |
| Y       | 114                               | 184                               |
| Z       | 39                                | 688                               |

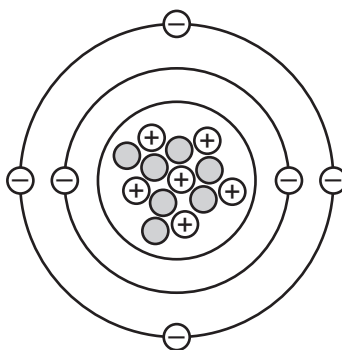
In which elements do the particles vibrate about fixed positions at  $0^{\circ}\text{C}$ ?

- A** W and X      **B** W and Z      **C** X and Y      **D** Y and Z
- 2 Which statements about clean, dry air are correct?

- 1 It is a mixture of elements only.
- 2 It is a mixture of elements and compounds.
- 3 It contains only non-metals.

- A** 1 and 3      **B** 1 only      **C** 2 and 3      **D** 2 only

- 3 A representation of an atom is shown.



What is the nucleon number of this atom?

- A** 6      **B** 7      **C** 12      **D** 13
- 4 Which statement describes isotopes of the same element?
- A** They have different electron arrangements.
- B** They have different nuclear charges.
- C** They have nuclei with masses that are the same.
- D** They have the same number of protons.

- 5 Potassium reacts with iodine to form potassium iodide.

Which statement about potassium iodide is correct?

- A Each potassium atom shares a pair of electrons with an iodine atom.
- B In potassium iodide, the particles of potassium have more protons than electrons.
- C Potassium iodide has a high melting point because it is a covalent compound.
- D Potassium iodide has a low melting point because it is an ionic compound.

- 6 Which row describes the properties of a simple molecular substance?

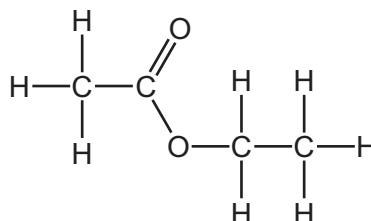
|   | boiling point | electrical conductivity when solid |
|---|---------------|------------------------------------|
| A | low           | poor                               |
| B | high          | poor                               |
| C | low           | good                               |
| D | high          | good                               |

- 7 Different forms of an element G are used as lubricants and in cutting tools.

What is the structure of G?

- A giant covalent
- B ionic
- C metallic
- D simple covalent

- 8 The diagram shows the structure of a molecule of ethyl ethanoate.



What is the molecular formula of a molecule of ethyl ethanoate?

- A CHO
- B C<sub>4</sub>H<sub>8</sub>O<sub>2</sub>
- C C<sub>4</sub>(H<sub>2</sub>)<sub>2</sub>(O<sub>2</sub>)
- D C<sub>2</sub>H<sub>4</sub>O

- 9 The formula of a compound containing element X is  $\text{Na}_2\text{X}_2\text{O}_3$ .

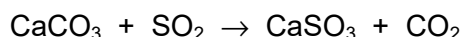
The relative formula mass of the compound is 158.

What is the relative atomic mass of X?

- A** 32                      **B** 59.5                      **C** 64                      **D** 119

- 10 Limestone is used to reduce sulfur dioxide emissions from coal-fired power stations.

The equation for the reaction is shown.



What is the smallest mass of  $\text{CaCO}_3$  required to remove 1 tonne of  $\text{SO}_2$ ?

- A** 1 tonne  
**B** 2 tonnes  
**C** 64 tonnes  
**D** 100 tonnes

- 11 Which statement about electrolysis is correct?

- A** Bromine and hydrogen are formed during the electrolysis of molten lead(II) bromide.  
**B** Metals are formed at the positive electrode.  
**C** Molten covalent compounds are broken down by electricity.  
**D** Platinum is used as an inert electrode.

- 12 Which statements about hydrogen-oxygen fuel cells are correct?

- 1 The reaction between hydrogen and oxygen is endothermic.
- 2 The waste product in a hydrogen-oxygen fuel cell is water.
- 3 A chemical reaction in the cell produces hydrogen which is used as the fuel.
- 4 A hydrogen-oxygen fuel cell is used to generate electricity.

- A** 1 and 2                      **B** 1 and 3                      **C** 2 and 4                      **D** 3 and 4

**13** The initial and final temperatures of four different reactions are measured.

Which reaction is the **least** exothermic?

|          | initial<br>temperature<br>/°C | final<br>temperature<br>/°C |
|----------|-------------------------------|-----------------------------|
| <b>A</b> | 19                            | 25                          |
| <b>B</b> | 21                            | 18                          |
| <b>C</b> | 22                            | 17                          |
| <b>D</b> | 22                            | 26                          |

**14** Solid calcium carbonate reacts with dilute hydrochloric acid.

Which changes to the reaction conditions increase the rate of reaction?

|          | concentration of<br>hydrochloric acid | surface area of<br>calcium carbonate |
|----------|---------------------------------------|--------------------------------------|
| <b>A</b> | decrease                              | decrease                             |
| <b>B</b> | decrease                              | increase                             |
| <b>C</b> | increase                              | decrease                             |
| <b>D</b> | increase                              | increase                             |

**15** Zinc reacts slowly with dilute sulfuric acid at room temperature.

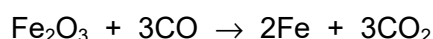
Bubbles of a gas, L, form on the surface of the zinc.

When a small amount of copper is added, the reaction is faster.

Which row identifies L and explains why the reaction is faster?

|          | gas formed<br>in reaction | reason the reaction is faster     |
|----------|---------------------------|-----------------------------------|
| <b>A</b> | hydrogen                  | copper acts as a catalyst         |
| <b>B</b> | hydrogen                  | copper is more reactive than zinc |
| <b>C</b> | oxygen                    | copper acts as a catalyst         |
| <b>D</b> | oxygen                    | copper is more reactive than zinc |

- 16 Which reaction shows a colour change from white to blue?
- A adding water to anhydrous copper(II) sulfate
  - B adding water to hydrated copper(II) sulfate
  - C heating anhydrous copper(II) sulfate
  - D heating hydrated copper(II) sulfate
- 17 In a blast furnace, iron(III) oxide is converted to iron and carbon monoxide is converted to carbon dioxide.



What happens to each of these reactants?

- A Both iron(III) oxide and carbon monoxide are oxidised.
  - B Both iron(III) oxide and carbon monoxide are reduced.
  - C Iron(III) oxide is oxidised and carbon monoxide is reduced.
  - D Iron(III) oxide is reduced and carbon monoxide is oxidised.
- 18 Which products are formed when magnesium carbonate reacts with dilute hydrochloric acid?
- A carbon dioxide, hydrogen and magnesium chloride
  - B carbon dioxide and magnesium chloride only
  - C carbon dioxide, water and magnesium chloride
  - D water and magnesium chloride only
- 19 Which element forms an oxide that reacts with an aqueous solution of a base?
- A argon
  - B sulfur
  - C magnesium
  - D copper
- 20 Which salt is insoluble?
- A barium sulfate
  - B lead(II) nitrate
  - C magnesium chloride
  - D sodium carbonate



**21** Some properties of element R are shown.

|                                  |                              |
|----------------------------------|------------------------------|
| melting point in °C              | 98                           |
| boiling point in °C              | 883                          |
| reaction with cold water         | gives off H <sub>2</sub> gas |
| reaction when heated with oxygen | burns to give a white solid  |

In which part of the Periodic Table is R found?

- A** Group I
- B** Group VII
- C** Group VIII
- D** transition elements

**22** Lithium, sodium and potassium are elements in Group I.

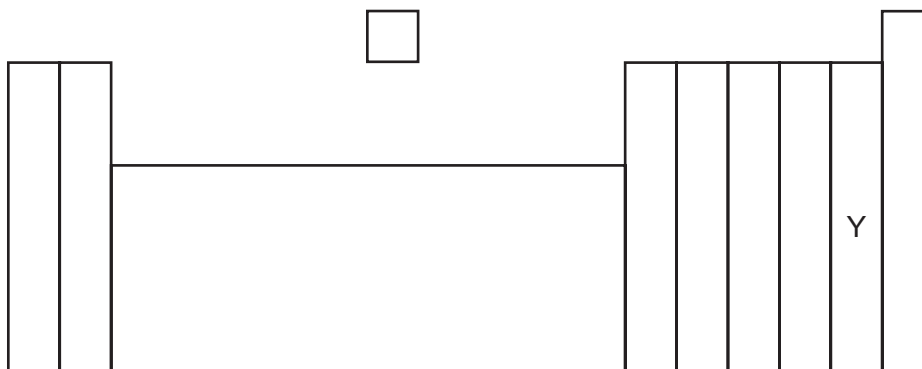
Statements about these elements are listed.

- 1 Lithium is more dense than sodium.
- 2 Sodium is more reactive than potassium.
- 3 They all conduct electricity at room temperature.
- 4 They all react with oxygen at room temperature.

Which statements are correct?

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

23 An outline of the Periodic Table is shown.



Which name is given to the elements in column Y?

- A** alkali metals
- B** halogens
- C** noble gases
- D** transition elements

24 Which row describes the properties of a metal that can be used in the manufacture of aircraft?

|          | strength | density | ease of corrosion |
|----------|----------|---------|-------------------|
| <b>A</b> | high     | high    | corrodes easily   |
| <b>B</b> | high     | low     | resists corrosion |
| <b>C</b> | low      | high    | corrodes easily   |
| <b>D</b> | low      | low     | resists corrosion |

25 Which metallic element is added to iron in the manufacture of stainless steel?

- A** carbon
- B** copper
- C** lead
- D** nickel

26 Which statement about the uses of metals is correct?

- A** Aluminium is used in the manufacture of overhead electrical cables as it has a high density.
- B** Aluminium is used to make food containers as it conducts electricity.
- C** Stainless steel is used in cutlery because it is resistant to rusting.
- D** Stainless steel is used to make chemical reactors because it is a soft alloy.

27 The list gives the order of some metals and hydrogen in the reactivity series.

Metal X is also included.

|                |    |
|----------------|----|
| most reactive  | K  |
|                | Mg |
|                | Zn |
|                | H  |
|                | X  |
| least reactive | Cu |

Which row shows the properties of metal X?

|          | reacts with dilute acids | oxide reduced by carbon |
|----------|--------------------------|-------------------------|
| <b>A</b> | no                       | no                      |
| <b>B</b> | no                       | yes                     |
| <b>C</b> | yes                      | no                      |
| <b>D</b> | yes                      | yes                     |

28 Which gas in the air is needed for iron to rust?

- A** argon
- B** carbon dioxide
- C** nitrogen
- D** oxygen

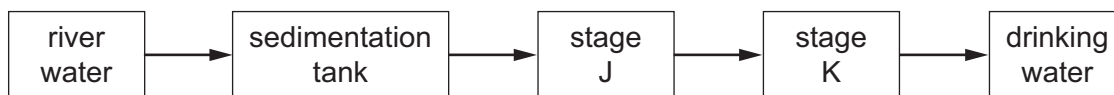
29 Why is limestone added to the blast furnace?

- A** It neutralises the molten slag produced.
- B** It reacts with impurities to form slag.
- C** It releases carbon dioxide which reduces the iron(III) oxide.
- D** It removes acidic gases such as carbon dioxide.

30 Which process removes carbon dioxide from the atmosphere?

- A** photosynthesis
- B** thermal decomposition of calcium carbonate
- C** combustion of fossil fuels
- D** reaction of sodium carbonate with an acid

- 31 The flow chart shows stages in the treatment of river water to produce drinking water.



What occurs at stages J and K?

|          | J            | K            |
|----------|--------------|--------------|
| <b>A</b> | distillation | chlorination |
| <b>B</b> | distillation | filtration   |
| <b>C</b> | filtration   | chlorination |
| <b>D</b> | filtration   | distillation |

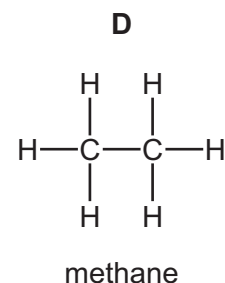
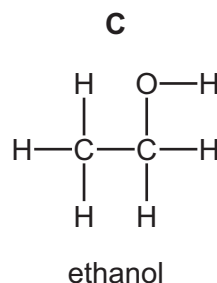
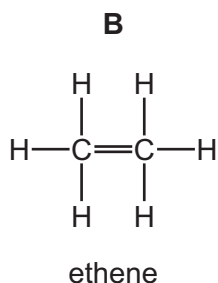
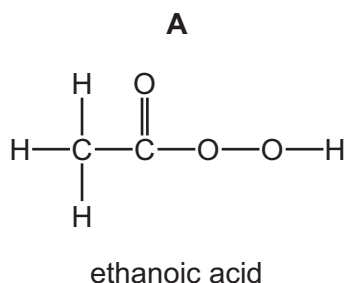
- 32 Which two compounds can be mixed together to form an NPK fertiliser?

- A** ammonium phosphate and calcium hydroxide
- B** calcium phosphate and ammonium nitrate
- C** potassium nitrate and calcium oxide
- D** potassium phosphate and ammonium nitrate

- 33 What are the main substances produced by the fractional distillation of liquid air?

- A** oxygen and carbon dioxide
- B** oxygen and nitrogen
- C** helium and nitrogen
- D** hydrogen and oxygen

- 34 Which diagram shows the displayed formula for the named organic compound?



**35** Poly(ethene) is formed from petroleum using three separate processes.

In which order are the processes used?

- A** cracking → fractional distillation → polymerisation
- B** cracking → polymerisation → fractional distillation
- C** fractional distillation → cracking → polymerisation
- D** fractional distillation → polymerisation → cracking

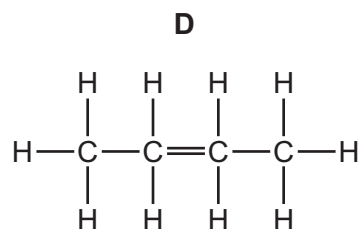
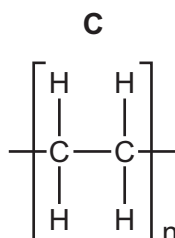
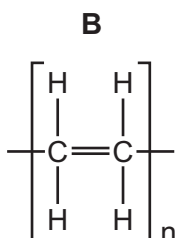
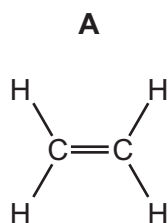
**36** Gas oil and naphtha are two fractions obtained from petroleum.

What are uses of gas oil and naphtha?

|          | gas oil            | naphtha          |
|----------|--------------------|------------------|
| <b>A</b> | jet fuel           | making chemicals |
| <b>B</b> | jet fuel           | making roads     |
| <b>C</b> | diesel engine fuel | making chemicals |
| <b>D</b> | diesel engine fuel | making roads     |

**37** Ethene can be polymerised.

Which diagram represents the structure of the product formed?



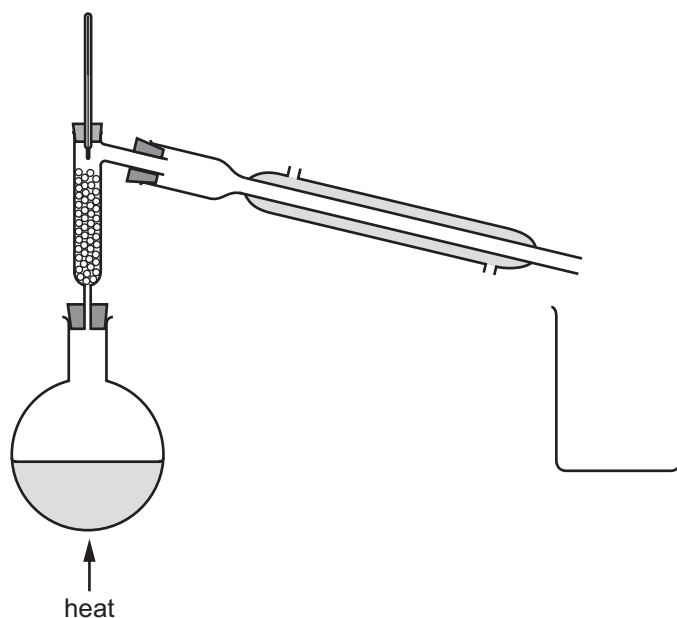
38 An acid–base titration is described.

- 25.0 cm<sup>3</sup> of dilute aqueous alkali is put into a conical flask.
- Indicator is added to the flask.
- Dilute acid is added to the aqueous alkali until the indicator changes colour.
- The volume of acid used is then recorded.

Which use of apparatus is correct?

- A** The 25.0 cm<sup>3</sup> of aqueous alkali is measured using a volumetric pipette.
- B** The 25.0 cm<sup>3</sup> of aqueous alkali is measured using the lines on the conical flask.
- C** The volume of acid is measured using a measuring cylinder.
- D** The volume of acid is measured using a volumetric pipette.

39 The apparatus shown is used to separate a mixture.



What is the mixture?

- A** anhydrous copper(II) sulfate and hydrated copper(II) sulfate
- B** sodium chloride and sand
- C** ethanol and methanol
- D** iron and steel

40 The results of tests on three gases, X, Y and Z, are shown.

| test                             | X                    | Y          | Z         |
|----------------------------------|----------------------|------------|-----------|
| aqueous potassium manganate(VII) | purple to colourless | no change  | no change |
| damp red litmus paper            | no change            | turns blue | no change |
| lighted splint                   | no change            | no change  | pops      |

What are X, Y and Z?

|          | X              | Y              | Z        |
|----------|----------------|----------------|----------|
| <b>A</b> | chlorine       | sulfur dioxide | hydrogen |
| <b>B</b> | chlorine       | sulfur dioxide | oxygen   |
| <b>C</b> | sulfur dioxide | ammonia        | oxygen   |
| <b>D</b> | sulfur dioxide | ammonia        | hydrogen |

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| Group                      |                             |                                                                                             |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                             |                             |                           |                            |                             |                           |                         |                           |                              |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
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| I                          | II                          |                                                                                             |                             |                             |                              |                              |                              |                            |                              |                            |                            | III                       | IV                         | V                           | VI                           | VII                      | VIII                     |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                             |                             |                           |                            |                             |                           |                         |                           |                              |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                             |                             |                              |                              |                              |                            |                              |                            |                            | 1<br>H<br>hydrogen<br>1   |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                             |                             |                           |                            |                             |                           |                         |                           |                              |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
|                            |                             |                                                                                             |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                             |                             |                           |                            |                             |                           |                         |                           |                              |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 | 13<br>Al<br>aluminium<br>27                                                                 | 14<br>Si<br>silicon<br>28   | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32      | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40      | 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40    | 21<br>Sc<br>scandium<br>45 | 22<br>Ti<br>titanium<br>48 | 23<br>V<br>vanadium<br>51 | 24<br>Cr<br>chromium<br>52 | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59 | 28<br>Ni<br>nickel<br>59 | 29<br>Cu<br>copper<br>64   | 30<br>Zn<br>zinc<br>65    | 31<br>Ga<br>gallium<br>70 | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79 | 35<br>Br<br>bromine<br>80  | 36<br>Kr<br>krypton<br>84 |                            |                             |                         |                             |                             |                           |                            |                             |                           |                         |                           |                              |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>—  | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108  | 48<br>Cd<br>cadmium<br>112 | 49<br>In<br>indium<br>115 | 50<br>Sn<br>tin<br>119     | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127 | 54<br>Xe<br>xenon<br>131 | 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137 | 57–71<br>lanthanoids      | 72<br>Hf<br>hafnium<br>178  | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184 | 75<br>Re<br>rhenium<br>186 | 76<br>Os<br>osmium<br>190 | 77<br>Ir<br>iridium<br>192 | 78<br>Pt<br>platinum<br>195 | 79<br>Au<br>gold<br>197 | 80<br>Hg<br>mercury<br>201  | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207   | 83<br>Bi<br>bismuth<br>209 | 84<br>Po<br>polonium<br>—   | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>—  | 87<br>Fr<br>francium<br>— | 88<br>Ra<br>radium<br>—      | 89–103<br>actinoids | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>— | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>— | 108<br>Hs<br>hassium<br>— | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>— | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganeson<br>—  |  |                        |  |                          |  |                          |  |                          |  |                        |  |                           |  |                             |  |                           |  |                            |  |                           |  |                           |  |                            |  |                             |  |                          |  |                             |  |                           |  |                              |  |                             |  |                              |  |                            |  |                              |  |                           |  |                            |  |                           |  |                        |  |                             |  |                              |  |                          |  |                          |  |                            |  |                           |  |                      |  |                            |  |                             |  |                            |  |                            |  |                           |  |                            |  |                             |  |                         |  |                            |  |                             |  |                         |  |                            |  |                           |  |                           |  |                        |  |                           |  |                         |  |                     |  |                                 |  |                           |  |                              |  |                           |  |                           |  |                              |  |                                |  |                               |  |                               |  |                            |  |                             |  |                             |  |                               |  |                              |  |                            |  |
| 2<br>He<br>helium<br>4     |                             |                                                                                             |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           | 9<br>F<br>fluorine<br>19    |                             | 10<br>Ne<br>neon<br>20     |                            | 11<br>Na<br>sodium<br>23  |                            | 12<br>Mg<br>magnesium<br>24 |                         | 13<br>Al<br>aluminium<br>27 |                             | 14<br>Si<br>silicon<br>28 |                            | 15<br>P<br>phosphorus<br>31 |                           | 16<br>S<br>sulfur<br>32 |                           | 17<br>Cl<br>chlorine<br>35.5 |                     | 18<br>Ar<br>argon<br>40         |                           | 19<br>K<br>potassium<br>39   |                           | 20<br>Ca<br>calcium<br>40 |                              | 21<br>Sc<br>scandium<br>45     |                               | 22<br>Ti<br>titanium<br>48    |                            | 23<br>V<br>vanadium<br>51   |                             | 24<br>Cr<br>chromium<br>52    |                              | 25<br>Mn<br>manganese<br>55 |  | 26<br>Fe<br>iron<br>56 |  | 27<br>Co<br>cobalt<br>59 |  | 28<br>Ni<br>nickel<br>59 |  | 29<br>Cu<br>copper<br>64 |  | 30<br>Zn<br>zinc<br>65 |  | 31<br>Ga<br>gallium<br>70 |  | 32<br>Ge<br>germanium<br>73 |  | 33<br>As<br>arsenic<br>75 |  | 34<br>Se<br>selenium<br>79 |  | 35<br>Br<br>bromine<br>80 |  | 36<br>Kr<br>krypton<br>84 |  | 37<br>Rb<br>rubidium<br>85 |  | 38<br>Sr<br>strontium<br>88 |  | 39<br>Y<br>yttrium<br>89 |  | 40<br>Zr<br>zirconium<br>91 |  | 41<br>Nb<br>niobium<br>93 |  | 42<br>Mo<br>molybdenum<br>96 |  | 43<br>Tc<br>technetium<br>— |  | 44<br>Ru<br>ruthenium<br>101 |  | 45<br>Rh<br>rhodium<br>103 |  | 46<br>Pd<br>palladium<br>106 |  | 47<br>Ag<br>silver<br>108 |  | 48<br>Cd<br>cadmium<br>112 |  | 49<br>In<br>indium<br>115 |  | 50<br>Sn<br>tin<br>119 |  | 51<br>Sb<br>antimony<br>122 |  | 52<br>Te<br>tellurium<br>128 |  | 53<br>I<br>iodine<br>127 |  | 54<br>Xe<br>xenon<br>131 |  | 55<br>Cs<br>caesium<br>133 |  | 56<br>Ba<br>barium<br>137 |  | 57–71<br>lanthanoids |  | 72<br>Hf<br>hafnium<br>178 |  | 73<br>Ta<br>tantalum<br>181 |  | 74<br>W<br>tungsten<br>184 |  | 75<br>Re<br>rhenium<br>186 |  | 76<br>Os<br>osmium<br>190 |  | 77<br>Ir<br>iridium<br>192 |  | 78<br>Pt<br>platinum<br>195 |  | 79<br>Au<br>gold<br>197 |  | 80<br>Hg<br>mercury<br>201 |  | 81<br>Tl<br>thallium<br>204 |  | 82<br>Pb<br>lead<br>207 |  | 83<br>Bi<br>bismuth<br>209 |  | 84<br>Po<br>polonium<br>— |  | 85<br>At<br>astatine<br>— |  | 86<br>Rn<br>radon<br>— |  | 87<br>Fr<br>francium<br>— |  | 88<br>Ra<br>radium<br>— |  | 89–103<br>actinoids |  | 104<br>Rf<br>rutherfordium<br>— |  | 105<br>Db<br>dubnium<br>— |  | 106<br>Sg<br>seaborgium<br>— |  | 107<br>Bh<br>bohrium<br>— |  | 108<br>Hs<br>hassium<br>— |  | 109<br>Mt<br>meitnerium<br>— |  | 110<br>Ds<br>darmstadtium<br>— |  | 111<br>Rg<br>roentgenium<br>— |  | 112<br>Cn<br>copernicium<br>— |  | 113<br>Nh<br>nihonium<br>— |  | 114<br>Fl<br>flerovium<br>— |  | 115<br>Mc<br>moscovium<br>— |  | 116<br>Lv<br>livermorium<br>— |  | 117<br>Ts<br>tennessine<br>— |  | 118<br>Og<br>oganeson<br>— |  |

|    |    |           |     |    |    |         |     |    |    |              |     |    |    |           |     |    |    |            |   |    |    |           |     |    |    |           |     |    |    |            |     |    |    |           |     |    |    |             |     |    |    |             |     |     |    |         |     |     |    |             |     |     |    |           |     |     |    |            |     |
|----|----|-----------|-----|----|----|---------|-----|----|----|--------------|-----|----|----|-----------|-----|----|----|------------|---|----|----|-----------|-----|----|----|-----------|-----|----|----|------------|-----|----|----|-----------|-----|----|----|-------------|-----|----|----|-------------|-----|-----|----|---------|-----|-----|----|-------------|-----|-----|----|-----------|-----|-----|----|------------|-----|
| 57 | La | lanthanum | 139 | 58 | Ce | cerium  | 140 | 59 | Pr | praseodymium | 141 | 60 | Nd | neodymium | 144 | 61 | Pm | promethium | — | 62 | Sm | samarium  | 150 | 63 | Eu | europlum  | 152 | 64 | Gd | gadolinium | 157 | 65 | Tb | terbium   | 159 | 66 | Dy | dysprosium  | 163 | 67 | Ho | holmium     | 165 | 68  | Er | erbium  | 167 | 69  | Tm | thulium     | 169 | 70  | Yb | ytterbium | 173 | 71  | Lu | lutetium   | 175 |
| 89 | Ac | actinium  | —   | 90 | Th | thorium | 232 | 91 | Pa | protactinium | 231 | 92 | U  | uranium   | 238 | 93 | Np | neptunium  | — | 94 | Pu | plutonium | —   | 95 | Am | americium | —   | 96 | Cm | curium     | —   | 97 | Bk | berkelium | —   | 98 | Cf | californium | —   | 99 | Es | einsteinium | —   | 100 | Fm | fermium | —   | 101 | Md | mendelevium | —   | 102 | No | nobelium  | —   | 103 | Lr | lawrencium | —   |



## **Cambridge IGCSE™ (9–1)**

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**CHEMISTRY**

**0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2023**

**MARK SCHEME**

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | D      | 1     |
| 2        | C      | 1     |
| 3        | D      | 1     |
| 4        | D      | 1     |
| 5        | B      | 1     |
| 6        | A      | 1     |
| 7        | A      | 1     |
| 8        | B      | 1     |
| 9        | A      | 1     |
| 10       | B      | 1     |
| 11       | D      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | D      | 1     |
| 15       | A      | 1     |
| 16       | A      | 1     |
| 17       | D      | 1     |
| 18       | C      | 1     |
| 19       | B      | 1     |
| 20       | A      | 1     |
| 21       | A      | 1     |
| 22       | D      | 1     |
| 23       | B      | 1     |
| 24       | B      | 1     |
| 25       | D      | 1     |
| 26       | C      | 1     |
| 27       | B      | 1     |
| 28       | D      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | B             | <b>1</b>     |
| 30              | A             | <b>1</b>     |
| 31              | C             | <b>1</b>     |
| 32              | D             | <b>1</b>     |
| 33              | B             | <b>1</b>     |
| 34              | C             | <b>1</b>     |
| 35              | C             | <b>1</b>     |
| 36              | C             | <b>1</b>     |
| 37              | C             | <b>1</b>     |
| 38              | A             | <b>1</b>     |
| 39              | C             | <b>1</b>     |
| 40              | D             | <b>1</b>     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/11

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 The boiling point of sodium is 890 °C.

What happens to sodium atoms as the temperature of a sample of sodium changes from 950 °C to 900 °C?

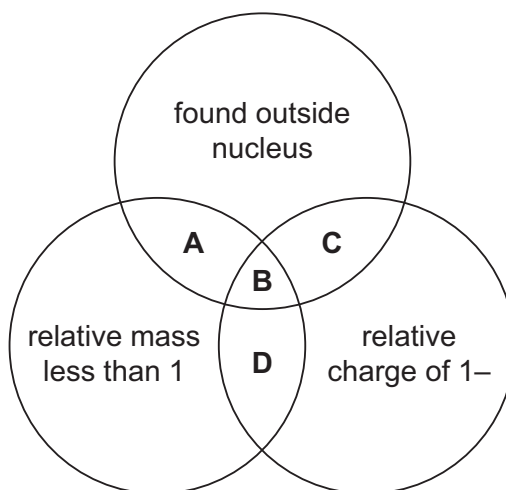
- A The atoms move more quickly and bonds are formed.
- B The atoms move more quickly and bonds are neither broken nor formed.
- C The atoms move more slowly and bonds are formed.
- D The atoms move more slowly and bonds are neither broken nor formed.

- 2 Which row shows the conditions for the particles of a gas colliding most frequently?

|          | pressure | temperature |
|----------|----------|-------------|
| <b>A</b> | high     | high        |
| <b>B</b> | high     | low         |
| <b>C</b> | low      | high        |
| <b>D</b> | low      | low         |

- 3 The diagram shows some properties of particles in an atom.

To which labelled part of the diagram do electrons belong?



- 4 Some properties of substances W, X, Y and Z are shown.

|   | melting point/ °C | electrical conductivity |
|---|-------------------|-------------------------|
| W | 801               | conducts when molten    |
| X | -182              | does not conduct        |
| Y | 840               | conducts when solid     |
| Z | 501               | conducts when molten    |

Which substances are ionic?

- A** W, X and Y      **B** W and Y only      **C** W and Z      **D** X and Z

- 5 Atoms lose or gain electrons to become ions.

Which row is correct?

|          | change to the atom      | type of ion | charge on ion |
|----------|-------------------------|-------------|---------------|
| <b>A</b> | loss of two electrons   | cation      | 2-            |
| <b>B</b> | loss of one electron    | anion       | 1-            |
| <b>C</b> | gain of three electrons | anion       | 3-            |
| <b>D</b> | gain of one electron    | cation      | 1-            |

- 6 A covalent molecule, M, contains four shared pairs of electrons.

What is M?

- A** ammonia,  $\text{NH}_3$   
**B** hydrogen chloride,  $\text{HCl}$   
**C** methane,  $\text{CH}_4$   
**D** water,  $\text{H}_2\text{O}$

- 7 Which substance has a giant covalent structure?

- A** sodium chloride  
**B** sodium  
**C** ethane  
**D** diamond

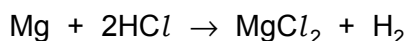


- 8 Iron(III) oxide is reduced by carbon monoxide to produce iron and carbon dioxide.

What is the balanced equation for this reaction?

- A**  $\text{Fe}_2\text{O}_3 + 2\text{CO} \rightarrow 2\text{Fe} + 2\text{CO}_2$   
**B**  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$   
**C**  $2\text{Fe}_2\text{O}_3 + 6\text{CO} \rightarrow 2\text{Fe} + 6\text{CO}_2$   
**D**  $2\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 4\text{Fe} + 3\text{CO}_2$

- 9 The equation for the reaction between magnesium and dilute hydrochloric acid is shown.



Which mass of magnesium chloride is formed when 48.0 g of magnesium completely reacts with excess dilute hydrochloric acid?

- A** 23.8 g                      **B** 47.5 g                      **C** 95.0 g                      **D** 190 g

- 10 Dilute sulfuric acid and lead(II) bromide are electrolysed separately.

Which statements are correct?

- 1 Colourless gases are produced when dilute sulfuric acid is electrolysed.
- 2 Lead(II) bromide can be electrolysed when molten.
- 3 Lead is formed at the positive electrode when lead(II) bromide is electrolysed.
- 4 Sulfate ions are produced at the negative electrode when dilute sulfuric acid is electrolysed.

- A** 1 and 2                      **B** 1 and 3                      **C** 2 and 3                      **D** 3 and 4

- 11 Which statements about a hydrogen–oxygen fuel cell are correct?

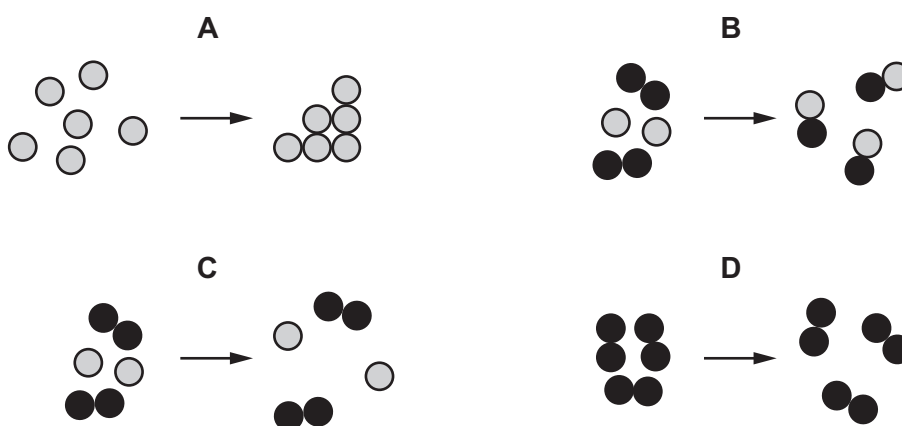
- 1 The main form of energy released by the fuel cell is heat.
- 2 The reaction is a redox reaction.
- 3 An acidic gas is produced.
- 4 Water is the only chemical product.

- A** 1 and 3                      **B** 1 and 4                      **C** 2 and 3                      **D** 2 and 4

12 Which row describes what happens during an endothermic reaction?

|          | thermal energy is transferred | change in temperature of the reaction mixture |
|----------|-------------------------------|-----------------------------------------------|
| <b>A</b> | from the surroundings         | decrease                                      |
| <b>B</b> | from the surroundings         | increase                                      |
| <b>C</b> | to the surroundings           | decrease                                      |
| <b>D</b> | to the surroundings           | increase                                      |

13 Which diagram represents a chemical change?



- 14** A method used to investigate the rate of reaction of calcium carbonate with dilute hydrochloric acid under different conditions is shown.

- Place 50 cm<sup>3</sup> of dilute hydrochloric acid in a conical flask.
- Add a known volume of water to the conical flask.
- Heat the conical flask to the required temperature.
- Add 1.0g of calcium carbonate to the conical flask.
- Measure the time taken for the reaction to finish.

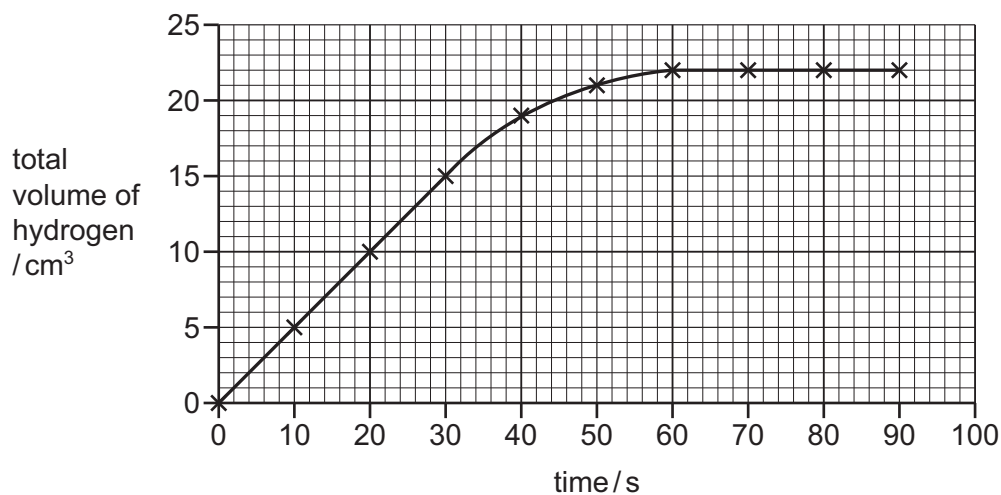
Which volume of water and which temperature give the shortest time taken for the reaction to finish?

|          | volume of water<br>added / cm <sup>3</sup> | temperature / °C |
|----------|--------------------------------------------|------------------|
| <b>A</b> | 10                                         | 30               |
| <b>B</b> | 10                                         | 50               |
| <b>C</b> | 40                                         | 30               |
| <b>D</b> | 40                                         | 50               |

**15** The rate of reaction between magnesium and hydrochloric acid is investigated.

The total volume of hydrogen given off is measured at different times.

A graph of the results is shown.



Which conclusions are correct?

- 1 The rate is fastest between 0 and 30 seconds.
- 2 The maximum volume of hydrogen given off is  $22\text{ cm}^3$ .
- 3 At 40 seconds,  $20\text{ cm}^3$  of hydrogen is given off.

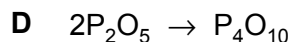
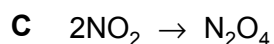
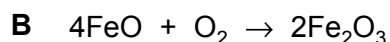
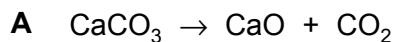
**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

**16** Water is added to anhydrous copper(II) sulfate.

Which row describes the direction of energy change and the colour change of the mixture during the reaction?

|          | direction of energy change     | colour change |
|----------|--------------------------------|---------------|
| <b>A</b> | absorbed from the surroundings | blue to white |
| <b>B</b> | absorbed from the surroundings | white to blue |
| <b>C</b> | released to the surroundings   | blue to white |
| <b>D</b> | released to the surroundings   | white to blue |

17 Which equation represents an oxidation reaction?



18 A farmer treats a field with calcium hydroxide to make it less acidic.

When the farmer adds ammonium nitrate fertiliser to the field immediately after the calcium hydroxide, the two substances react.

Why does this reaction make the fertiliser less effective?

A It makes ammonia gas, so less nitrogen is absorbed by the soil.

B It makes an acid, making the soil acidic again.

C It makes nitrogen gas, so less nitrogen is absorbed by the soil.

D It makes the fertiliser too strong, stopping the plants growing well.

19 Which statement about sodium oxide or nitrogen dioxide is correct?

A Nitrogen dioxide is a solid at room temperature.

B Nitrogen dioxide is acidic.

C Sodium oxide has a lower melting point than nitrogen dioxide.

D Sodium oxide is covalently bonded.

20 A titration method is used to prepare a pure soluble sulfate salt from dilute sulfuric acid.

What is the other reagent?

A copper(II) oxide

B magnesium

C sodium hydroxide

D zinc carbonate

21 Which row about elements in the Periodic Table is correct?

|          | statement 1                                                      | statement 2                          |
|----------|------------------------------------------------------------------|--------------------------------------|
| <b>A</b> | two elements in the same group have similar chemical properties  | metals are on the left of the table  |
| <b>B</b> | two elements in the same group have similar chemical properties  | metals are on the right of the table |
| <b>C</b> | two elements in the same period have similar chemical properties | metals are on the left of the table  |
| <b>D</b> | two elements in the same period have similar chemical properties | metals are on the right of the table |

22 The table gives some information about three elements in Group I of the Periodic Table.

| element  | atomic number | melting point<br>in °C | density<br>in g/cm <sup>3</sup> |
|----------|---------------|------------------------|---------------------------------|
| lithium  | 3             | 181                    | 0.53                            |
| sodium   | 11            | 98                     | 0.97                            |
| rubidium | 37            | X                      | X                               |

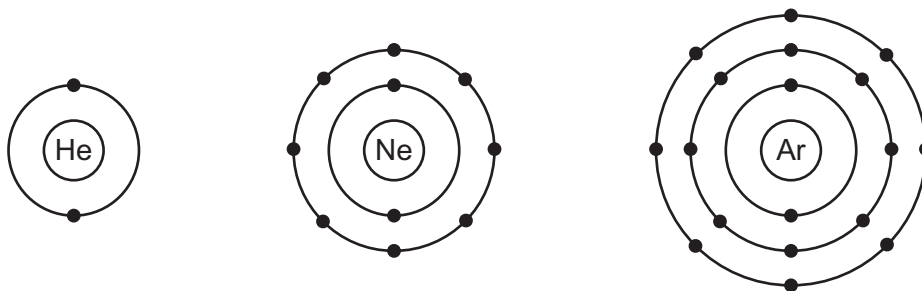
Which row identifies the melting point and the density of rubidium?

|          | melting point<br>in °C | density<br>in g/cm <sup>3</sup> |
|----------|------------------------|---------------------------------|
| <b>A</b> | 39                     | 0.38                            |
| <b>B</b> | 39                     | 1.53                            |
| <b>C</b> | 253                    | 0.38                            |
| <b>D</b> | 253                    | 1.53                            |

23 Which statement describes a transition element?

- A** It is a dull grey metal that only forms white compounds.
- B** It is a high-density metal with a high melting point that is used as a catalyst.
- C** It is a low-density metal with a high melting point that reacts with steam to make hydrogen.
- D** It is a soft, shiny silver metal that reacts vigorously with water.

**24** The electronic configurations of helium, neon and argon are shown.



Which row describes these gases?

|          | reactivity | form of the gas | electronic configuration            |
|----------|------------|-----------------|-------------------------------------|
| <b>A</b> | reactive   | monatomic       | incomplete outer shell of electrons |
| <b>B</b> | unreactive | diatomic        | complete outer shell of electrons   |
| <b>C</b> | unreactive | diatomic        | incomplete outer shell of electrons |
| <b>D</b> | unreactive | monatomic       | complete outer shell of electrons   |

**25** X is a shiny silver-coloured solid at room temperature and pressure.

X is a good conductor of heat and electricity when solid.

Which statement about X is correct?

- A** X is an ionic compound or a metallic element.
- B** X is a metallic element or a non-metallic element.
- C** X is an alloy or a metallic element.
- D** X is an alloy or a non-metallic element.

**26** Which elements can be combined to produce an alloy?

- 1 magnesium and aluminium
- 2 nitrogen and oxygen
- 3 iron and carbon
- 4 copper and zinc

- A** 1, 3 and 4      **B** 1 and 2      **C** 2 and 3      **D** 4 only

**27** Three metals, L, M and N, are added separately to dilute hydrochloric acid and cold water.

The results are shown.

| metal | reaction with dilute hydrochloric acid | reaction with cold water |
|-------|----------------------------------------|--------------------------|
| L     | hydrogen forms                         | no reaction              |
| M     | hydrogen forms                         | hydrogen forms           |
| N     | no reaction                            | no reaction              |

What is the order of reactivity of the metals?

|          | least reactive | → | most reactive |
|----------|----------------|---|---------------|
| <b>A</b> | L              | N | M             |
| <b>B</b> | M              | L | N             |
| <b>C</b> | N              | L | M             |
| <b>D</b> | N              | M | L             |

**28** Which reaction produces carbon dioxide?

- A** cracking of large hydrocarbon molecules
- B** photosynthesis
- C** reaction of a base with a carbonate
- D** thermal decomposition of calcium carbonate



**29** A sample of air containing four gases only is analysed.

99.0% of the sample contains the two main gases in the same percentages as in clean, dry air.

The remaining 1.0% of the sample contains argon and carbon dioxide.

The gas that makes up 0.1% of the sample turns limewater milky.

Which row shows the percentage composition of the sample of air?

|          | 99.0% of the sample          | 0.9% of the sample | 0.1% of the sample |
|----------|------------------------------|--------------------|--------------------|
| <b>A</b> | 78.0% nitrogen, 21.0% oxygen | argon              | carbon dioxide     |
| <b>B</b> | 78.0% nitrogen, 21.0% oxygen | carbon dioxide     | argon              |
| <b>C</b> | 78.0% oxygen, 21.0% nitrogen | argon              | carbon dioxide     |
| <b>D</b> | 78.0% oxygen, 21.0% nitrogen | carbon dioxide     | argon              |

**30** Which substance contains **two** elements that are found in NPK fertilisers?

- A** ammonium chloride
- B** calcium hydroxide
- C** potassium nitrate
- D** sodium phosphate

**31** Which statement about sulfur is correct?

- A** When sulfur is burned, it produces a substance that causes acid rain.
- B** Sulfur is produced by the thermal decomposition of limestone.
- C** Compounds of sulfur make up approximately 1% of unpolluted air.
- D** Sulfur is a member of the family of elements called halogens.

**32** What are **two** adverse effects of particulates in the air?

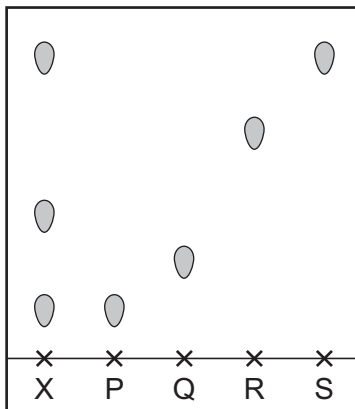
- 1 acid rain
- 2 cancer
- 3 photochemical smog
- 4 respiratory problems

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

- 33 Which formula represents a compound that is a member of the homologous series of alkanes?
- A  $C_2H_4$                       B  $C_3H_6$                       C  $C_4H_8$                       D  $C_5H_{12}$
- 34 Which statement about ethane is correct?
- A It rapidly decolourises aqueous bromine.  
B It does **not** burn.  
C It forms long-chain compounds called polymers.  
D It only contains single bonds between its atoms.
- 35 Which raw material is used to make ethanol by fermentation?
- A carbon dioxide  
B ethene  
C glucose  
D natural gas
- 36 Which statement about ethanoic acid is correct?
- A It contains a  $-COOH$  group.  
B It has a pH greater than pH 7.  
C It reacts with sodium carbonate to form hydrogen gas.  
D It reacts with copper to form copper(II) ethanoate.
- 37 Which statement explains why the disposal of plastic waste leads to environmental problems?
- A Plastic waste forms toxic gases when it is burned.  
B Plastic waste contains many small molecules.  
C Plastic waste rapidly dissolves in the oceans.  
D Plastic waste reacts with both acids and bases.

- 38 Substance X and four known substances, P, Q, R and S, are analysed by chromatography.

The chromatogram produced is shown.



Which statement about X is correct?

- A It is a mixture of P, Q and S.
  - B It contains P and S only.
  - C It contains P, S and another unknown substance.
  - D It is a mixture of Q, R and S.
- 39 Copper is insoluble in water.

Copper(II) oxide is a solid that is insoluble in water but reacts with dilute hydrochloric acid.

Which method is used to separate copper from a mixture of copper and copper(II) oxide?

- A dissolve the mixture in water then filter
- B dissolve the mixture in water then crystallise
- C react the mixture with dilute hydrochloric acid then filter
- D react the mixture with dilute hydrochloric acid then crystallise

**40** A salt, S, is dissolved in water and three tests are carried out on the solution formed.

|   | test                                                        | result                                                        |
|---|-------------------------------------------------------------|---------------------------------------------------------------|
| 1 | aqueous sodium hydroxide is added                           | green precipitate forms, insoluble in excess sodium hydroxide |
| 2 | dilute nitric acid is added                                 | no reaction                                                   |
| 3 | aqueous barium nitrate is added to the solution from test 2 | white precipitate forms                                       |

What is the identity of S?

- A** copper(II) chloride
- B** copper(II) sulfate
- C** iron(II) chloride
- D** iron(II) sulfate

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The Periodic Table of Elements

| Group                      |                             |                                                                                      |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |
|----------------------------|-----------------------------|--------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|------------------------------|-----------------------------|
| I                          | II                          |                                                                                      |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                          | V                           | VI                            | VII                          | VIII                        |
|                            |                             | <div>1<br/>H<br/>hydrogen<br/>1</div>                                                |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |
|                            |                             | <div>Key<br/>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   |                                                                                      |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 |                                                                                      |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                           | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79    | 35<br>Br<br>bromine<br>80    | 36<br>Kr<br>krypton<br>84   |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                             | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127     | 54<br>Xe<br>xenon<br>131    |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                 | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—     | 85<br>At<br>astatine<br>—    | 86<br>Rn<br>radon<br>—      |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                  | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>—  | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesson<br>— |

lanthanoids

|                              |                           |                                 |                              |                             |                             |                             |                               |                            |                               |                            |                           |                            |                              |                             |
|------------------------------|---------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|----------------------------|---------------------------|----------------------------|------------------------------|-----------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140 | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165 | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169 | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175 |
|------------------------------|---------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|----------------------------|---------------------------|----------------------------|------------------------------|-----------------------------|

actinoids

|                           |                            |                                 |                           |                            |                            |                            |                         |                            |                              |                              |                           |                               |                            |                              |
|---------------------------|----------------------------|---------------------------------|---------------------------|----------------------------|----------------------------|----------------------------|-------------------------|----------------------------|------------------------------|------------------------------|---------------------------|-------------------------------|----------------------------|------------------------------|
| 89<br>Ac<br>actinium<br>— | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238 | 93<br>Np<br>neptunium<br>— | 94<br>Pu<br>plutonium<br>— | 95<br>Am<br>americium<br>— | 96<br>Cm<br>curium<br>— | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>— | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>— | 103<br>Lr<br>lawrencium<br>— |
|---------------------------|----------------------------|---------------------------------|---------------------------|----------------------------|----------------------------|----------------------------|-------------------------|----------------------------|------------------------------|------------------------------|---------------------------|-------------------------------|----------------------------|------------------------------|

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).

# Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)****0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2024**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | D      | 1     |
| 2        | A      | 1     |
| 3        | B      | 1     |
| 4        | C      | 1     |
| 5        | C      | 1     |
| 6        | C      | 1     |
| 7        | D      | 1     |
| 8        | B      | 1     |
| 9        | D      | 1     |
| 10       | A      | 1     |
| 11       | D      | 1     |
| 12       | A      | 1     |
| 13       | B      | 1     |
| 14       | B      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | B      | 1     |
| 20       | C      | 1     |
| 21       | A      | 1     |
| 22       | B      | 1     |
| 23       | B      | 1     |
| 24       | D      | 1     |
| 25       | C      | 1     |
| 26       | A      | 1     |
| 27       | C      | 1     |
| 28       | D      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | A      | 1     |
| 30       | C      | 1     |
| 31       | A      | 1     |
| 32       | D      | 1     |
| 33       | D      | 1     |
| 34       | D      | 1     |
| 35       | C      | 1     |
| 36       | A      | 1     |
| 37       | A      | 1     |
| 38       | C      | 1     |
| 39       | C      | 1     |
| 40       | D      | 1     |





# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/12

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



1 Which statement about gases is correct?

- A Gases are difficult to compress when pressure is applied.
- B The particles in gases are close together.
- C The particles in gases have a random arrangement.
- D The particles in gases move slowly past each other.

2 A sample of argon gas is heated in a closed container.

Which row describes what happens to the pressure and the size of the argon atoms?

|   | pressure  | size           |
|---|-----------|----------------|
| A | decreases | increases      |
| B | decreases | stays the same |
| C | increases | increases      |
| D | increases | stays the same |

3 Which statement is correct?

- A Air is a mixture of gaseous elements only.
- B Alloys are formed when a metal is ionically bonded to other elements.
- C Carbon dioxide is a mixture of carbon and oxygen.
- D Potassium bromide is an ionic compound.

4 Which row identifies methods of testing the purity of a compound?

|   | melting point | boiling point | chromatography |
|---|---------------|---------------|----------------|
| A | ✓             | ✓             | ✓              |
| B | ✓             | x             | x              |
| C | ✓             | ✓             | x              |
| D | x             | ✓             | ✓              |

- 5 Which row shows the number of protons, neutrons and electrons in the ion  $^{18}_8\text{O}^{2-}$ ?

|          | protons | neutrons | electrons |
|----------|---------|----------|-----------|
| <b>A</b> | 10      | 8        | 8         |
| <b>B</b> | 8       | 10       | 8         |
| <b>C</b> | 10      | 8        | 10        |
| <b>D</b> | 8       | 10       | 10        |

- 6 Isotopes of the same element have some features that are the same and some that are different.

Which row shows the features that are the same and those that are different?

|          | nucleon number | proton number | number of outer shell electrons |
|----------|----------------|---------------|---------------------------------|
| <b>A</b> | ✓              | ✓             | ✗                               |
| <b>B</b> | ✗              | ✓             | ✓                               |
| <b>C</b> | ✓              | ✗             | ✗                               |
| <b>D</b> | ✗              | ✗             | ✓                               |

key

✓ = same

✗ = different

- 7 The electronic configurations of four atoms, W, X, Y and Z, are shown.

| atom | electronic configuration |
|------|--------------------------|
| W    | 2                        |
| X    | 2,6                      |
| Y    | 2,8,2                    |
| Z    | 2,8,6                    |

Which atoms form an ion with a charge of 2– when they react?

- A** W and Y      **B** W only      **C** X and Z      **D** Y only

8 Which statement about bonding is correct?

- A All the atoms in CH<sub>4</sub>, NH<sub>3</sub> and H<sub>2</sub>O molecules have noble gas electronic configurations.
- B Calcium chloride is a covalent molecule.
- C Group I metals gain electrons when they bond with Group VII elements.
- D Oxide ions in calcium oxide are positively charged.

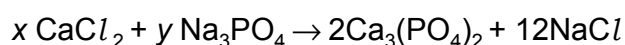
9 Which statement about diamond is correct?

- A It is a giant covalent structure consisting of carbon atoms and each atom is bonded to four other atoms.
- B It is a giant covalent structure consisting of flat sheets of carbon atoms.
- C It is a structure held together by ionic bonds and each ion is bonded to four other ions.
- D It is a structure held together by ionic bonds and each ion is bonded to three other ions.

10 Which row shows the correct formulae of lead(II) bromide and butane?

|   | lead(II) bromide   | butane                         |
|---|--------------------|--------------------------------|
| A | PbBr <sub>2</sub>  | C <sub>4</sub> H <sub>8</sub>  |
| B | PbBr <sub>2</sub>  | C <sub>4</sub> H <sub>10</sub> |
| C | Pb <sub>2</sub> Br | C <sub>4</sub> H <sub>8</sub>  |
| D | Pb <sub>2</sub> Br | C <sub>4</sub> H <sub>10</sub> |

11 Calcium phosphate forms when calcium chloride and sodium phosphate solutions react together.



Which values of  $x$  and  $y$  balance the equation?

|   | $x$ | $y$ |
|---|-----|-----|
| A | 2   | 2   |
| B | 3   | 4   |
| C | 6   | 3   |
| D | 6   | 4   |

**12** What is the definition of relative molecular mass,  $M_r$ ?

- A** It is the average mass of the isotopes in a compound.
- B** It is the sum of the atomic numbers in a compound.
- C** It is the sum of the relative atomic masses in a compound.
- D** It is the total number of atoms in a compound.

**13** In an experiment, a molten compound is broken down using electricity.

Which row identifies the negative electrode and the general name for the molten compound being broken down?

|          | negative electrode | general name for the molten compound |
|----------|--------------------|--------------------------------------|
| <b>A</b> | anode              | electrolysis                         |
| <b>B</b> | anode              | electrolyte                          |
| <b>C</b> | cathode            | electrolyte                          |
| <b>D</b> | cathode            | electrolysis                         |

**14** Hydrogen–oxygen fuel cells can be used to power vehicles.

What is produced by the fuel cells?

- 1 carbon dioxide
- 2 electricity
- 3 water

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**15** Three reactions are described.

- 1 An acid is added to substance H. Rapid fizzing happens and the temperature decreases.
- 2 When substance J is ignited, it produces large quantities of heat.
- 3 Substance K reacts slowly with air and becomes warmer.

Which reactions are endothermic?

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 only      **D** 2 and 3 only

16 Which of the processes produces a physical change?

- A thermal decomposition of calcium carbonate
- B addition of sodium chloride to water
- C addition of magnesium to hydrochloric acid
- D combustion of sodium

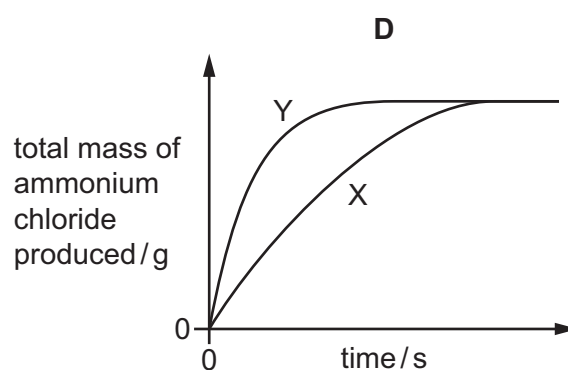
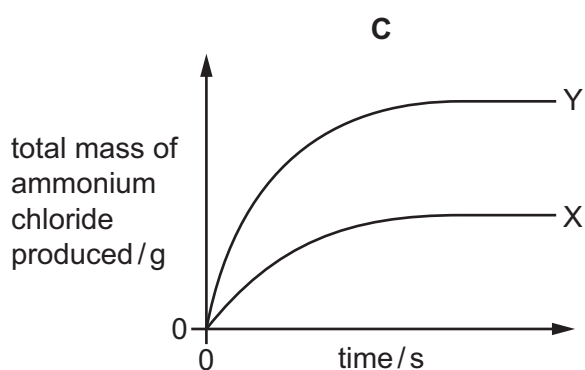
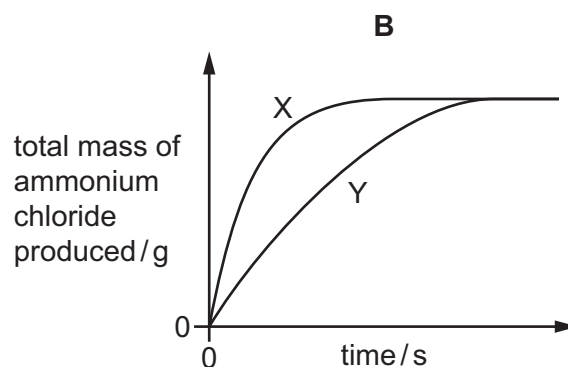
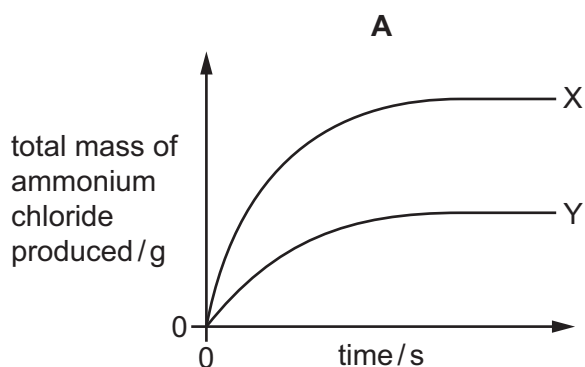
17 A known mass of gaseous ammonia and excess gaseous hydrogen chloride react together to make solid ammonium chloride.

Line X shows the total mass of ammonium chloride produced over time.

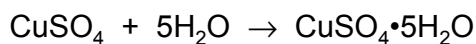
The reaction is repeated at a higher pressure. All other conditions are kept the same.

Line Y shows the total mass of ammonium chloride produced over time at the higher pressure.

Which diagram is correct?

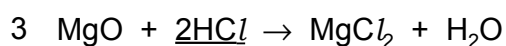
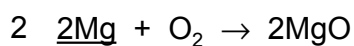
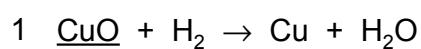


- 18 The equation for the hydration of anhydrous copper(II) sulfate is shown.



Which colour change is observed in this reaction?

- A blue to white  
 B white to blue  
 C pink to blue  
 D blue to pink
- 19 Which of the reactions show the underlined substance being reduced?



- A 1 and 2      B 1 only      C 2 and 3      D 3 only
- 20 Four different solutions are separately tested with blue litmus and with methyl orange. Each solution is known to be either acidic or alkaline. The results are shown.

| solution | result with blue litmus | result with methyl orange |
|----------|-------------------------|---------------------------|
| 1        | red                     | red                       |
| 2        | red                     | yellow                    |
| 3        | blue                    | yellow                    |
| 4        | blue                    | yellow                    |

Which statement is correct?

- A Solutions 1 and 4 are acidic.  
 B Solutions 1 and 2 are alkaline.  
 C Solutions 3 and 4 are acidic.  
 D Solutions 3 and 4 are alkaline.

**21** Which statement about sulfur dioxide or calcium oxide is correct?

- A** Calcium oxide is an acid.
- B** Calcium oxide turns thymolphthalein yellow.
- C** Sulfur dioxide is a base.
- D** Sulfur dioxide turns thymolphthalein colourless.

**22** Which substances can be used to make pure crystals of sodium sulfate?

- A** potassium sulfate and sodium hydroxide  
**B** sodium carbonate and sulfuric acid  
**C** sodium nitrate and magnesium sulfate  
**D** sulfuric acid and sodium chloride

**23** Which statements describe changes that occur from left to right across a period of the Periodic Table?

- 1 The atomic number of the elements increases.
- 2 The metallic character of the elements decreases.
- 3 The physical state of the elements changes from gas to solid.

- A** 2 only      **B** 1 and 2      **C** 1 and 3      **D** 2 and 3

**24** Part of the Periodic Table is shown.

Which element is the **most** reactive non-metal?

A 10x5 grid representing a periodic table. The grid is divided into four main groups of elements:

- Group 1 (Leftmost):** A vertical column of 5 cells. The second cell from the bottom is labeled **A**.
- Group 2 (Second from left):** A vertical column of 5 cells.
- Group 3 (Third from left):** A vertical column of 5 cells.
- Group 4 (Fourth from left):** A vertical column of 5 cells.
- Group 5 (Fifth from left):** A vertical column of 5 cells.
- Group 6 (Sixth from left):** A vertical column of 5 cells.
- Group 7 (Seventh from left):** A vertical column of 5 cells.
- Group 8 (Eighth from left):** A vertical column of 5 cells.
- Group 9 (Ninth from left):** A vertical column of 5 cells.
- Group 10 (Tenth from left):** A vertical column of 5 cells.
- Group 11 (Eleventh from left):** A vertical column of 5 cells.
- Group 12 (Twelfth from left):** A vertical column of 5 cells.
- Group 13 (Thirteenth from left):** A vertical column of 5 cells.
- Group 14 (Fourteenth from left):** A vertical column of 5 cells.
- Group 15 (Fifteenth from left):** A vertical column of 5 cells.
- Group 16 (Sixteenth from left):** A vertical column of 5 cells.
- Group 17 (Seventeenth from left):** A vertical column of 5 cells.
- Group 18 (Eighteenth from left):** A vertical column of 5 cells.

The labels are positioned as follows:

- A:** In the second cell from the bottom of the first column.
- B:** In the second cell from the bottom of the fifth column.
- C:** In the second cell from the bottom of the eighteenth column.
- D:** In the second cell from the bottom of the seventeenth column.



- 25** An alloy contains aluminium, copper, magnesium, manganese, silver and zirconium.

Which row identifies the number of transition elements in the alloy and the relative density of the transition elements compared to sodium?

|          | number of transition elements in the alloy | relative density of transition elements compared to sodium |
|----------|--------------------------------------------|------------------------------------------------------------|
| <b>A</b> | 4                                          | higher                                                     |
| <b>B</b> | 4                                          | lower                                                      |
| <b>C</b> | 5                                          | higher                                                     |
| <b>D</b> | 5                                          | lower                                                      |

- 26** Which statement about the halogens and their compounds is correct?

- A** The colour of the element gets lighter going down Group VII.
- B** The elements get less dense going down Group VII.
- C** When chlorine is added to sodium iodide solution, iodine is formed.
- D** When iodine is added to sodium bromide solution, bromine is formed.

- 27** Which row describes an element in Group VIII of the Periodic Table?

|          | boiling point / °C | structure |
|----------|--------------------|-----------|
| <b>A</b> | -107               | diatomic  |
| <b>B</b> | -107               | monatomic |
| <b>C</b> | 107                | diatomic  |
| <b>D</b> | 107                | monatomic |

- 28** Magnesium reacts with dilute hydrochloric acid.

Which gas is given off in this reaction?

- A** carbon dioxide
- B** chlorine
- C** hydrogen
- D** oxygen

29 Which statement about aluminium is correct?

- A Aluminium is easy to extract from its ore because it is near the bottom of the reactivity series.
- B Aluminium is formed when aluminium oxide is heated with carbon.
- C Bauxite is an important ore of aluminium.
- D Hematite is an important ore of aluminium.

30 Which statement shows that a liquid is pure water?

- A It boils at 100 °C.
- B It has a pH value of 7.
- C It turns blue cobalt(II) chloride pink.
- D It turns white copper(II) sulfate blue.

31 Which compound can be added to ammonium sulfate to make an NPK fertiliser?

- A  $(\text{NH}_4)_3\text{PO}_4$
- B  $\text{KNO}_3$
- C  $\text{K}_3\text{PO}_4$
- D  $\text{CO}(\text{NH}_2)_2$

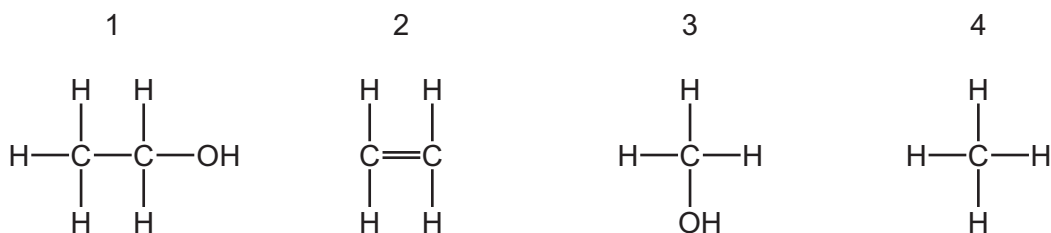
32 Four gases found in air are listed.

- 1 carbon dioxide
- 2 carbon monoxide
- 3 methane
- 4 sulfur dioxide

Which gases lead directly to global warming when their concentrations are increased?

- A 1 and 3      B 1 and 4      C 2 and 3      D 2 and 4

33 The structures of four organic compounds are shown.



Which compounds are members of the same homologous series?

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 3 and 4

34 Which of the statements about propene are correct?

- 1 Propene contains only single bonds.
- 2 Propene decolourises aqueous bromine.
- 3 Propene is obtained by cracking large alkanes.
- 4 Propene is a hydrocarbon.

- A** 1, 3 and 4      **B** 1 only      **C** 2, 3 and 4      **D** 2 and 4 only

35 Petroleum is a mixture.

Which method is used to separate petroleum into its components?

- A** chromatography  
**B** cracking  
**C** filtration  
**D** fractional distillation

36 The equation for a reaction that produces ethanol is shown.



Which type of reaction does the equation represent?

- A** addition  
**B** combustion  
**C** fermentation  
**D** polymerisation

**37** Four statements about ethene or poly(ethene) are listed.

- 1 Poly(ethene) is produced by an addition reaction.
- 2 Ethene is a monomer.
- 3 Poly(ethene) is a monomer.
- 4 Poly(ethene) decolourises aqueous bromine.

Which statements are correct?

- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

**38** Sea water contains dissolved sodium chloride.

Which method is used to obtain pure water from sea water?

- A** chromatography  
**B** distillation  
**C** evaporation  
**D** filtration

**39** Which piece of apparatus is used to measure exactly  $21.4\text{ cm}^3$  of water?

- A** a  $25\text{ cm}^3$  beaker  
**B** a  $25\text{ cm}^3$  pipette  
**C** a  $50\text{ cm}^3$  burette  
**D** a  $50\text{ cm}^3$  measuring cylinder

**40** An aqueous solution of compound M is tested.

The results are shown.

- a lilac colour in a flame test
- a white precipitate when tested with acidified barium nitrate

What is compound M?

- A** copper(II) chloride  
**B** copper(II) sulfate  
**C** potassium carbonate  
**D** potassium sulfate

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# Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)****0971/12**

Paper 1 Multiple Choice (Core)

**May/June 2024**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | D      | 1     |
| 3        | D      | 1     |
| 4        | A      | 1     |
| 5        | D      | 1     |
| 6        | B      | 1     |
| 7        | C      | 1     |
| 8        | A      | 1     |
| 9        | A      | 1     |
| 10       | B      | 1     |
| 11       | D      | 1     |
| 12       | C      | 1     |
| 13       | C      | 1     |
| 14       | D      | 1     |
| 15       | C      | 1     |
| 16       | B      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | B      | 1     |
| 20       | D      | 1     |
| 21       | D      | 1     |
| 22       | B      | 1     |
| 23       | B      | 1     |
| 24       | D      | 1     |
| 25       | A      | 1     |
| 26       | C      | 1     |
| 27       | B      | 1     |
| 28       | C      | 1     |

| <b>Question</b> | <b>Answer</b> | <b>Marks</b> |
|-----------------|---------------|--------------|
| 29              | C             | 1            |
| 30              | A             | 1            |
| 31              | C             | 1            |
| 32              | A             | 1            |
| 33              | C             | 1            |
| 34              | C             | 1            |
| 35              | D             | 1            |
| 36              | A             | 1            |
| 37              | A             | 1            |
| 38              | B             | 1            |
| 39              | C             | 1            |
| 40              | D             | 1            |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The table shows some information about the three states of matter.

|   | particle separation                                     | particle arrangement | type of motion                     |
|---|---------------------------------------------------------|----------------------|------------------------------------|
| 1 | touching with some particles having spaces between them | random               | slide past each other at low speed |
| 2 | particles are far apart                                 | random               | rapid motion in straight lines     |
| 3 | touching with very little space between the particles   | regular              | vibration only                     |

Which row is correct?

|          | 1      | 2      | 3      |
|----------|--------|--------|--------|
| <b>A</b> | gas    | liquid | solid  |
| <b>B</b> | liquid | solid  | gas    |
| <b>C</b> | liquid | gas    | solid  |
| <b>D</b> | solid  | gas    | liquid |

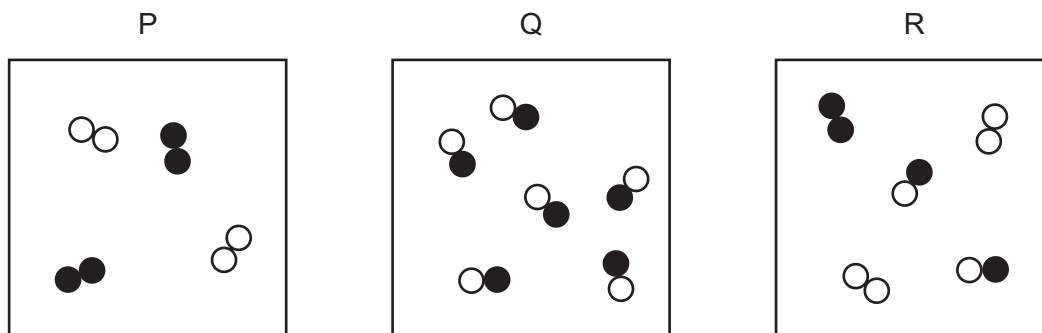
- 2 Which arrow represents evaporation?



- 3 In which states of matter does diffusion occur readily?

- A** gases and liquids
- B** gases only
- C** liquids and solids
- D** solids only

- 4 Which statement about the boxes P, Q and R is correct?



- A** Box P contains two compounds, and box R contains two elements.  
**B** Box P contains two elements, and box Q contains a mixture.  
**C** Box P contains two elements, and box Q contains one compound.  
**D** Box Q contains two compounds, and box R contains a mixture.
- 5 Which information about an element is given by its atomic number?
- A** the number of protons in the nucleus of an atom of an element  
**B** the number of particles in the nucleus of an atom of an element  
**C** the relative mass of one atom of an element  
**D** the total number of particles in one atom of an element
- 6 The symbols represent four atoms. The letters used are **not** the usual atomic symbols.



Which atoms are isotopes of the same element?

- A** W and X      **B** W and Y      **C** X and Y      **D** Y and Z
- 7 Covalent bonds are formed when electrons are .....1..... .

Most covalent compounds have .....2..... electrical conductivity.

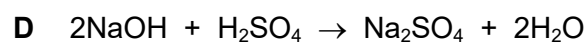
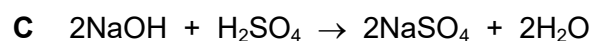
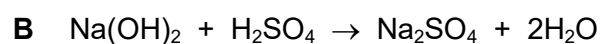
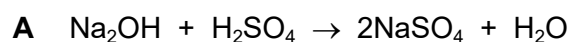
Which words correctly complete gaps 1 and 2?

|          | 1           | 2    |
|----------|-------------|------|
| <b>A</b> | shared      | high |
| <b>B</b> | shared      | low  |
| <b>C</b> | transferred | high |
| <b>D</b> | transferred | low  |

8 Which row describes the structure and a use of diamond?

|          | structure      | use              |
|----------|----------------|------------------|
| <b>A</b> | ionic          | in cutting tools |
| <b>B</b> | ionic          | as a lubricant   |
| <b>C</b> | giant covalent | in cutting tools |
| <b>D</b> | giant covalent | as a lubricant   |

9 Which symbol equation represents the reaction between aqueous sodium hydroxide and dilute sulfuric acid?



10 What is the relative formula mass of magnesium bromide?

**A** 47

**B** 82

**C** 104

**D** 184

11 Three substances are listed.

1 solid copper

2 aqueous sodium bromide

3 solid lead(II) bromide

Which substances conduct electricity?

**A** 1, 2 and 3

**B** 1 and 2 only

**C** 1 and 3 only

**D** 2 and 3 only

12 Hydrogen–oxygen fuel cells can be used to power cars.

Which processes produce the fuel of a hydrogen–oxygen fuel cell?

1 the cracking of hydrocarbons

2 the electrolysis of dilute sulfuric acid

3 photosynthesis

4 the electrolysis of molten aluminium oxide

**A** 1 and 2

**B** 1 and 4

**C** 2 and 3

**D** 3 and 4

- 13** Molten sodium sulfide,  $\text{Na}_2\text{S}$ , is electrolysed using inert electrodes.

Which row identifies the product at each electrode?

|          | cathode  | anode    |
|----------|----------|----------|
| <b>A</b> | sodium   | sulfur   |
| <b>B</b> | sulfur   | sodium   |
| <b>C</b> | hydrogen | sulfur   |
| <b>D</b> | sodium   | hydrogen |

- 14** The temperature of the water in two beakers, X and Y, is measured as  $21.5^\circ\text{C}$ .

5 g of sodium chloride is dissolved in the water in beaker X. The temperature changes to  $18.0^\circ\text{C}$ .

5 g of calcium oxide is dissolved in the water in beaker Y. The temperature changes to  $29.4^\circ\text{C}$ .

Which types of process are occurring in beakers X and Y?

|          | X           | Y           |
|----------|-------------|-------------|
| <b>A</b> | endothermic | endothermic |
| <b>B</b> | endothermic | exothermic  |
| <b>C</b> | exothermic  | endothermic |
| <b>D</b> | exothermic  | exothermic  |

- 15** Which process involves a chemical change?

- A** adding sodium chloride to water
- B** adding magnesium to hydrochloric acid
- C** heating solid iodine until it turns into a gas
- D** melting lead

- 16** Which two pieces of apparatus are most useful to measure the rate of a reaction in which a gas is given off?

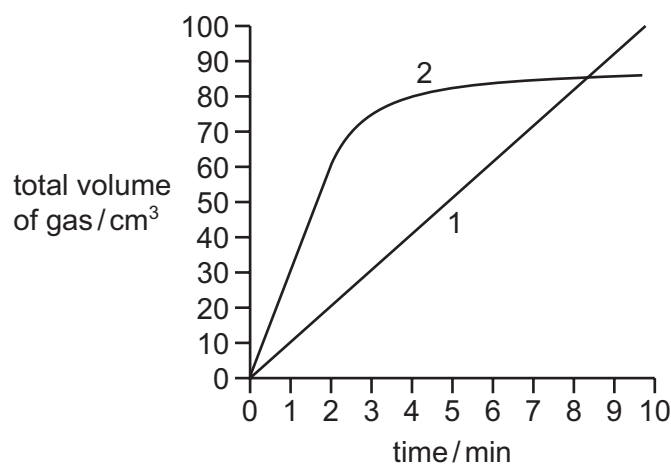
- A** accurate balance and gas syringe
- B** accurate balance and thermometer
- C** gas syringe and stop-watch
- D** stop-watch and thermometer



17 Reaction 1 and reaction 2 both produce a gas.

The total volume of gas produced in each reaction is measured every minute for 10 minutes.

A graph of the results is shown.



Which row describes how the rate of reaction changes, if at all, during each reaction?

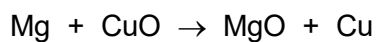
|          | reaction 1           | reaction 2                         |
|----------|----------------------|------------------------------------|
| <b>A</b> | the rate is constant | the rate decreases after 2 minutes |
| <b>B</b> | the rate increases   | the rate increases                 |
| <b>C</b> | the rate increases   | the rate decreases after 2 minutes |
| <b>D</b> | the rate is constant | the rate increases                 |

18 When a few drops of water are added to a solid, E, the colour changes from blue to pink.

What is E?

- A** anhydrous cobalt(II) chloride
- B** anhydrous copper(II) sulfate
- C** hydrated cobalt(II) chloride
- D** hydrated copper(II) sulfate

- 19 The equation for the reaction of magnesium with copper(II) oxide is shown.



Which word describes this reaction?

- A combustion
- B decomposition
- C neutralisation
- D redox

- 20 Compound M contains calcium.

Two reactions of M are listed.

- M reacts with dilute hydrochloric acid to form a salt and water only.
- M reacts with aqueous ammonium chloride to form a gas that turns damp red litmus paper blue.

What is M?

- A CaOH              B  $\text{Ca(OH)}_2$               C  $\text{CaCO}_3$               D  $\text{Ca(CO}_3)_2$
- 21 The diagram shows one period of the Periodic Table.

|    |    |   |   |   |   |   |    |
|----|----|---|---|---|---|---|----|
| Li | Be | B | C | N | O | F | Ne |
|----|----|---|---|---|---|---|----|

Which two elements form acidic oxides?

- A beryllium and lithium
  - B carbon and neon
  - C carbon and nitrogen
  - D nitrogen and neon
- 22 A student tests four solutions with universal indicator.

Which colour identifies the solution containing the greatest concentration of  $\text{OH}^-$  ions?

- A red
- B yellow
- C green
- D blue

**23** The following steps are done to prepare solid magnesium sulfate.

- 1    filtration
- 2    measurement of 20 cm<sup>3</sup> of dilute sulfuric acid using a measuring cylinder
- 3    evaporation
- 4    addition of an excess of solid magnesium carbonate to dilute sulfuric acid

What is the correct order for these steps?

- A**   2 → 4 → 3 → 1
- B**   2 → 4 → 1 → 3
- C**   4 → 2 → 1 → 3
- D**   4 → 2 → 3 → 1

**24** Which statement about the Periodic Table is correct?

- A**   All the metals in the Periodic Table are transition elements.
- B**   The halogens are elements in Group I of the Periodic Table.
- C**   The elements become more metallic across a period from Group I to Group VII.
- D**   The Periodic Table can be used to predict the properties of the elements.

**25** Zinc is formed when zinc oxide is heated with carbon.



Which substance is oxidised in this reaction?

- A**   carbon
- B**   carbon monoxide
- C**   zinc
- D**   zinc oxide

**26** Which word equation represents the rusting of iron?

- A**   iron + oxygen + water → anhydrous iron(II) hydroxide
- B**   iron + oxygen → hydrated iron(II) oxide
- C**   iron + oxygen + water → anhydrous iron(III) hydroxide
- D**   iron + oxygen + water → hydrated iron(III) oxide

**27** Which option describes the electronic configurations of three different elements from the same group of the Periodic Table?

- A** 2    2,2    2,8,8,2
- B** 2    2,8    2,8,2
- C** 2,1    2,8,1    2,8,8,1
- D** 2,1    2,2    2,3

**28** Which metal forms compounds that can be used to colour glass?

- A** aluminium
- B** calcium
- C** chromium
- D** sodium

**29** Two properties of element R are listed.

- It is a dark solid at room temperature.
- It is a diatomic molecule.

Where on the Periodic Table is R placed?

- A** Group I
- B** Group VII
- C** Group VIII
- D** transition elements

**30** Four metals, W, X, Y and Z, are tested with either cold water, steam or both.

The observations are shown.

| metal | observations                                                |
|-------|-------------------------------------------------------------|
| W     | reacts slowly with cold water                               |
| X     | reacts rapidly with cold water                              |
| Y     | does <b>not</b> react with cold water but reacts with steam |
| Z     | does <b>not</b> react with cold water or steam              |

What is the order of reactivity of the metals from the least reactive to the most reactive?

|   | least reactive  most reactive |   |   |   |
|---|----------------------------------------------------------------------------------------------------------------|---|---|---|
| A | W                                                                                                              | X | Y | Z |
| B | W                                                                                                              | Y | X | Z |
| C | Z                                                                                                              | Y | X | W |
| D | Z                                                                                                              | Y | W | X |

**31** Which statement about the displacement reactions of the halogens is correct?

- A** Iodine displaces bromine from aqueous sodium bromide.
- B** Bromine displaces chlorine from aqueous potassium chloride.
- C** Iodine displaces chlorine from aqueous potassium chloride.
- D** Chlorine displaces bromine from aqueous sodium bromide.

**32** Which substances in water from natural sources are beneficial to aquatic animals?

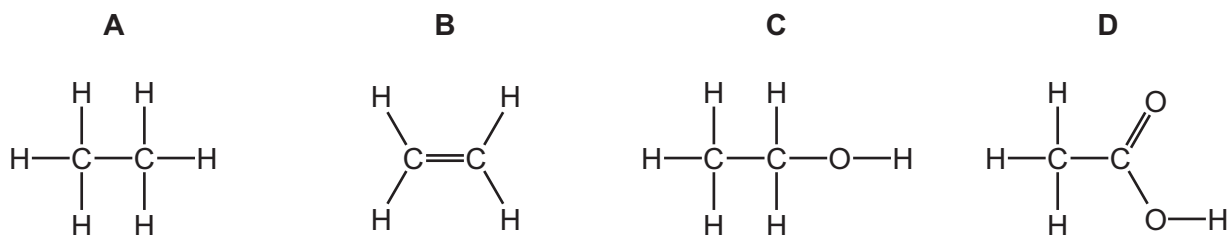
- 1 metal compounds
- 2 plastics
- 3 phosphates
- 4 dissolved oxygen

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

33 What are the products formed when glucose is fermented?

- A ethanol and carbon dioxide
- B ethanol and oxygen
- C ethene and carbon dioxide
- D ethene and oxygen

34 Which structure represents a molecule of ethanol?



35 Which statement describes a homologous series?

- A a family of elements in the same group of the Periodic Table
- B a family of elements with similar chemical properties
- C a family of compounds with the same functional group
- D a family of compounds with similar physical properties

36 What are the properties of aqueous ethanoic acid?

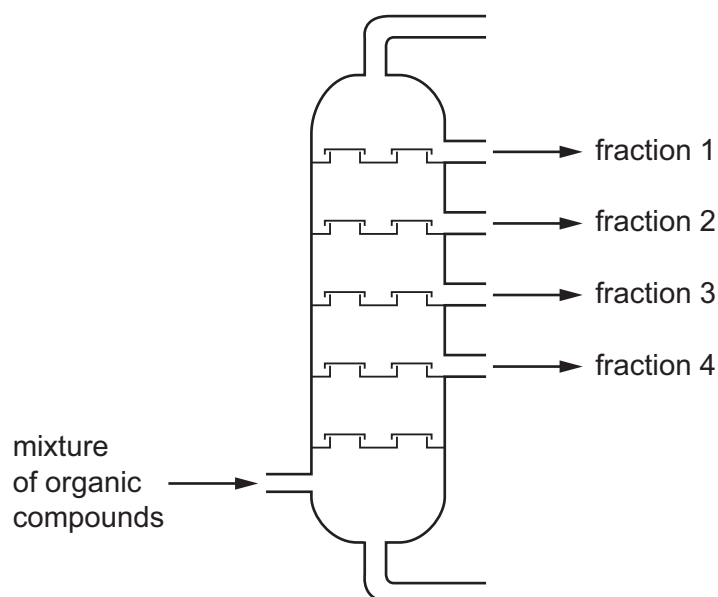
|          | decolourises aqueous bromine | reacts with calcium carbonate to make carbon dioxide | turns damp red litmus paper blue |
|----------|------------------------------|------------------------------------------------------|----------------------------------|
| <b>A</b> | ✓                            | ✓                                                    | x                                |
| <b>B</b> | ✓                            | x                                                    | ✓                                |
| <b>C</b> | x                            | ✓                                                    | x                                |
| <b>D</b> | x                            | x                                                    | ✓                                |

**37** Four different organic compounds are separated by a fractionating column.

The table shows the boiling points of the compounds.

The diagram shows the position in the fractionating column where they are separated.

| compound | boiling point / °C |
|----------|--------------------|
| Q        | 69                 |
| R        | 196                |
| S        | 90                 |
| T        | 125                |



Which row identifies the compound in each fraction?

|          | fraction 1 | fraction 2 | fraction 3 | fraction 4 |
|----------|------------|------------|------------|------------|
| <b>A</b> | Q          | S          | T          | R          |
| <b>B</b> | Q          | T          | S          | R          |
| <b>C</b> | R          | T          | S          | Q          |
| <b>D</b> | R          | S          | T          | Q          |

**38** Which piece of apparatus is used to measure exactly 21.50 cm<sup>3</sup> of dilute sulfuric acid?

- A** beaker
- B** burette
- C** measuring cylinder
- D** volumetric pipette

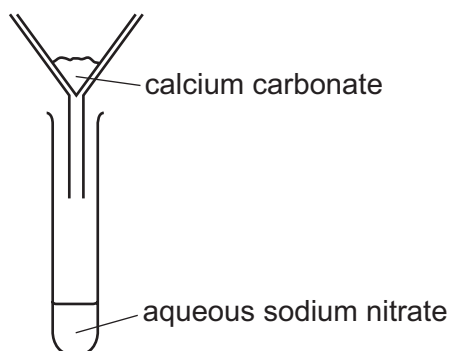
39 Which row shows an advantage and a disadvantage for the stated apparatus used in a titration?

|          | apparatus                             | advantage                                     | disadvantage                                                   |
|----------|---------------------------------------|-----------------------------------------------|----------------------------------------------------------------|
| <b>A</b> | 25 cm <sup>3</sup> volumetric pipette | measures volume accurately                    | can only be used to measure 25 cm <sup>3</sup> of solution     |
| <b>B</b> | 50 cm <sup>3</sup> burette            | measures volume accurately                    | can only be used to measure 50 cm <sup>3</sup> of solution     |
| <b>C</b> | 100 cm <sup>3</sup> beaker            | suitable for filling burette                  | can only be used to fill a 100 cm <sup>3</sup> burette         |
| <b>D</b> | 250 cm <sup>3</sup> conical flask     | allows solutions to be mixed without spilling | not suitable for holding volumes less than 250 cm <sup>3</sup> |

40 Sample M contains calcium carbonate and sodium nitrate.

The result of adding water to M, stirring and filtering is shown.

No chemical reaction occurs.



Which terms describe M, calcium carbonate and aqueous sodium nitrate?

|          | sample M | calcium carbonate | aqueous sodium nitrate |
|----------|----------|-------------------|------------------------|
| <b>A</b> | compound | filtrate          | residue                |
| <b>B</b> | compound | residue           | filtrate               |
| <b>C</b> | mixture  | filtrate          | residue                |
| <b>D</b> | mixture  | residue           | filtrate               |



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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|------------------------------|-----------------------------|------------------------|
| I                          | II                          |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                          | V                           | VI                            | VII                          | VIII                        |                        |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 1<br>H<br>hydrogen<br>1     | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12      | 7<br>N<br>nitrogen<br>14      | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19    | 10<br>Ne<br>neon<br>20 |
|                            |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
|                            |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28   | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32       | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40     |                        |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79    | 35<br>Br<br>bromine<br>80    | 36<br>Kr<br>krypton<br>84   |                        |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127     | 54<br>Xe<br>xenon<br>131    |                        |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—     | 85<br>At<br>astatine<br>—    | 86<br>Rn<br>radon<br>—      |                        |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>—  | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesson<br>— |                        |

lanthanoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

actinoids

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).

# Cambridge IGCSE™ (9–1)

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**CHEMISTRY****0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2024****MARK SCHEME**Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | B      | 1     |
| 3        | A      | 1     |
| 4        | C      | 1     |
| 5        | A      | 1     |
| 6        | B      | 1     |
| 7        | B      | 1     |
| 8        | C      | 1     |
| 9        | D      | 1     |
| 10       | D      | 1     |
| 11       | B      | 1     |
| 12       | A      | 1     |
| 13       | A      | 1     |
| 14       | B      | 1     |
| 15       | B      | 1     |
| 16       | C      | 1     |
| 17       | A      | 1     |
| 18       | A      | 1     |
| 19       | D      | 1     |
| 20       | B      | 1     |
| 21       | C      | 1     |
| 22       | D      | 1     |
| 23       | B      | 1     |
| 24       | D      | 1     |
| 25       | A      | 1     |
| 26       | D      | 1     |
| 27       | C      | 1     |
| 28       | C      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | B      | 1     |
| 30       | D      | 1     |
| 31       | D      | 1     |
| 32       | B      | 1     |
| 33       | A      | 1     |
| 34       | C      | 1     |
| 35       | C      | 1     |
| 36       | C      | 1     |
| 37       | A      | 1     |
| 38       | B      | 1     |
| 39       | A      | 1     |
| 40       | D      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/11

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

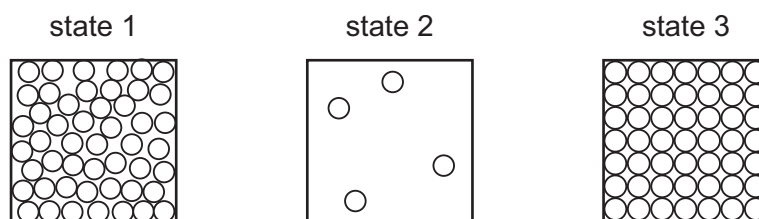
This document has **16** pages. Any blank pages are indicated.



1 Which process happens when water vapour changes to rain?

- A boiling
- B condensing
- C evaporating
- D freezing

2 The diagrams show the arrangement of particles in three different states of matter.



Which row describes the change in energy of the particles and in particle motion for the given change in state?

|          | change in state | energy of particles | particle motion |
|----------|-----------------|---------------------|-----------------|
| <b>A</b> | 1 → 2           | decreases           | decreases       |
| <b>B</b> | 2 → 1           | decreases           | increases       |
| <b>C</b> | 3 → 1           | increases           | increases       |
| <b>D</b> | 1 → 3           | increases           | decreases       |

3 An atom of element Q contains 19 electrons, 19 protons and 20 neutrons.

What is Q?

- A calcium
- B potassium
- C strontium
- D yttrium

4 Which part of an atom has a relative mass of 1 and a relative charge of 0?

- A electron
- B neutron
- C nucleus
- D proton



- 5 Which row identifies the number of electrons, neutrons and protons in a particle which is an isotope of  $^{11}_5\text{B}$ ?

|          | electrons | neutrons | protons |
|----------|-----------|----------|---------|
| <b>A</b> | 5         | 5        | 5       |
| <b>B</b> | 5         | 6        | 5       |
| <b>C</b> | 6         | 5        | 6       |
| <b>D</b> | 6         | 6        | 6       |

- 6 Which statements about potassium iodide are correct?

- 1 It is formed from potassium anions and iodide cations.
- 2 It is a good electrical conductor when molten or in aqueous solution.
- 3 Potassium atoms share electrons with iodine atoms.

**A** 1 and 3      **B** 1 only      **C** 2 and 3      **D** 2 only

- 7 Which substances contain one or more shared pairs of electrons?

- 1 argon
- 2 methane
- 3 iron(III) oxide
- 4 chlorine

**A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

- 8 Which substance has a giant covalent structure at room temperature and pressure?

- A** ammonia  
**B** carbon dioxide  
**C** diamond  
**D** water

- 9 Which row shows the correct formula for the named substance?

|          | substance           | formula                  |
|----------|---------------------|--------------------------|
| <b>A</b> | cobalt(II) chloride | $\text{Cu}_2\text{Cl}$   |
| <b>B</b> | sodium carbonate    | $\text{Na}_2\text{CO}_3$ |
| <b>C</b> | xenon               | $\text{Xe}_2$            |
| <b>D</b> | ammonium sulfate    | $\text{NH}_4\text{SO}_4$ |

- 10 The equation shows the thermal decomposition of magnesium carbonate.



[ $M_r$ :  $\text{MgCO}_3$ , 84]

Which mass of magnesium oxide is formed when 21.0 g of magnesium carbonate is completely decomposed?

- A** 1.9 g                      **B** 4.0 g                      **C** 10.0 g                      **D** 40.0 g
- 11 Concentrated aqueous sodium chloride is electrolysed using inert electrodes.

What is the main product formed at the positive electrode (anode)?

- A** chlorine  
**B** hydrogen  
**C** oxygen  
**D** sodium

- 12 A hydrogen–oxygen fuel cell uses  $630 \text{ dm}^3$  of oxygen.

The oxygen for the reaction is extracted from clean, dry air.

What is the minimum volume of clean, dry air needed to provide this volume of oxygen?

- A**  $788 \text{ dm}^3$                       **B**  $808 \text{ dm}^3$                       **C**  $3000 \text{ dm}^3$                       **D**  $3316 \text{ dm}^3$

**13** Three statements about energy changes in chemical reactions are listed.

- 1 In an endothermic reaction, the temperature of the surroundings increases.
- 2 In an exothermic reaction, thermal energy is taken in from the surroundings.
- 3 In the reaction pathway diagram for an exothermic reaction, the energy level of the products is lower than the energy level of the reactants.

Which statements are correct?

- A** 1 and 2      **B** 1 only      **C** 2 and 3      **D** 3 only

**14** When a small piece of a Group I metal is placed into a large beaker of cold water, a reaction occurs.

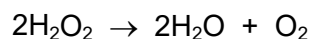
Four statements about this reaction are listed.

- 1 The metal melts.
- 2 Hydrogen is produced.
- 3 Steam is produced.
- 4 The pH of the solution increases.

Which statements about this reaction describe a physical change?

- A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

**15** The equation for the decomposition of hydrogen peroxide is shown.



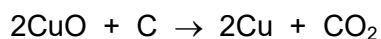
The reaction is exothermic.

When a small amount of a catalyst is added, the oxygen is produced more quickly.

Which statement about the catalyst is correct?

- A** The catalyst makes the reaction more exothermic.
- B** The mass of catalyst is the same before and after the reaction.
- C** The catalyst increases the final volume of oxygen produced.
- D** All of the catalyst is used up in the reaction.

- 16 The equation for the reaction between copper(II) oxide and carbon is shown.



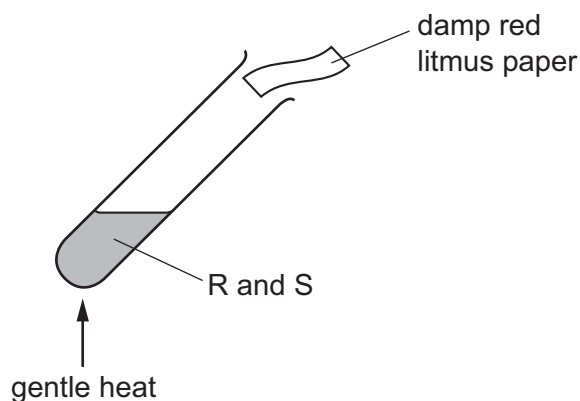
Which statement about this reaction is correct?

- A** CuO is reduced.  
**B** CO<sub>2</sub> is oxidised.  
**C** Cu is oxidised.  
**D** C is reduced.
- 17 Which row gives the colours observed when thymolphthalein and methyl orange are added separately to the named solution?

|          | solution     | colour with thymolphthalein | colour with methyl orange |
|----------|--------------|-----------------------------|---------------------------|
| <b>A</b> | dilute HCl   | colourless                  | yellow                    |
| <b>B</b> | dilute HCl   | blue                        | red                       |
| <b>C</b> | aqueous NaOH | colourless                  | red                       |
| <b>D</b> | aqueous NaOH | blue                        | yellow                    |

- 18 A mixture of two substances, R and S, is heated gently.

The damp red litmus paper turns blue.



What are R and S?

|          | R               | S                 |
|----------|-----------------|-------------------|
| <b>A</b> | a basic oxide   | ammonium chloride |
| <b>B</b> | a basic oxide   | sodium nitrate    |
| <b>C</b> | an acidic oxide | ammonium chloride |
| <b>D</b> | an acidic oxide | sodium nitrate    |

**19** Which reactants are used to make the salt copper(II) sulfate?

- A** dilute acid + alkali  
**B** dilute acid + metal carbonate  
**C** dilute acid + metal  
**D** dilute acid + non-metal oxide

**20** Which statement about the Periodic Table is correct?

- A** The elements are arranged in order of increasing relative atomic mass.
- B** The reactivity of the elements in Group I and in Group VII increases as the groups are descended.
- C** Elements in the same period have similar chemical properties.
- D** Elements in Group II form ions with a 2+ charge.

**21** Part of the Periodic Table is shown.

Which element shows the most metallic character?

A blank periodic table grid is shown, consisting of a 4x16 grid of squares. The grid is divided into four main sections by a large gap in the middle. The sections are labeled A, B, C, and D. Section A is a 2x2 grid in the top-left. Section B is a 2x4 grid in the top-right. Section C is a 2x2 grid in the bottom-right. Section D is a 2x2 grid in the bottom-right. The labels A, B, C, and D are placed in the bottom-right corner of each section's top row.

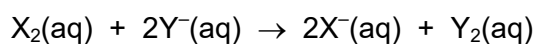
**22** E is an element from Group I of the Periodic Table. Two properties of E are listed.

- It has a higher melting point than caesium.
- It has a lower density than sodium.

What is the identity of E?

- A** lithium  
**B** potassium  
**C** rubidium  
**D** francium

- 23** An equation for the displacement reaction between aqueous halogen,  $X_2$ , and aqueous halide ions,  $Y^-$ , is shown.



Which row identifies  $X_2$  and  $Y^-$  and explains why the reaction takes place?

|          | $X_2$    | $Y^-$    | explanation                           |
|----------|----------|----------|---------------------------------------|
| <b>A</b> | chlorine | iodide   | chlorine is less reactive than iodine |
| <b>B</b> | chlorine | iodide   | chlorine is more reactive than iodine |
| <b>C</b> | iodine   | chloride | chlorine is less reactive than iodine |
| <b>D</b> | iodine   | chloride | chlorine is more reactive than iodine |

- 24** Some information about an element is shown.

|                    |               |
|--------------------|---------------|
| melting point / °C | 1555          |
| boiling point / °C | 2963          |
| colour of oxide    | brown         |
| use of element     | as a catalyst |

What is the position of this element in the Periodic Table?

- A** Group I
- B** Group VII
- C** Group VIII
- D** transition elements

- 25 The table shows the observations when four metals, Q, R, S and T, are added separately to cold water and to dilute hydrochloric acid.

| metal | observation with cold water | observation with dilute hydrochloric acid |
|-------|-----------------------------|-------------------------------------------|
| Q     | slow fizzing                | fizzing                                   |
| R     | no reaction                 | fizzing                                   |
| S     | no reaction                 | no reaction                               |
| T     | vigorous fizzing            | vigorous fizzing                          |

Which row gives the order of reactivity of the metals?

|          | least reactive | → |   |   | most reactive |
|----------|----------------|---|---|---|---------------|
| <b>A</b> | S              | Q | R | T |               |
| <b>B</b> | T              | R | Q | S |               |
| <b>C</b> | S              | R | Q | T |               |
| <b>D</b> | T              | Q | R | S |               |

- 26 Magnesium is reacted separately with dilute sulfuric acid and with steam.

Which row correctly identifies if hydrogen is formed as a product in each reaction?

|          | reaction with dilute sulfuric acid | reaction with steam |
|----------|------------------------------------|---------------------|
| <b>A</b> | ✓                                  | ✓                   |
| <b>B</b> | ✓                                  | ✗                   |
| <b>C</b> | ✗                                  | ✓                   |
| <b>D</b> | ✗                                  | ✗                   |

key

✓ = hydrogen is formed

✗ = hydrogen is **not** formed

- 27 Which statements about aluminium are correct?

- 1 It is more reactive than calcium.
- 2 The main ore of aluminium is bauxite.
- 3 It can be extracted from its oxide using carbon.
- 4 Brass is an alloy of aluminium and copper.

**A** 1 and 2

**B** 1 and 3

**C** 2 only

**D** 3 and 4

**28** Steel is a mixture of iron and one or more other elements.

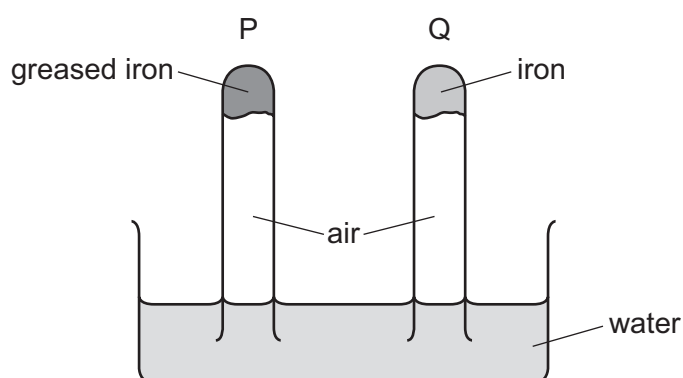
The table gives some information about three common types of steel.

|   | type of steel     | elements added to iron      | properties                              |
|---|-------------------|-----------------------------|-----------------------------------------|
| 1 | high-carbon steel | carbon only                 | strong, brittle and corrodes            |
| 2 | low-carbon steel  | carbon only                 | soft, easily shaped and corrodes slowly |
| 3 | stainless steel   | carbon, chromium and nickel | hard and resistant to corrosion         |

Which rows identify a type of steel that is suitable to make cutlery?

- A** 1 and 2      **B** 1 only      **C** 2 and 3      **D** 3 only

**29** The diagram shows an experiment to investigate the corrosion of iron.



What happens to the water level in tubes P and Q?

|          | tube P    | tube Q    |
|----------|-----------|-----------|
| <b>A</b> | rises     | falls     |
| <b>B</b> | no change | rises     |
| <b>C</b> | falls     | rises     |
| <b>D</b> | no change | no change |



**30** Fertilisers are used to provide the three elements needed for improved plant growth.

Which two compounds would provide **all three** of these elements?

- A** ammonium nitrate and calcium phosphate
- B** ammonium nitrate and potassium sulfate
- C** potassium nitrate and calcium phosphate
- D** potassium phosphate and potassium sulfate

**31** Which substances can be used to detect the presence of water?

- 1 anhydrous cobalt(II) chloride
- 2 anhydrous copper(II) sulfate
- 3 litmus
- 4 methyl orange

- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

**32** What is produced by the incomplete combustion of methane?

- A** carbon monoxide
- B** hydrogen
- C** lead compounds
- D** sulfur dioxide

**33** Which row identifies compounds in the same homologous series?

|          | chemical properties | functional group |
|----------|---------------------|------------------|
| <b>A</b> | different           | different        |
| <b>B</b> | different           | same             |
| <b>C</b> | similar             | different        |
| <b>D</b> | similar             | same             |

**34** The molecular formula of compound Z is  $C_4H_{10}$ .

Which row identifies the substance that reacts with Z and describes the type of reaction that occurs?

|          | substance | type of reaction |
|----------|-----------|------------------|
| <b>A</b> | chlorine  | addition         |
| <b>B</b> | chlorine  | substitution     |
| <b>C</b> | steam     | addition         |
| <b>D</b> | steam     | substitution     |

**35** Which list shows the fractions obtained from the fractional distillation of petroleum, in order of increasing boiling point?

- A** bitumen → diesel oil → fuel oil → lubricating oil
- B** diesel oil → gasoline → naphtha → kerosene
- C** gasoline → naphtha → kerosene → diesel oil
- D** kerosene → lubricating oil → naphtha → refinery gas

**36** Three statements about cracking of larger alkane molecules are listed.

- 1 Cracking produces petrol for cars.
- 2 Cracking only produces short-chain alkenes.
- 3 Cracking produces alkenes used to make polymers.

Which statements are correct?

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**37** Some words used to describe organic compounds are listed.

- 1 hydrocarbon
- 2 monomer
- 3 saturated
- 4 unreactive

Which words describe ethene?

- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

- 38** A student measures  $25.00\text{ cm}^3$  of dilute hydrochloric acid accurately.

Which piece of apparatus is most suitable?

- A** beaker
- B** measuring cylinder
- C** burette
- D** dropping pipette

- 39** Excess solid magnesium oxide is added to dilute nitric acid.

Which separation technique is used to remove the excess solid magnesium oxide after the reaction finishes?

- A** chromatography
- B** crystallisation
- C** distillation
- D** filtration

- 40** Four different aqueous metal nitrates are tested separately with aqueous sodium hydroxide, with dilute sulfuric acid and with a flame test.

Which row shows the correct set of results for the named aqueous metal nitrate?

|          | aqueous metal nitrate | with aqueous sodium hydroxide | with dilute sulfuric acid | flame test       |
|----------|-----------------------|-------------------------------|---------------------------|------------------|
| <b>A</b> | sodium nitrate        | no visible change             | white precipitate         | yellow flame     |
| <b>B</b> | copper(II) nitrate    | blue precipitate              | no visible change         | blue-green flame |
| <b>C</b> | barium nitrate        | white precipitate             | no visible change         | blue-green flame |
| <b>D</b> | calcium nitrate       | no visible change             | white precipitate         | yellow flame     |

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The Periodic Table of Elements

| Group                      |                             |                                                                       |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                            |
|----------------------------|-----------------------------|-----------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|------------------------------|----------------------------|
| I                          | II                          | 1<br>H<br>hydrogen<br>1                                               |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                          | V                           | VI                            | VII                          | VIII                       |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | Key<br>atomic number<br>atomic symbol<br>name<br>relative atomic mass |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12      | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16        | 9<br>F<br>fluorine<br>19     | 10<br>Ne<br>neon<br>20     |
|                            |                             |                                                                       |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                            |
|                            |                             |                                                                       |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                            |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 |                                                                       |                                 |                             |                              |                             |                              |                              |                                |                               | 13<br>Al<br>aluminium<br>27   | 14<br>Si<br>silicon<br>28   | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32     | 17<br>Cl<br>chlorine<br>35.5  | 18<br>Ar<br>argon<br>40      |                            |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                            | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79    | 35<br>Br<br>bromine<br>80    | 36<br>Kr<br>krypton<br>84  |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                              | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127     | 54<br>Xe<br>xenon<br>131   |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                  | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—     | 85<br>At<br>astatine<br>—    | 86<br>Rn<br>radon<br>—     |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                   | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>—  | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganeson<br>— |

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).

# Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)****0971/11**

Paper 1 Multiple Choice (Core)

**May/June 2025**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | B      | 1     |
| 2        | C      | 1     |
| 3        | B      | 1     |
| 4        | B      | 1     |
| 5        | A      | 1     |
| 6        | D      | 1     |
| 7        | D      | 1     |
| 8        | C      | 1     |
| 9        | B      | 1     |
| 10       | C      | 1     |
| 11       | A      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | A      | 1     |
| 15       | B      | 1     |
| 16       | A      | 1     |
| 17       | D      | 1     |
| 18       | A      | 1     |
| 19       | B      | 1     |
| 20       | D      | 1     |
| 21       | A      | 1     |
| 22       | A      | 1     |
| 23       | B      | 1     |
| 24       | D      | 1     |
| 25       | C      | 1     |
| 26       | A      | 1     |
| 27       | C      | 1     |
| 28       | D      | 1     |



| Question | Answer | Marks |
|----------|--------|-------|
| 29       | B      | 1     |
| 30       | C      | 1     |
| 31       | A      | 1     |
| 32       | A      | 1     |
| 33       | D      | 1     |
| 34       | B      | 1     |
| 35       | C      | 1     |
| 36       | C      | 1     |
| 37       | A      | 1     |
| 38       | C      | 1     |
| 39       | D      | 1     |
| 40       | B      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

Paper 1 Multiple Choice (Core)

0971/12

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Substance L melts at  $-7^{\circ}\text{C}$  and is a brown liquid at room temperature.

What is the boiling point of pure L?

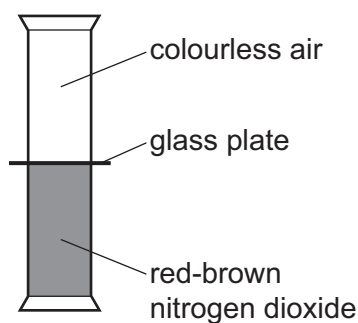
- A**  $-77^{\circ}\text{C}$   
**B**  $-7^{\circ}\text{C}$  to  $+7^{\circ}\text{C}$   
**C**  $59^{\circ}\text{C}$   
**D**  $107^{\circ}\text{C}$  to  $117^{\circ}\text{C}$

- 2 Which row describes how the volume of a gas changes when its temperature is increased at constant pressure and when its pressure is increased at constant temperature?

|          | temperature is increased<br>at constant pressure | pressure is increased<br>at constant temperature |
|----------|--------------------------------------------------|--------------------------------------------------|
| <b>A</b> | volume decreases                                 | volume decreases                                 |
| <b>B</b> | volume decreases                                 | volume increases                                 |
| <b>C</b> | volume increases                                 | volume decreases                                 |
| <b>D</b> | volume increases                                 | volume increases                                 |

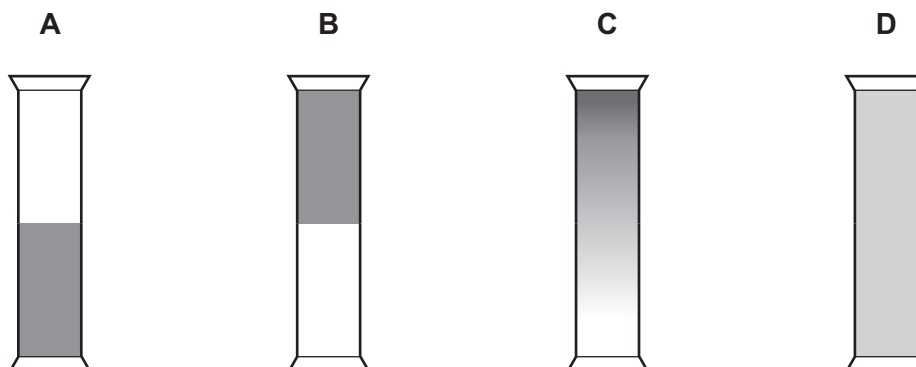
- 3 Nitrogen dioxide is a red-brown gas which is more dense than air.

The diagram shows the arrangement of two gas jars which contain nitrogen dioxide and air separated with a glass plate.



The glass plate is removed, and the gas jars are left for 24 hours.

Which diagram shows the gas jars after 24 hours?



4 Which substance is a mixture of **one** element and **one** compound?

- A aqueous glucose
- B clean, dry air
- C oxygen dissolved in distilled water
- D stainless steel

5 Which statement defines the atomic number of an element?

- A the group number of the element
- B the mass of one atom of the element
- C the number of particles in the nucleus of one atom of the element
- D the number of protons in the nucleus of one atom of the element

6 What is different for isotopes of the same element?

- A nucleon number
- B number of electron shells
- C number of electrons in the outer electron shell
- D proton number

7 When sodium chloride is formed from its elements, each chlorine atom .....1..... one .....2..... .

Which words correctly complete gaps 1 and 2?

|          | 1     | 2        |
|----------|-------|----------|
| <b>A</b> | gains | electron |
| <b>B</b> | gains | proton   |
| <b>C</b> | loses | electron |
| <b>D</b> | loses | proton   |

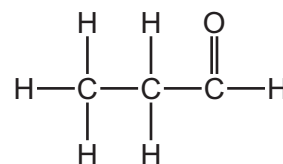
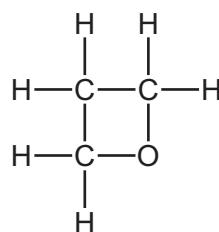
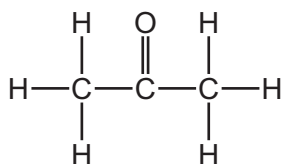
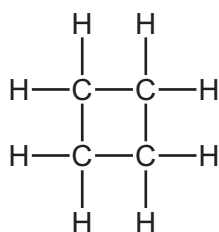
8 In the molecules  $\text{CH}_4$ ,  $\text{HCl}$  and  $\text{H}_2\text{O}$ , which atoms use **all** of their outer shell electrons in bonding?

- A C and Cl
- B C and H
- C Cl and H
- D H and O

9 Which substance has a giant covalent structure?

- A sodium chloride
- B ammonia
- C magnesium chloride
- D graphite

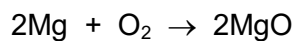
10 The diagrams show the structures of four molecules.



Which statement about these molecules is correct?

- A All four molecules have different molecular formulae.
- B All four molecules have the same molecular formula.
- C Only two molecules have the same molecular formula.
- D Only three molecules have the same molecular formula.

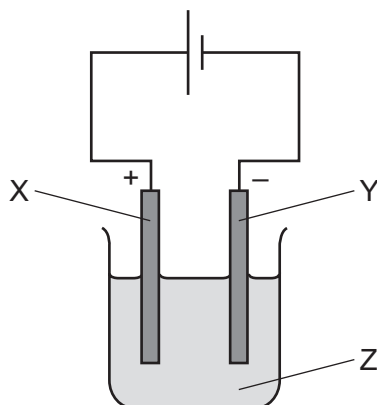
11 The equation for the reaction of magnesium with oxygen is shown.



Which mass of oxygen reacts exactly with 6 g of magnesium?

- A 2 g
- B 4 g
- C 8 g
- D 16 g

12 The diagram shows a simple electrolytic cell for an electrolysis experiment.



Three parts of the electrolytic cell are labelled X, Y and Z.

Which row shows the correct labels to replace X, Y and Z in the diagram?

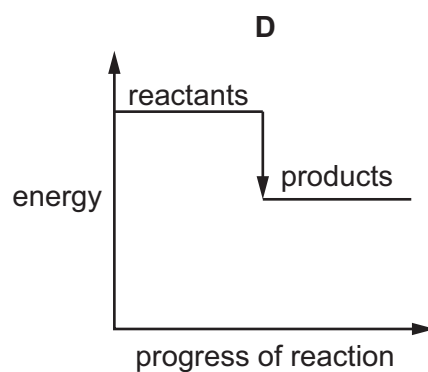
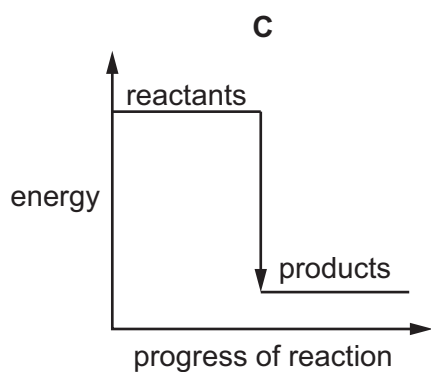
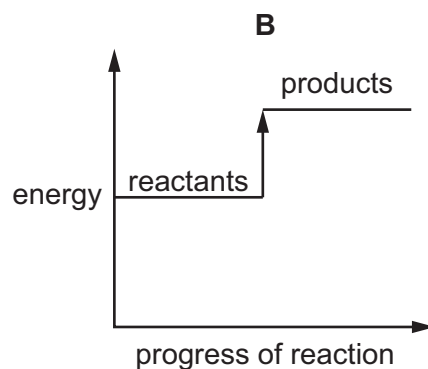
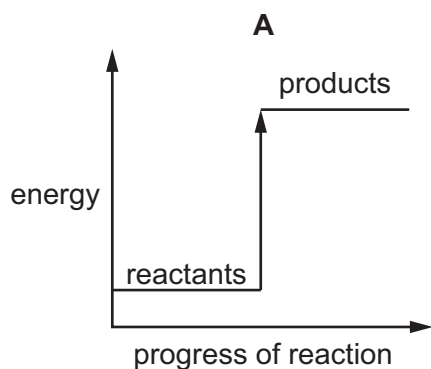
|          | X       | Y       | Z            |
|----------|---------|---------|--------------|
| <b>A</b> | anode   | cathode | aqueous salt |
| <b>B</b> | anode   | cathode | solid salt   |
| <b>C</b> | cathode | anode   | aqueous salt |
| <b>D</b> | cathode | anode   | solid salt   |

13 Which equation represents the overall reaction that produces electricity in a hydrogen–oxygen fuel cell?

- A**  $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$
- B**  $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$
- C**  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- D**  $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$

- 14 Which reaction pathway diagram shows the largest increase in the temperature of the surroundings?

(The scale on the y-axis is the same in each diagram.)



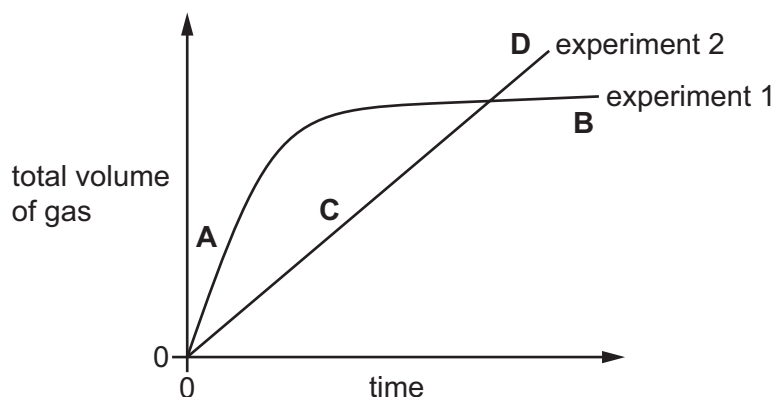
- 15 Which process involves a chemical change?

- A** dissolving copper(II) sulfate
- B** distilling ethanol
- C** freezing water
- D** neutralising copper(II) oxide

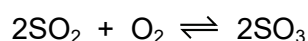
- 16 The total volume of gas produced in two different experiments is measured against time.

A graph of the results of the two experiments is shown.

Which point shows the fastest rate of reaction?



- 17 The equation for the reaction of sulfur dioxide with oxygen is shown.



What is meant by the symbol  $\rightleftharpoons$ ?

- A** redox reaction  
**B** reversible reaction  
**C** displacement reaction  
**D** thermal decomposition reaction
- 18 Which reactions are redox reactions?
- 1 the incomplete combustion of propane
  - 2 the rusting of iron
  - 3 the reaction of aqueous chloride ions with acidified aqueous silver nitrate
  - 4 the thermal decomposition of calcium carbonate
- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4
- 19 Which row describes an aqueous solution for the pH shown?

|          | pH | description                      |
|----------|----|----------------------------------|
| <b>A</b> | 1  | has a very high alkalinity       |
| <b>B</b> | 5  | is neutral                       |
| <b>C</b> | 9  | turns universal indicator yellow |
| <b>D</b> | 14 | contains $\text{OH}^-$ ions      |



20 Which parts of the Periodic Table contain elements that **only** form basic oxides?

- A Group I and Period 3
- B Group I only
- C Group VIII and Period 3
- D Group VIII only

21 Which statement about elements in the Periodic Table is correct?

- A Elements in the same group of the Periodic Table have the same electronic configuration.
- B The atoms of the elements in Group VII form ions with a charge equal to the number of outer shell electrons minus 8.
- C The metallic character of the elements increases from left to right across a period.
- D The Periodic Table is an arrangement of elements in order of increasing nucleon number.

22 Which row identifies the trends in properties of the elements lithium to sodium to potassium?

|   | melting point | density    | reactivity |
|---|---------------|------------|------------|
| A | decreasing    | increasing | increasing |
| B | increasing    | increasing | decreasing |
| C | increasing    | decreasing | increasing |
| D | decreasing    | decreasing | increasing |

23 Which statements about the halogens are correct?

- 1 They are all diatomic molecules.
- 2 They are all gases at room temperature and pressure.
- 3 The density of the elements increases down the group.
- 4 The reactivity of the elements increases down the group.

- A 1 and 2      B 1 and 3      C 2 and 4      D 3 and 4

24 Zirconium is a transition element.

What is a property of zirconium?

- A high density
- B does **not** conduct heat
- C forms white compounds only
- D low melting point

- 25** Some gases exist as single atoms, are unreactive and have eight electrons in their outer electron shell.

Which gases fit this description?

- 1 helium
- 2 argon
- 3 oxygen
- 4 krypton

**A** 1, 2 and 4      **B** 1 and 3      **C** 2 and 4 only      **D** 3 only

- 26** Which statements about metals are correct?

- 1 Calcium reacts faster than magnesium with cold water.
- 2 Aluminium is more easily extracted than iron from its ore.
- 3 Copper reacts faster than iron with dilute hydrochloric acid.
- 4 Potassium and silver are both good conductors of electricity.

**A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

- 27** Four metals, P, Q, R and S, are separately reacted with dilute hydrochloric acid.

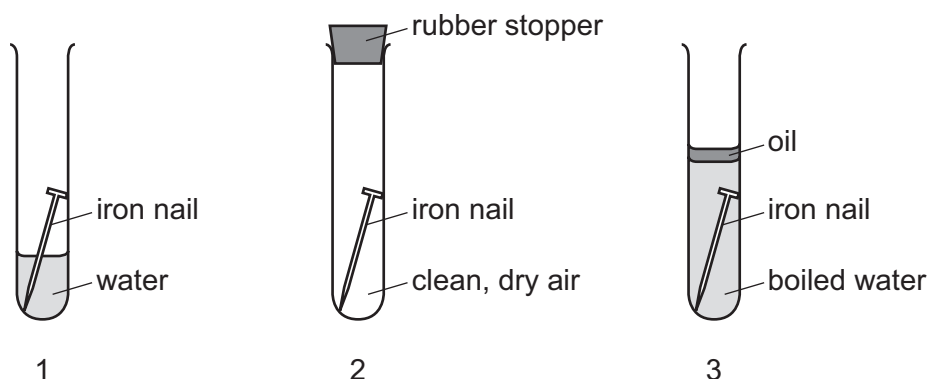
The table shows the observation for each metal.

| metal | observation               |
|-------|---------------------------|
| P     | lots of bubbles produced  |
| Q     | few bubbles produced      |
| R     | no bubbles produced       |
| S     | very few bubbles produced |

What is the order of reactivity of these four metals, from most reactive to least reactive?

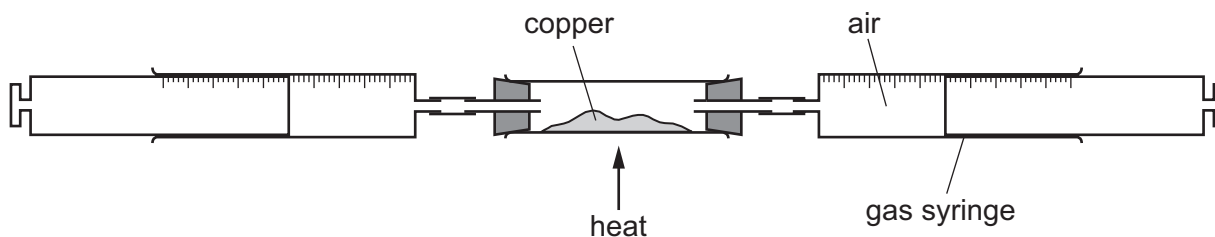
- A**  $R \rightarrow S \rightarrow Q \rightarrow P$
- B**  $P \rightarrow S \rightarrow Q \rightarrow R$
- C**  $P \rightarrow Q \rightarrow S \rightarrow R$
- D**  $Q \rightarrow R \rightarrow P \rightarrow S$

- 28 The diagram shows iron nails which are left for one week in separate test-tubes with the conditions shown.



Which test-tubes show evidence of rusting after one week?

- A** 1 and 3      **B** 1 only      **C** 2 and 3      **D** 2 only
- 29 Pure metals can be combined to make alloys.
- Why are the alloys sometimes used rather than the metals they are made from?
- A** Alloys are easier to extract and cheaper.  
**B** Alloys are easier to hammer into different shapes.  
**C** Alloys are harder and keep their shape better.  
**D** Alloys contain only atoms of the same size but pure metals do **not**.
- 30 In the experiment shown,  $100\text{ cm}^3$  of clean, dry air is passed backwards and forwards through a tube containing heated copper until there is no further decrease in the volume of the air.



All of the gas is pushed into **one** syringe.

What is the final reading on this gas syringe?

- A**  $0\text{ cm}^3$       **B**  $21\text{ cm}^3$       **C**  $79\text{ cm}^3$       **D**  $100\text{ cm}^3$

- 31 Sulfur dioxide, carbon monoxide and oxides of nitrogen are gaseous pollutants found in the air.

Which pollutants contribute to acid rain?

- A carbon monoxide and sulfur dioxide
- B oxides of nitrogen and sulfur dioxide
- C oxides of nitrogen only
- D sulfur dioxide only

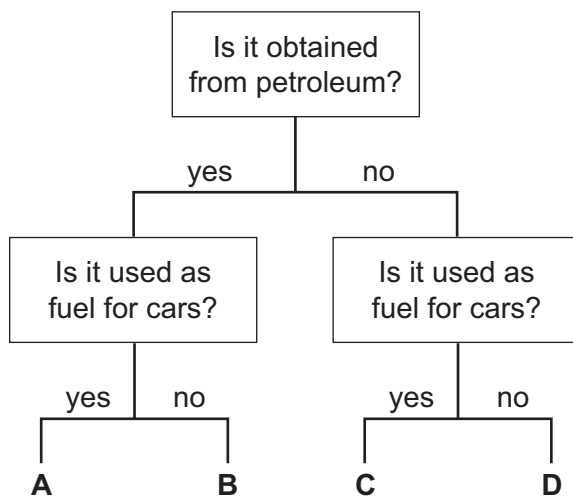
- 32 The formulae for three organic compounds are listed.

- 1  $\text{C}_2\text{H}_4$
- 2  $\text{C}_4\text{H}_8$
- 3  $\text{C}_4\text{H}_{10}$

Which compounds belong to the same homologous series?

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

- 33 Which fuel could be gasoline?



34 Three hydrocarbons are separately mixed with aqueous bromine.

Which row gives the correct observation for each hydrocarbon?

|          | $\begin{array}{c} \text{H} \\   \\ \text{H}-\text{C}-\text{H} \\   \\ \text{H} \end{array}$ | $\begin{array}{c} \text{H} \quad \text{H} \\   \quad   \\ \text{H}-\text{C}-\text{C}-\text{H} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$ | $\begin{array}{c} \text{H} \quad \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{H} \end{array}$ |
|----------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | colour changes from orange to colourless                                                    | colour stays unchanged                                                                                                                             | colour changes from orange to colourless                                                                                                                                 |
| <b>B</b> | colour changes from orange to colourless                                                    | colour changes from orange to colourless                                                                                                           | colour stays unchanged                                                                                                                                                   |
| <b>C</b> | colour stays unchanged                                                                      | colour stays unchanged                                                                                                                             | colour changes from orange to colourless                                                                                                                                 |
| <b>D</b> | colour stays unchanged                                                                      | colour changes from orange to colourless                                                                                                           | colour stays unchanged                                                                                                                                                   |

35 Which compound is formed when ethene reacts with steam?

- A** ethane
- B** ethanoic acid
- C** ethanol
- D** poly(ethene)

36 Which statements about ethanoic acid are correct?

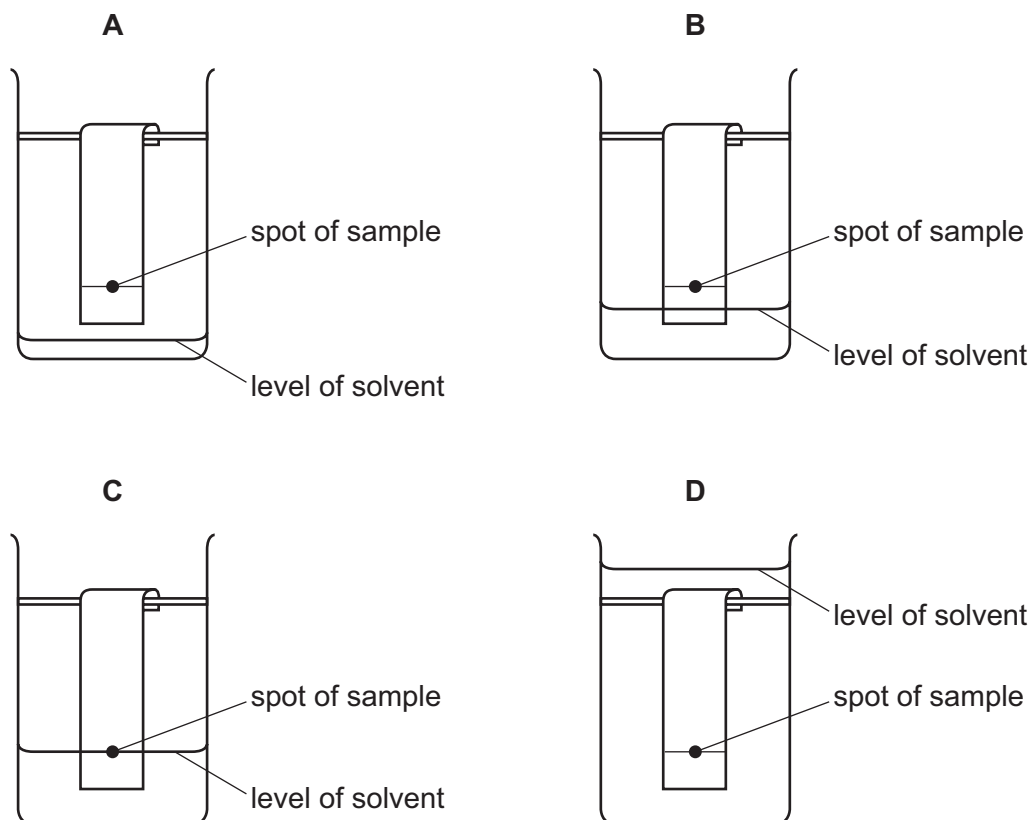
- 1 Its aqueous solution has a pH value of 10.
- 2 It reacts with metal carbonates to produce carbon dioxide gas.
- 3 It reacts with magnesium metal to produce hydrogen gas.

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

37 Which statement explains why plastic bags made from poly(ethene) accumulate in oceans?

- A** Poly(ethene) is a saturated molecule.
- B** Poly(ethene) is made from hydrocarbon monomers.
- C** Poly(ethene) reacts with water.
- D** Poly(ethene) does **not** decompose in water.

38 Which diagram shows the correct level of solvent at the start of a chromatography experiment?



39 Which process is used to separate a mixture of liquids with different boiling points?

- A dissolving
- B crystallisation
- C filtration
- D fractional distillation

40 Four different colourless solutions are each tested separately with aqueous sodium hydroxide and with acidified aqueous silver nitrate.

Which row shows the results for sodium chloride?

|   | aqueous sodium hydroxide | acidified aqueous silver nitrate |
|---|--------------------------|----------------------------------|
| A | no visible reaction      | white precipitate                |
| B | no visible reaction      | no visible reaction              |
| C | white precipitate        | no visible reaction              |
| D | white precipitate        | white precipitate                |



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The Periodic Table of Elements

| Group                      |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|------------------------------|-----------------------------|------------------------|
| I                          | II                          |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               | III                         | IV                          | V                           | VI                            | VII                          | VIII                        |                        |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 1<br>H<br>hydrogen<br>1     | 5<br>B<br>boron<br>11       | 6<br>C<br>carbon<br>12      | 7<br>N<br>nitrogen<br>14      | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19    | 10<br>Ne<br>neon<br>20 |
|                            |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
|                            |                             |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               |                             |                             |                             |                               |                              |                             |                        |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 |                                                                                             |                                 |                             |                              |                             |                              |                              |                                |                               |                               | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28   | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32       | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40     |                        |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40   | 21<br>Sc<br>scandium<br>45                                                                  | 22<br>Ti<br>titanium<br>48      | 23<br>V<br>vanadium<br>51   | 24<br>Cr<br>chromium<br>52   | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59     | 28<br>Ni<br>nickel<br>59       | 29<br>Cu<br>copper<br>64      | 30<br>Zn<br>zinc<br>65        | 31<br>Ga<br>gallium<br>70   | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79    | 35<br>Br<br>bromine<br>80    | 36<br>Kr<br>krypton<br>84   |                        |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                                                    | 40<br>Zr<br>zirconium<br>91     | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103   | 46<br>Pd<br>palladium<br>106   | 47<br>Ag<br>silver<br>108     | 48<br>Cd<br>cadmium<br>112    | 49<br>In<br>indium<br>115   | 50<br>Sn<br>tin<br>119      | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128  | 53<br>I<br>iodine<br>127     | 54<br>Xe<br>xenon<br>131    |                        |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137   | 57–71<br>lanthanoids                                                                        | 72<br>Hf<br>hafnium<br>178      | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184   | 75<br>Re<br>rhenium<br>186  | 76<br>Os<br>osmium<br>190    | 77<br>Ir<br>iridium<br>192   | 78<br>Pt<br>platinum<br>195    | 79<br>Au<br>gold<br>197       | 80<br>Hg<br>mercury<br>201    | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207     | 83<br>Bi<br>bismuth<br>209  | 84<br>Po<br>polonium<br>—     | 85<br>At<br>astatine<br>—    | 86<br>Rn<br>radon<br>—      |                        |
| 87<br>Fr<br>francium<br>—  | 88<br>Ra<br>radium<br>—     | 89–103<br>actinoids                                                                         | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>—   | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>—   | 108<br>Hs<br>hassium<br>—    | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>—  | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesson<br>— |                        |

|             |                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|-------------|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| lanthanoids | 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| actinoids   | 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).

# Cambridge IGCSE™ (9–1)

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**CHEMISTRY (9–1)****0971/12**

Paper 1 Multiple Choice (Core)

**May/June 2025**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | C      | 1     |
| 3        | D      | 1     |
| 4        | C      | 1     |
| 5        | D      | 1     |
| 6        | A      | 1     |
| 7        | A      | 1     |
| 8        | B      | 1     |
| 9        | D      | 1     |
| 10       | D      | 1     |
| 11       | B      | 1     |
| 12       | A      | 1     |
| 13       | C      | 1     |
| 14       | C      | 1     |
| 15       | D      | 1     |
| 16       | A      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | D      | 1     |
| 20       | B      | 1     |
| 21       | B      | 1     |
| 22       | A      | 1     |
| 23       | B      | 1     |
| 24       | A      | 1     |
| 25       | C      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | C      | 1     |
| 31       | B      | 1     |
| 32       | B      | 1     |
| 33       | A      | 1     |
| 34       | C      | 1     |
| 35       | C      | 1     |
| 36       | D      | 1     |
| 37       | D      | 1     |
| 38       | B      | 1     |
| 39       | D      | 1     |
| 40       | A      | 1     |



# Cambridge IGCSE™ (9–1)

## CHEMISTRY

0971/12

Paper 1 Multiple Choice (Core)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Water exists as solid ice, liquid water and steam.

In which states can water easily be compressed to half its original volume?

- A ice, liquid water and steam
- B liquid water and steam only
- C liquid water only
- D steam only

- 2 The melting point of sulfur is  $113^{\circ}\text{C}$ .

Which row describes the arrangement and movement of sulfur particles at  $25^{\circ}\text{C}$ ?

|   | arrangement of particles | movement of particles         |
|---|--------------------------|-------------------------------|
| A | regular                  | vibrate about fixed positions |
| B | regular                  | random                        |
| C | random                   | vibrate about fixed positions |
| D | random                   | random                        |

- 3 Solid Y is added to an excess of hot water.

A blue solution forms and a brown solid remains.

The brown solid is filtered off and dried.

The brown solid conducts electricity.

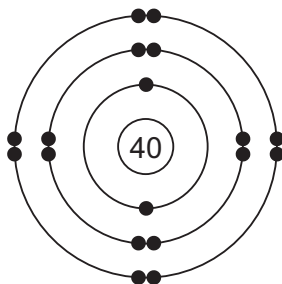
What is Y?

- A a compound that contains a metal
- B a mixture that contains a metal
- C a pure substance that is a metal
- D a pure substance that is a non-metal

- 4 Which statement about the structure of an atom is correct?

- A Electrons and protons each have a relative mass of 1, and neutrons have no charge.
- B Electrons have a relative mass of 1 and a negative charge.
- C Protons have a relative mass of 1 and a negative charge.
- D Neutrons have a relative mass of 1 and no charge.

- 5 The diagram shows the electronic configuration of a particle with a nucleon number of 40.

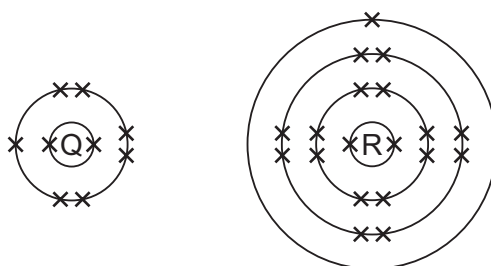


The table shows the suggestions that three students, 1, 2 and 3, made to identify the particle.

|          | student |    |                  |
|----------|---------|----|------------------|
|          | 1       | 2  | 3                |
| particle | Ar      | Cl | Ca <sup>2+</sup> |

Which students are correct?

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only
- 6 What is defined as different atoms of the same element with the same number of protons but different numbers of neutrons?
- A** homologous series  
**B** isotopes  
**C** mass number  
**D** the group number
- 7 The electronic configurations of atoms Q and R are shown.



Q and R form an ionic compound.

What is the formula of the compound?

- A** R<sub>7</sub>Q      **B** R<sub>4</sub>Q<sub>2</sub>      **C** RQ      **D** RQ<sub>7</sub>

8 Which properties are typical of simple molecules?

- 1 high melting point
- 2 good electrical conductivity
- 3 low boiling point
- 4 high malleability

**A** 1 and 2      **B** 2 and 4      **C** 3 and 4      **D** 3 only

9 Diamond is extremely hard and does **not** conduct electricity.

Which statement explains these properties?

- A** It has positive carbon ions surrounded by delocalised electrons.
- B** It has strong ionic bonds between each carbon atom.
- C** It has delocalised electrons, and each carbon atom forms three covalent bonds with other carbon atoms.
- D** It has **no** delocalised electrons, and each carbon atom forms four covalent bonds with other carbon atoms.

10 Which equation represents the complete combustion of butane,  $C_4H_{10}$ ?

- A**  $2C_4H_{10} + 5O_2 \rightarrow 8C + 10H_2O$
- B**  $2C_4H_{10} + 9O_2 \rightarrow 8CO + 10H_2O$
- C**  $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$
- D**  $C_4H_{10} + 4O_2 \rightarrow 4CO_2 + 5H_2$

11 What is the relative formula mass of calcium sulfate,  $CaSO_4$ ?

**A** 88      **B** 92      **C** 136      **D** 232

12 What is the product at each electrode when molten silver iodide is electrolysed using inert electrodes?

|          | cathode  | anode  |
|----------|----------|--------|
| <b>A</b> | hydrogen | iodine |
| <b>B</b> | iodine   | silver |
| <b>C</b> | silver   | iodine |
| <b>D</b> | silver   | oxygen |



- 13 Which row describes a product at each electrode when dilute sulfuric acid is electrolysed using inert electrodes?

|          | product at anode                       | product at cathode                     |
|----------|----------------------------------------|----------------------------------------|
| <b>A</b> | 10 cm <sup>3</sup> of a colourless gas | 10 cm <sup>3</sup> of a brown gas      |
| <b>B</b> | 10 cm <sup>3</sup> of a colourless gas | 20 cm <sup>3</sup> of a colourless gas |
| <b>C</b> | 20 cm <sup>3</sup> of a colourless gas | 10 cm <sup>3</sup> of a colourless gas |
| <b>D</b> | 20 cm <sup>3</sup> of a brown gas      | 10 cm <sup>3</sup> of a colourless gas |

- 14 Which experiment is the most exothermic?

|          | initial temperature / °C | final temperature / °C |
|----------|--------------------------|------------------------|
| <b>A</b> | 20                       | 5                      |
| <b>B</b> | 20                       | 32                     |
| <b>C</b> | 25                       | 12                     |
| <b>D</b> | 25                       | 34                     |

- 15 In which test-tube is a physical change taking place?

**A**

water

sodium chloride

**B**

water

iron nail

**C**

dilute hydrochloric acid

zinc

**D**

dilute hydrochloric acid

calcium carbonate

- 16 Which items of apparatus are essential when measuring the effect of temperature on the rate of any reaction?

- 1 balance
- 2 gas syringe
- 3 stop-watch
- 4 thermometer

**A** 1 and 2

**B** 1 and 3

**C** 2 and 4

**D** 3 and 4

**17** When pink cobalt(II) chloride is heated, it forms a blue solid and water.

When water is added to the blue solid, it releases heat and a pink solid is formed.

Which row describes the type of reaction that occurs and the pink cobalt(II) chloride?

|          | type of reaction | pink cobalt(II) chloride |
|----------|------------------|--------------------------|
| <b>A</b> | non-reversible   | anhydrous                |
| <b>B</b> | non-reversible   | hydrated                 |
| <b>C</b> | reversible       | anhydrous                |
| <b>D</b> | reversible       | hydrated                 |

**18** What is the oxidation number of iron in iron(III) oxide,  $\text{Fe}_2\text{O}_3$ ?

**A** +2

**B** +3

**C** -2

**D** -3

**19** Which substances react with dilute hydrochloric acid to form aqueous copper(II) chloride?

1 copper

2 copper(II) oxide

3 copper(II) carbonate

4 copper(II) hydroxide

**A** 1, 2, 3 and 4

**B** 1, 2 and 3 only

**C** 2, 3 and 4 only

**D** 2 and 3 only

- 20** A few drops of thymolphthalein and methyl orange are added separately to two test-tubes containing solution X.

Solution Y is then added to both test-tubes until present in excess.

The colours of the indicators before and after addition of solution Y are shown.

|                                             | colour of thymolphthalein | colour of methyl orange |
|---------------------------------------------|---------------------------|-------------------------|
| solution X only                             | colourless                | red                     |
| mixture of solution X and excess solution Y | blue                      | yellow                  |

What is solution Y?

- A**  $\text{HCl(aq)}$       **B**  $\text{NaOH(aq)}$       **C**  $\text{H}_2\text{O(l)}$       **D**  $\text{NaCl(aq)}$
- 21** Which mixture of solids is separated by adding water, stirring, and then using filtration?
- A** ammonium chloride and silver chloride  
**B** barium sulfate and calcium sulfate  
**C** copper(II) hydroxide and iron(III) hydroxide  
**D** potassium nitrate and sodium nitrate
- 22** The character of the elements and charges on the ions of the elements change across the Periodic Table.

Which row describes the elements on the left and the elements on the right of the Periodic Table?

|          | elements on the left |               | elements on the right |               |
|----------|----------------------|---------------|-----------------------|---------------|
|          | character            | charge on ion | character             | charge on ion |
| <b>A</b> | metallic             | positive      | non-metallic          | negative      |
| <b>B</b> | metallic             | negative      | non-metallic          | positive      |
| <b>C</b> | non-metallic         | positive      | metallic              | negative      |
| <b>D</b> | non-metallic         | negative      | metallic              | positive      |

- 23** Which statement describes a general trend down Group I?
- A** The melting point increases.  
**B** The reactivity decreases.  
**C** The density increases.  
**D** The total number of electrons decreases.

**24** Which statement describes the appearance of the named halogen at room temperature and pressure?

- A** Chlorine is a colourless gas.
- B** Chlorine is a red-brown liquid.
- C** Iodine is a purple gas.
- D** Iodine is a grey-black solid.

**25** Osmium is a transition element.

Which row gives the expected properties of osmium?

|          | melting point | density at r.t.p. | compounds formed |
|----------|---------------|-------------------|------------------|
| <b>A</b> | high          | high              | coloured         |
| <b>B</b> | high          | high              | white            |
| <b>C</b> | high          | low               | white            |
| <b>D</b> | low           | high              | coloured         |

**26** Which statement about the element neon, Ne, is correct?

- A** It is in Group VII of the Periodic Table.
- B** It reacts rapidly with oxygen.
- C** It exists as diatomic molecules.
- D** Its atoms have a full outer shell of electrons.

**27** Substance G reacts with:

- dilute acids to produce hydrogen
- steam to produce a basic oxide
- oxygen to produce a basic oxide.

What is substance G?

- A** magnesium
- B** phosphorus
- C** silicon
- D** silver

28 Which row describes two properties and a use of the named metal?

|          | metal     | property 1                   | property 2                   | use                   |
|----------|-----------|------------------------------|------------------------------|-----------------------|
| <b>A</b> | aluminium | high density                 | resistant to corrosion       | food containers       |
| <b>B</b> | aluminium | low density                  | good electrical conductivity | overhead power cables |
| <b>C</b> | copper    | good electrical conductivity | low ductility                | electrical wiring     |
| <b>D</b> | copper    | high melting point           | low density                  | aircraft manufacture  |

29 Four methods to prevent the rusting of iron are listed.

- 1 alloying
- 2 coating with plastic
- 3 greasing
- 4 painting

Which methods are barrier methods?

- A** 1, 2 and 3      **B** 1 and 4      **C** 2, 3 and 4      **D** 3 and 4 only

30 Iron is extracted from hematite in a blast furnace.

Which reaction involves thermal decomposition?

- A** Calcium carbonate reacts to form calcium oxide and carbon dioxide.  
**B** Carbon (coke) reacts with oxygen to form carbon dioxide.  
**C** Carbon dioxide reacts with carbon to form carbon monoxide.  
**D** Iron(III) oxide reacts with carbon monoxide to form iron and carbon dioxide.

- 31** Clean, dry air contains nitrogen, oxygen and small amounts of other gases.

The noble gases have been left out of the table.

Which row shows the composition of clean, dry air?

|          | nitrogen / % | oxygen / % | other gases                     |
|----------|--------------|------------|---------------------------------|
| <b>A</b> | 21           | 78         | small amount of carbon dioxide  |
| <b>B</b> | 21           | 78         | small amount of carbon monoxide |
| <b>C</b> | 78           | 21         | small amount of carbon dioxide  |
| <b>D</b> | 78           | 21         | small amount of carbon monoxide |

- 32** Some substances found in water from natural sources are potentially harmful.

Which substance contains microbes that cause disease?

- A** metal compounds
- B** nitrates
- C** phosphates
- D** sewage

- 33** Gas X is a waste gas from digestion in animals.

Gas Y is formed when gas X is burned with an insufficient amount of oxygen.

Gas Z is formed when gas X is burned with an excess of oxygen.

What are X, Y and Z?

|          | X               | Y               | Z               |
|----------|-----------------|-----------------|-----------------|
| <b>A</b> | carbon dioxide  | methane         | carbon monoxide |
| <b>B</b> | carbon monoxide | methane         | carbon dioxide  |
| <b>C</b> | methane         | carbon dioxide  | carbon monoxide |
| <b>D</b> | methane         | carbon monoxide | carbon dioxide  |

- 34** Which statement about photosynthesis is correct?

- A** It decreases the amount of oxygen in the atmosphere.
- B** It is a reaction between carbon dioxide and water.
- C** It produces a gas that contributes to increased global warming.
- D** It requires a temperature greater than 50 °C.

35 L and M are different homologous series.

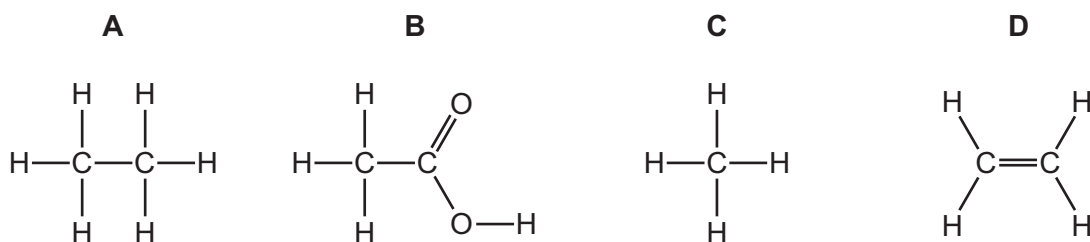
The first three members of homologous series L all decolourise aqueous bromine.

The first three members of homologous series M all dissolve in water to give solutions with a pH below 7.

What are the general formulae for homologous series L and M?

|          | homologous series L | homologous series M |
|----------|---------------------|---------------------|
| <b>A</b> | $C_nH_{2n}$         | $C_nH_{2n+1}COOH$   |
| <b>B</b> | $C_nH_{2n}$         | $C_nH_{2n+1}OH$     |
| <b>C</b> | $C_nH_{2n+2}$       | $C_nH_{2n+1}OH$     |
| <b>D</b> | $C_nH_{2n+2}$       | $C_nH_{2n+1}COOH$   |

36 Which structure represents an unsaturated hydrocarbon?



37 Petroleum is a mixture of compounds.

Which elements are present in all of these compounds?

- 1 carbon
- 2 hydrogen
- 3 oxygen

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

38 Which statement about ethanol explains why it is used as a fuel?

- A** It is a colourless liquid.
- B** It has a low density.
- C** It forms carbon dioxide when burned.
- D** It releases energy when burned.

- 39 The table shows the observations of two tests on an aqueous solution of a solid.

| test                         | observations                                                |
|------------------------------|-------------------------------------------------------------|
| add acidified silver nitrate | white precipitate                                           |
| add aqueous sodium hydroxide | green precipitate that dissolves in excess sodium hydroxide |

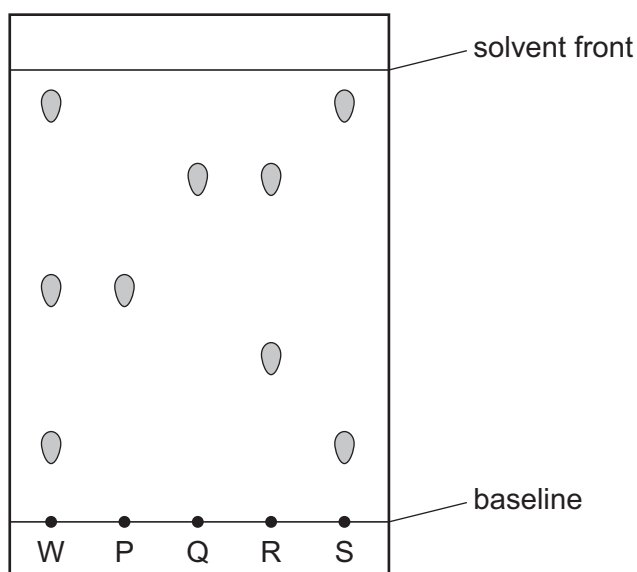
What is the solid?

- A chromium(III) bromide
- B chromium(III) chloride
- C iron(II) bromide
- D iron(II) chloride

- 40 W is a mixture of coloured dyes. It is thought to contain dyes P, Q, R or S.

A sample of W is separated by chromatography.

The chromatogram is shown.



What does the chromatogram show?

- A W contains P and S only.
- B W contains P and R only.
- C W contains P, Q and S.
- D W contains P, R and S.



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The Periodic Table of Elements

| Group                    |                             |                                                                                             |                           |                             |                         |                              |                         |                            |                           |                            |                            |                           |                            |                             |                          |                          |                          |                          |                         |                           |                             |                           |                            |                           |                             |                              |                             |                              |                             |                              |                               |                             |                              |                            |                              |                           |                            |                           |                              |                             |                                 |                           |                               |                               |                              |                             |                            |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
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| I                        | II                          |                                                                                             |                           |                             |                         |                              |                         |                            |                           |                            |                            | III                       | IV                         | V                           | VI                       | VII                      | VIII                     |                          |                         |                           |                             |                           |                            |                           |                             |                              |                             |                              |                             |                              |                               |                             |                              |                            |                              |                           |                            |                           |                              |                             |                                 |                           |                               |                               |                              |                             |                            |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
| 3<br>Li<br>lithium<br>7  | 4<br>Be<br>beryllium<br>9   | <div>Key</div> <div>atomic number<br/>atomic symbol<br/>name<br/>relative atomic mass</div> |                           |                             |                         |                              |                         |                            |                           |                            |                            | 1<br>H<br>hydrogen<br>1   | 5<br>B<br>boron<br>11      | 6<br>C<br>carbon<br>12      | 7<br>N<br>nitrogen<br>14 | 8<br>O<br>oxygen<br>16   | 9<br>F<br>fluorine<br>19 | 10<br>Ne<br>neon<br>20   | 18<br>Ar<br>argon<br>40 | 20<br>Ca<br>calcium<br>40 | 24<br>Mg<br>magnesium<br>24 | 28<br>Fe<br>iron<br>56    | 32<br>Zn<br>zinc<br>65     | 36<br>Kr<br>krypton<br>84 | 40<br>Sr<br>strontium<br>88 | 44<br>Ru<br>ruthenium<br>101 | 48<br>Cd<br>cadmium<br>112  | 52<br>Te<br>tellurium<br>128 | 56<br>Ba<br>barium<br>137   | 60<br>Nd<br>neodymium<br>144 | 64<br>Gd<br>gadolinium<br>157 | 68<br>Er<br>erbium<br>167   | 72<br>Hf<br>hafnium<br>178   | 76<br>Os<br>osmium<br>190  | 80<br>Hg<br>mercury<br>201   | 84<br>Po<br>polonium<br>— | 88<br>Ra<br>radium<br>—    | 92<br>U<br>uranium<br>238 | 96<br>Pu<br>plutonium<br>244 | 100<br>Fm<br>fermium<br>257 | 104<br>Rf<br>rutherfordium<br>— | 108<br>Hs<br>hassium<br>— | 112<br>Cn<br>copernicium<br>— | 116<br>Lv<br>livermorium<br>— | 120<br>Ts<br>tennessine<br>— | 124<br>Og<br>oganesson<br>— |                            |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
|                          |                             |                                                                                             |                           |                             |                         |                              |                         |                            |                           |                            |                            |                           |                            |                             |                          |                          |                          |                          |                         |                           |                             |                           |                            |                           |                             |                              |                             |                              |                             |                              |                               |                             |                              |                            |                              |                           |                            |                           |                              |                             |                                 |                           |                               |                               |                              |                             |                            |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
|                          |                             |                                                                                             |                           |                             |                         |                              |                         |                            |                           |                            |                            |                           |                            |                             |                          |                          |                          |                          |                         |                           |                             |                           |                            |                           |                             |                              |                             |                              |                             |                              |                               |                             |                              |                            |                              |                           |                            |                           |                              |                             |                                 |                           |                               |                               |                              |                             |                            |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
| 11<br>Na<br>sodium<br>23 | 12<br>Mg<br>magnesium<br>24 | 13<br>Al<br>aluminium<br>27                                                                 | 14<br>Si<br>silicon<br>28 | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32 | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40 | 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40 | 21<br>Sc<br>scandium<br>45 | 22<br>Ti<br>titanium<br>48 | 23<br>V<br>vanadium<br>51 | 24<br>Cr<br>chromium<br>52 | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56   | 27<br>Co<br>cobalt<br>59 | 28<br>Ni<br>nickel<br>59 | 29<br>Cu<br>copper<br>64 | 30<br>Zn<br>zinc<br>65  | 31<br>Ga<br>gallium<br>70 | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75 | 34<br>Se<br>selenium<br>79 | 35<br>Br<br>bromine<br>80 | 36<br>Kr<br>krypton<br>84   | 37<br>Rb<br>rubidium<br>85   | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89     | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93    | 42<br>Mo<br>molybdenum<br>96  | 43<br>Tc<br>technetium<br>— | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108 | 48<br>Cd<br>cadmium<br>112 | 49<br>In<br>indium<br>115 | 50<br>Sn<br>tin<br>119       | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128    | 53<br>I<br>iodine<br>127  | 54<br>Xe<br>xenon<br>131      | 55<br>Cs<br>caesium<br>133    | 56<br>Ba<br>barium<br>137    | 57–71<br>lanthanoids        | 72<br>Hf<br>hafnium<br>178 | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184 | 75<br>Re<br>rhenium<br>186 | 76<br>Os<br>osmium<br>190 | 77<br>Ir<br>iridium<br>192 | 78<br>Pt<br>platinum<br>195 | 79<br>Au<br>gold<br>197 | 80<br>Hg<br>mercury<br>201 | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207 | 83<br>Bi<br>bismuth<br>209 | 84<br>Po<br>polonium<br>— | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>— | 87<br>Fr<br>francium<br>— | 88<br>Ra<br>radium<br>— | 89–103<br>actinoids | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>— | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>— | 108<br>Hs<br>hassium<br>— | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>— | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesson<br>— |

lanthanoids

|                              |                            |                                 |                              |                             |                             |                             |                               |                            |                               |                              |                           |                               |                              |                              |
|------------------------------|----------------------------|---------------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------|-------------------------------|------------------------------|------------------------------|
| 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140  | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>— | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165   | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169    | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175  |
| 89<br>Ac<br>actinium<br>—    | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238    | 93<br>Np<br>neptunium<br>—  | 94<br>Pu<br>plutonium<br>—  | 95<br>Am<br>americium<br>—  | 96<br>Cm<br>curium<br>—       | 97<br>Bk<br>berkelium<br>— | 98<br>Cf<br>californium<br>—  | 99<br>Es<br>einsteinium<br>— | 100<br>Fm<br>fermium<br>— | 101<br>Md<br>mendelevium<br>— | 102<br>No<br>nobelium<br>—   | 103<br>Lr<br>lawrencium<br>— |

actinoids

The volume of one mole of any gas is 24 dm<sup>3</sup> at room temperature and pressure (r.t.p.).

# Cambridge IGCSE™ (9–1)

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**CHEMISTRY****0971/12**

Paper 1 Multiple Choice (Core)

**October/November 2025****MARK SCHEME**Maximum Mark: 40

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | D      | 1     |
| 2        | A      | 1     |
| 3        | B      | 1     |
| 4        | D      | 1     |
| 5        | C      | 1     |
| 6        | B      | 1     |
| 7        | C      | 1     |
| 8        | D      | 1     |
| 9        | D      | 1     |
| 10       | C      | 1     |
| 11       | C      | 1     |
| 12       | C      | 1     |
| 13       | B      | 1     |
| 14       | B      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | C      | 1     |
| 20       | B      | 1     |
| 21       | A      | 1     |
| 22       | A      | 1     |
| 23       | C      | 1     |
| 24       | D      | 1     |
| 25       | A      | 1     |
| 26       | D      | 1     |
| 27       | A      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | A      | 1     |
| 31       | C      | 1     |
| 32       | D      | 1     |
| 33       | D      | 1     |
| 34       | B      | 1     |
| 35       | A      | 1     |
| 36       | D      | 1     |
| 37       | B      | 1     |
| 38       | D      | 1     |
| 39       | B      | 1     |
| 40       | A      | 1     |